DNX16

Hardware Design Guidelines for StrataDNX™ 16-nm Devices

Application Note
BCM88790
BCM88690
BCM88800/BCM88820
BCM88480
BCM88280
BCM88290

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Table of Contents

Chapter 1: Introduction 9

Chapter 2: Blackhawk SerDes Interface 10

2.1 Blackhawk Performance 10

2.1.1 Blackhawk Electrical Specification Compliance 11

2.1.2 Channel Compliance Validation 11

2.1.3 Blackhawk Modeling and Simulations 13

2.1.3.1 Blackhawk IBIS-AMI Model 13

2.1.3.2 Device Package Model 13

2.1.3.3 Pass/Fail Criteria 14

2.2 Blackhawk P/N Polarity Inversion 15

2.3 Blackhawk Lane Mapping (Swapping) 16

2.3.1 Blackhawk Lane Mapping – Fabric 16

2.3.1.1 Overview and Terminology 16

2.3.1.2 Fabric Lane-Mapping Rules 18

2.3.2 Blackhawk Lane Mapping – NIF Ethernet 20

2.3.2.1 Overview and Terminology 20

2.3.2.2 NIF Ethernet Lane-Mapping Rules 21

2.3.3 Blackhawk Lane Mapping – BCM88690 ELK-over-ILKN 22

2.3.3.1 Overview and Terminology 22

2.3.3.2 ELK-over-ILKN Lane-Mapping Rules 23

2.3.4 Blackhawk Lane Mapping – BCM88800 ILKN 24

2.3.4.1 Overview and Terminology 24

2.3.4.2 ILKN Lane-Mapping Rules 24

2.3.4.3 FEC Rules 25

2.3.5 Blackhawk Lane Mapping – BCM88480 ILKN 26

2.3.5.1 Overview and Terminology 26

2.3.5.2 ILKN Lane-Mapping Rules 26

2.3.5.3 FEC Rules 26

2.4 Unused Blackhawk Links 27

2.5 Blackhawk AC Coupling 27

2.5.1 Blackhawk RX 27

2.5.2 Blackhawk TX 27

2.6 Blackhawk Hot Plug 29

2.7 Blackhawk Layout Guidelines 30

2.7.1 Trace Impedance 30

2.7.2 Differential Pair Geometry and Stack-Up 30

2.7.2.1 Loosely and Tightly Coupled Differential Pair Structure 30

2.7.2.2 Stripline and Reference Planes 30
2.7.2.3 Trace Clearances 31
2.7.2.4 Ground Fills and Pours 31
2.7.3 Vias and PTHs 32
2.7.3.1 Via Drill Diameter 32
2.7.3.2 Keep Two Vias per SerDes Lane 32
2.7.3.3 Remove Pads on Unused Layers 33
2.7.3.4 Controlling the Via or PTH Stub 33
2.7.3.5 Power and GND Via Stub 34
2.7.3.6 Special Anti-Pad Shapes 34
2.7.3.7 Back-Drilling PTH of Press-Fit Connectors 35
2.7.3.8 GND-Shadow Vias 36
2.7.3.9 Adding GND Vias between Pairs of Differential Signal Vias 36
2.7.3.10 Connector Return-loss and Crosstalk 36
2.7.4 Maintaining Low Crosstalk 36
2.7.5 PCB Material 37
2.7.5.1 Microstrip Plating and Finish 37
2.7.5.2 Dielectric Material Adjacent to Routing Layers 37
2.7.5.3 Dielectric Glass Weave 38
2.7.6 Blackhawk Skew and Length Matching 39
2.7.6.1 Pair P/N Skew 39
2.7.6.2 Lane (Pair-to-Pair) Skew 40
2.7.7 AC Coupling Capacitors 41
2.8 External Re-Timer Device Guidelines 42
2.8.1 Fabric FEC 42
2.8.2 Link-Training and Force TX FIR Settings 43
2.8.2.1 Link-Training Connectivity Scheme – General Discussion 43
2.8.2.2 Blackhawk Link-Training vs. Force TX FIR 44
2.8.2.3 Blackhawk PAM-4-Supported Schemes 44
2.8.2.4 Blackhawk NRZ-Supported Schemes 45
2.8.3 Fabric Latency 46
2.8.4 Clock Recovery 46
2.8.5 Electrical Interface to the DNX16 Device 47
2.9 Distributed Disaggregated Chassis Considerations 48
2.9.1 DNX Fabric Links Are Not Ethernet 48
2.9.2 Each Front-Panel Connector Is Connected to More Than One Device 48
2.9.3 Considerations for the Fabric Interconnect on a DDC 49
2.9.3.1 Example 1: Clock Synchronization of the Interconnect Lanes 49
2.9.3.2 Example 2: 400GBASE-DR4 Modules 49
2.9.3.3 Example 3: Lane Status 50

Chapter 3: Falcon16 SerDes Interface 51
3.1 Falcon16 Performance 51
3.2 Falcon16 PLL Frequency and Lane Rate 51
3.3 Falcon16 P/N Polarity Inversion 51
3.4 Falcon16 Lane Mapping 52
3.4.1 Falcon16 Lane Mapping – NIF Ethernet 52
3.4.2 Falcon16 Lane Mapping – BCM88800 ILKN 52
3.4.3 Falcon16 Lane Mapping – BCM88480 ILKN 52
3.5 Unused Falcon16 Links 52
3.6 Falcon16 AC Coupling 52
3.7 Falcon16 Hot Plug 52
3.8 Falcon16 Layout Guidelines 53
3.9 External Re-Timer Device Guidelines 53
Chapter 4: Statistics Interface SerDes Limitations 54
Chapter 5: PCIe Interface 55
5.1 DNX16 PCIe Interface Overview 55
5.2 PCIe Link and Lanes 56
5.2.1 PCIe Link 56
5.2.2 PCIe Polarity Swap 56
5.2.3 Link Forming and Lane Configuration 56
5.2.4 Unused Lanes 56
5.2.5 PCIe AC Coupling 57
5.3 PCIe QSPI Flash 58
5.3.1 QSPI Controller External Interface (BSPI and MSPI Modes) 58
5.3.2 QSPI Flash Device Selection Guidelines 59
5.3.3 QSPI Flash Device Part Numbers for Reference 60
5.3.4 QSPI Interface Connectivity 61
5.3.5 DNX16 QSPI Flash Programming 62
5.4 PCIe Layout Guidelines 66
5.4.1 Trace Impedance 66
5.4.2 Skew and Length Match 66
5.5 Calibration Resistors 66
5.6 PCIe Modeling and Simulations 67
Chapter 6: High Bandwidth Memory 68
6.1 HBM DA Test Port 68
6.2 X-Ray Inspection Considerations 68
Chapter 7: GDDR6 Interfaces (BCM88480 and BCM88290) 69
7.1 Introduction 69
7.2 Logical Connectivity 71

7.3 GDDR6 Device Selection 73
7.4 GDDR6 DRAM Interface Signal Description 74
7.4.1 GDDR6 Part Number Dependent Signals 76
7.4.1.1 WCK 76
7.4.1.2 CA[10] 76
7.5 Special Signals 78
7.5.1 BCM88480 VREF_MON Pins 78
7.5.2 BCM88480/BCM88290 Calibration Pins 78
7.5.3 GDDR6 Device VREFC Pin 79
7.5.4 GDDR6 Device ZQ Calibration Pins 79
7.5.5 GDDR6 Device JTAG 80
7.5.6 GDDR6 RESET_n 80
7.6 PCB Routing Guidelines 81
7.6.1 Trace Impedance 81
7.6.2 Reference Planes 81
7.6.3 CK and WCK 81
7.6.4 Analog Signals 82
7.6.5 Layer Partitioning 82
7.6.6 Signal Vias 83
7.6.7 Power Vias and Vias Related to Decoupling 83
7.6.8 Power Planes 83
7.6.9 Trace Clearances 84
7.7 GDDR6 Signal Length and Skew Matching 85
7.8 Channel Swapping 86
7.9 DQ Swapping 86
7.9.1 BCM88480 DQ Swap SOC Example 87
7.9.2 BCM88290 DQ Swap SOC Example 89
7.10 BCM88480 GDDR6 Connectivity and Swapping 90
7.11 BCM88290 GDDR6 Connectivity and Swapping 94
Chapter 8: Clock and PLL 97
8.1 Differential Clock Driving 97
8.2 FAB/NIF REFCLK 102
8.2.1 FAB/NIF REFCLK Routing 102
8.2.2 Suggested Part Numbers 102
8.2.3 BCM88790 Re-Timer Mode – Special Requirements 103
8.3 BCM88480 FlexE Reference Clocks 104
Chapter 9: Power Aspects 106
9.1 VDDC Core Supply 106
9.1.1 VDDC Core Recommended Operation Voltage 107

9.1.2 VDDC Core Power Supply Voltage Accuracy 107
9.1.3 VDDC Core IR Drop and Sense Compensation 108
9.1.4 Measure and Validate VDDC 108
9.1.5 VDDC Supply Filtering and Decoupling 109
9.2 Blackhawk SerDes Supply 110
9.2.1 Blackhawk SerDes Supply – PVDD0P8 110
9.2.2 Blackhawk SerDes Supply – PVDD1P8 112
9.2.3 Blackhawk SerDes Supply – TVDD0P8 and RVDD0P8 113
9.2.4 Blackhawk SerDes Supply – TVDD1P2 114
9.2.5 SRD_AGND 114
9.3 Falcon16 SerDes Supply 115
9.4 PCIe SerDes Supply 115
9.5 HBM Power Supply 116
9.5.1 BCM88690 HBM Power Supply 116
9.5.2 BCM88800 HBM Power Supply 118
9.6 PLL AVDD1P8 Supply 118
9.7 AVS_VDD1P8 Supply 119
9.8 Supply Pull-Down Load Resistors 119
9.9 I/O Fail-Safe Considerations 120
9.10 GDDR6-Related Power Supply 122
9.10.1 BCM88480 GDDR6-Related Power Supply 122
9.10.1.1 BCM88480 DDR_VDDO and GDDR6 VDDQ 123
9.10.1.2 BCM88480 DDR_VDDC 124
9.10.1.3 BCM88480 DDR#_AVDD1P8 125
9.10.1.4 BCM88480 DDR#_V_ANALOG 126
9.10.1.5 GDDR6 VDD 127
9.10.1.6 GDDR6 VPP 128
9.10.2 BCM88290 GDDR6-Related Power Supply 129
Chapter 10: Miscellaneous Interfaces 130
10.1 RESCAL 130
10.2 LCPLL Calibration Resistors 130
10.3 DISCONNECT 130
10.4 INT_N 130
10.5 I^2C 131
10.5.1 Using I^2C (Software Aspects) 131
10.5.2 I^2C NACK Response 131
Chapter 11: Hardware Connectivity for IEEE 1588 and Time-of-Day User Applications 132
11.1 Ordinary/Boundary Clock – External IEEE 1588 Engine 133
11.2 Ordinary/Boundary Clock – DNX BroadPTP (Pizza-Box) 135

11.3 Transparent Clock 137
11.4 ToD for User Applications 138
Chapter 12: Thermal Aspects 139
12.1 Heat Sink Considerations 139
12.1.1 Heat Sink Selection 139
12.1.2 Heat Sink Attachment 139
12.2 Thermal Monitoring 142
12.2.1 Continuous-Operation Thermal Monitoring 142
12.2.2 Power-Down or Pre-SDK Temperature Reading 142
12.3 BCM88690 and BCM88800 HBM Thermal Design Considerations 143
12.3.1 Temperature Operating Regions 143
12.3.2 Thermal Simulations 144
12.3.3 Thermal Supervision 144
12.3.3.1 HBM Temperature Monitor Enabled 145
12.3.3.2 Temperature Thresholds 145
12.3.4 CATTRIP 146
12.4 Industrial-Grade Devices 147
12.4.1 BCM88800 Industrial-Grade Information 147
12.4.2 BCM88790 Industrial-Grade Information 148
12.4.3 BCM88480/BCM88280/BCM88290 Industrial-Grade Information 148
Chapter 13: PCB Land Pattern and Reflow Process Recommendations 149
Appendix A: Known Issues for SVK Platforms 150
A.1 BCM88790 SVK 150
A.2 BCM88690 SVK 150
A.3 BCM88480 SVK 151
A.4 BCM88290 SVK 152
Appendix B: DNX Hardware Reviews 153
Related Documents 155

Chapter 1: Introduction

This document describes the hardware design guidelines for the Broadcom® StrataDNX™ 16-nm family, referred to as DNX16 throughout this document. The Broadcom DNX16 family includes the following devices:

- BCM88790 (Ramon)
- BCM88690 (Jericho2)
- BCM88800/BCM88820 (Jericho2c/Qumran2c)
- BCM88480 (Qumran2a)
- BCM88280 (Qumran2u)
- BCM88290 (Qumran2n)

NOTE:

- BCM88800 is used throughout this document for both the BCM88800 and BCM88820 devices, and all information that applies to the BCM88800 device also applies to the BCM88820 device.
- BCM88480 is used throughout this document for both the BCM88480 and BCM88280 devices, and all information that applies to the BCM88480 device also applies to the BCM88280 device.

This document focuses on the external interfaces, power handling, and the hardware environment required by these devices. Most guidelines are common to all devices or multiple devices, but some guidelines are device specific.

Chapter 2: Blackhawk SerDes Interface

On the DNX16 devices, the Blackhawk SerDes is used for fabric and network interfaces. Blackhawk is a SerDes block supporting eight serial links. Each Blackhawk core consists of eight SerDes (referred to as an *octet*) that is made up of two *quads*. A quad includes four TX/RX SerDes pairs.

NOTE: For supported rates, refer to the data sheet for the specific device.

2.1 Blackhawk Performance

Blackhawk is designed to achieve a BER of 10^{-15} in the following modes:

- NRZ (Non-Return to Zero) mode without FEC
- PAM-4 (Pulse Amplitude Modulation) mode with RS-FEC (Reed-Solomon Forward Error Correction)

The target BER is achieved through design and validation, and by supporting a specific set of electrical specification standards. See Section 2.1.1, Blackhawk Electrical Specification Compliance.

To validate that the system can meet the target BER, perform the following two required procedures:

1. Validate channel compliance as defined in the relevant IEEE or OIF standard by using the COM method and the channel masks, as described in Section 2.1.2, Channel Compliance Validation.
2. Run IBIS-AMI simulations for BER performance validation, as described in Section 2.1.3, Blackhawk Modeling and Simulations.

NOTE: Performing both procedures is mandatory, as each procedure covers different aspects of the standards compliance.

The following table summaries the requirements that are detailed in the following sections.

Table 1: Electrical Specification Compliance Summary

Validation Type	What to Validate	Includes	Required/Reference
Channel compliance validation	Meeting the relevant standard (IEEE or OIF) channel requirements	COM[a]	Required
		IL and RL masks	Required
	Meeting the Blackhawk-specific masks	ILD and ICN masks	Required
IBIS-AMI simulations	Meeting the target BER, end-to-end	IBIS-AMI simulations[a]	Required

a. The target BER should be 1.5×10^{-5} or better for KR and CR, and 1×10^{-5} or better for C2C.

For example, in the IEEE 802.3cd CL137 KR specification:

- The COM is described in Section 137.10, Channel characteristics and Table 137-5-COM parameter values.
- The IL is described in Section 137.10.1, Channel insertion loss.
- The RL is described in Section 137.10.2, Channel return loss.

Other system-level considerations include the following:

- Optical modules at a rate above 10G without a CDR/re-timer are not supported by the DNX16 devices. These modules include (but are not limited to) 100GBASE-SR4/LR4 without a CDR/re-timer.
- Do not use modules that provide invalid signals toward the Blackhawk, in case of link-down (that is, modules that are not equipped with RX squelch). Support of such modules is limited, and the appropriate operation of the DNX16 device is not guaranteed (mainly on link-down and link-up events).

2.1.1 Blackhawk Electrical Specification Compliance

Each device data sheet includes a table describing the electrical specifications the Blackhawk core in the DNX16 is targeted to support.

There are separate tables for fabric, network interface (NIF) Ethernet, and External Lookup (ELK) over Interlaken (ILKN).

2.1.2 Channel Compliance Validation

To validate channel compliance, the procedures and requirements detailed in the relevant standard (IEEE or OIF) must be fulfilled. Validating that a channel is compliant usually includes meeting the Channel Operating Margin (COM) method as well as meeting the Insertion-Loss (IL) and Return-Loss (RL) masks.

For details, refer to the relevant IEEE/OIF standard.

NOTE: The requirement for PAM-4 simulation is to have a pre-FEC BER of 1.5×10^{-5} for KR and CR links (as opposed to default IEEE of 1×10^{-4}) and of 1×10^{-5} for C2C (the same as the IEEE standard). This value guarantees a post-FEC BER of 10^{-15}.

IMPORTANT: Performing the Channel Operating Margin (COM) is required.

What is the *Channel Operating Margin (COM)*?

The COM is the channel compliance method adopted by the IEEE. It provides a figure that relates to the ratio of the calculated signal amplitude to the calculated noise amplitude. As an output, the COM generates a single number, in dB, that reflects the system margin for operating in the required BER.

As an input to the process, the COM requires the channel to have S-parameters. The S-parameters can be acquired by modeling tools or by actual measurements.

The COM makes assumptions on the SerDes behavior (RX and TX), cross-talk aggressors, and on the device package model. In other words, the COM does not use the specific device properties (unlike IBIS-AMI simulations) but relates only to the channel (ball-to-ball).

In addition to meeting the relevant standard (IEEE or OIF) requirement (COM, IL mask and RL mask), the following two Blackhawk masks must be met:

- Insertion Loss Deviation (ILD) – See Figure 1
- Integrated Crosstalk Noise (ICN) – See Figure 2

Figure 1: Insertion Loss Deviation Specification (53.125G PAM-4)

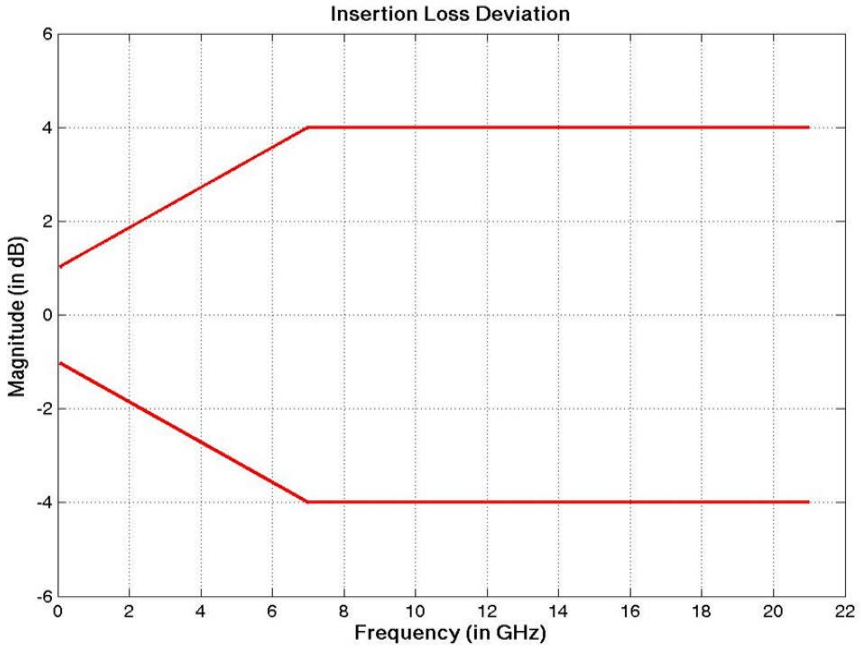
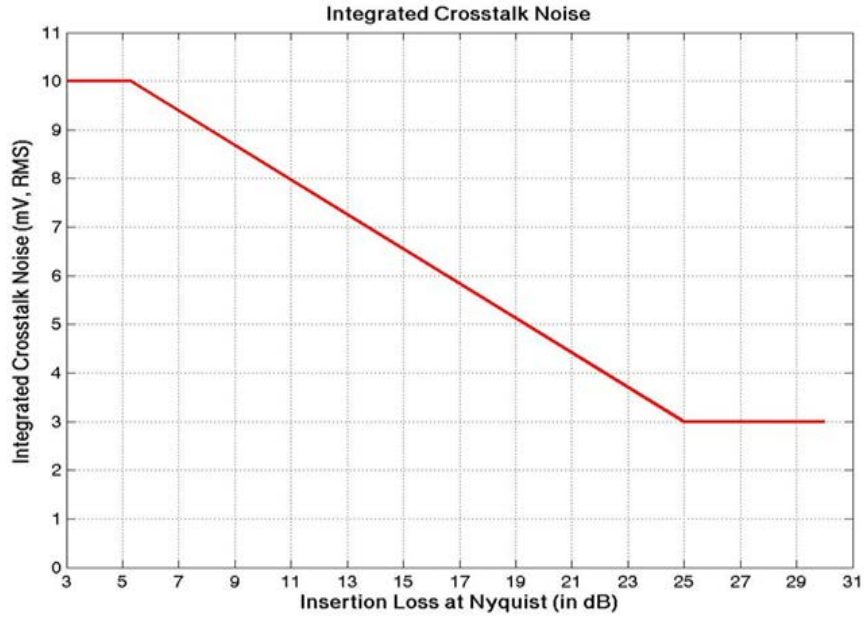


Figure 2: Integrated Crosstalk Noise (ICN) Specification (53.125G PAM-4)



NOTE: All masks are ball-to-ball.

NOTE: Performing full-channel IBIS-AMI simulations is always required for Blackhawk links. Even if the ICN value seems low (even as low as 0.5 mV RMS), performing such simulations is still mandatory.

2.1.3 Blackhawk Modeling and Simulations

For a high-speed SerDes, such as Blackhawk, simulations are required to validate the system design and to provide performance estimations in the form of BER.

The simulations should include the worst-case channel, and a detailed analysis of the system should be performed to define the worst-case channel for each scenario (insertion loss, return loss, NEXT, FEXT). Pending the specific system design, the scenario can be the shortest link, the longest link, a long link that resides near short link, and so on. .

2.1.3.1 Blackhawk IBIS-AMI Model

The Blackhawk IBIS-AMI model (available on the Engineering Support Portal) includes the following:

- TX path:
  - Analog-Front-End (AFE), modeled by a separate S-parameter (for example: blackhawk_onchip_tx_v1.s4p)
  - TX preemphasis equalizer
- RX path:
  - AFE modeled by a separate S-parameter (for example: blackhawk_onchip_rx_v1.s4p)
  - RX equalizer (PF, VGA, and DFE)

NOTE:

- The Blackhawk SerDes is not supported by the Broadcom LinkEye™ tool
- In addition to the system-level simulations, it is necessary to simulate and optimize each element of the design at early design stages, including BGA breakout, differential via structure, differential routing, connectors, and so on.

2.1.3.2 Device Package Model

On the IBIS-AMI simulations, use the specific package model of the device being used. The package model is in an s64p file that represents the full SerDes octet. The s64p file is available on the Engineering Support Portal (ESP) on the Broadcom website. See Table 51, IBIS-AMI Models.

Two models are usually delivered (depending on the package size): one for the octet with the longest traces in the package, and one for the octet with the shortest traces in the package.

The package model file is especially useful for crosstalk simulations.

NOTE: The Insertion-Loss values available in the model are applicable for operation at 20°C. When the device is operating at 110°C, add ~5% to those values.

2.1.3.3 Pass/Fail Criteria

The `rx.ami` file includes the *RX Sensitivity* parameter. The EDA tool uses this parameter to determine the simulated link-eye height and width openings at a given BER.

Because the eye opening reported by the EDA tool already takes into account the RX sensitivity, do not compare the eye opening reported by the EDA tool to the *RX Sensitivity* and determine pass/fail.

NOTE: For reference, the Blackhawk *RX Sensitivity* is ±10 mV. Refer to the current IBIS-AMI model on the Engineering Support Portal (ESP) for the most up-to-date information.

Determining pass/fail is different between NRZ and PAM-4 modes, as described in the following sections.

2.1.3.3.1 NRZ

Use an RX eye opening of 20 mVpp and 2 ps as the recommended pass/fail criteria.

The simulated BER of the system is the BER in which the eye opening is at least at the size of this recommended eye opening.

Usually, the minimum requirement for 10G or 25G SerDes is a BER of 10^{-15}.

Based on the eye opening at a BER of 10^{-12} and 10^{-15}, the user can calculate the Signal-to-Noise Ratio (SNR) margin at other points.

The following formulas can be used for calculating the SNR margin at specific BER:

$$SNR_{margin_vertical}[dB] = \frac{height@BER - 20\ mV}{(height@10^{-12}) - (height@10^{-15})}$$

$$SNR_{margin_horizontal}[dB] = \frac{width@BER - 2\ ps}{(width@10^{-12}) - (width@10^{-15})}$$

NOTE: This calculation assumes that the difference between 10^{-12} and 10^{-15} yields a 1-dB SNR margin.

2.1.3.3.2 PAM-4

The Blackhawk PAM-4 IBIS-AMI is a pre-FEC model, and so the BER at which eye-height and eye-width are evaluated is very high (for example 10^{-4} or 10^{-5}). There is no RX eye opening as recommended pass/fail criteria, but instead, the overall BER is the only measurement of the RX performance. The EDA tool evaluates RX performance on all eyes (based on the RX Sensitivity) and should be used for pass/fail criteria.

In other words, the eye provided by the EDA tool should not be compared to any pass/fail criteria mask as in NRZ. The BER reported by the EDA tool is the pre-FEC BER of the link. This pre-BER should be converted to post-FEC BER.

NOTE: The PAM-4 simulation must have a pre-FEC BER of 1.5×10^{-5} for KR and CR links (as opposed to the default IEEE requirement of 1×10^{-4}) and of 1×10^{-5} for C2C (the same as the IEEE standard). This value guarantees a post-FEC BER of 10^{-15}.

2.2 Blackhawk P/N Polarity Inversion

The Blackhawk core supports SerDes polarity inversion. The polarity of the P and N taps of each pair can be individually inverted.

This feature is useful in addressing PCB layout issues related to P/N deskewing.

Software configuration is controlled as follows:

- During device initialization, polarity inversion is supported (using the appropriate SOC property) for all device and interface types.
- After the device has been initialized, polarity inversion (without removing the port) is supported only on the NIF interfaces and is not supported on the fabric interfaces.

2.3 Blackhawk Lane Mapping (Swapping)

Lane mapping helps ease and optimize the PCB routing. Lane mapping may have different rules and considerations based on the application being used.

The information in this section identifies the DNX16 Blackhawk requirements for lane mapping.

NOTE: Lane mapping (swapping) can be set during device initialization. When setting lane mapping after the device has been initialized, all ports must first be removed from the 8-lane core SerDes. This feature is supported for all DNX16 devices and interfaces.

2.3.1 Blackhawk Lane Mapping – Fabric

2.3.1.1 Overview and Terminology

The following figure shows the basic structure of the fabric interface, including the fabric MAC (FMAC) and the Blackhawk SerDes. On the Blackhawk SerDes, no RX or TX mapping (swapping) exists. On the FMAC, only TX mapping (swapping) exists.

The term *quad* is used throughout this section to refer to a set of four RX/TX pairs. Two quads form the Blackhawk SerDes octet, which supports eight lanes. In each Blackhawk SerDes block, lanes 0 to 3 belong to one quad, and lanes 4 to 7 belong to the other quad.

Like the Blackhawk SerDes, the FMAC is also made out of two quads.

Figure 3: Fabric Interface Basic Structure

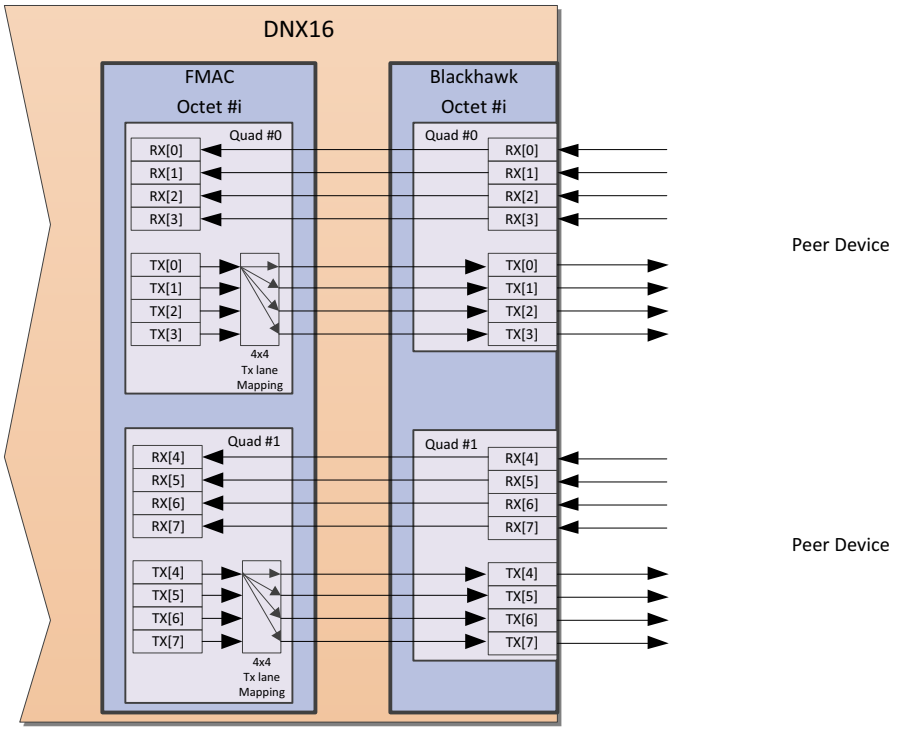
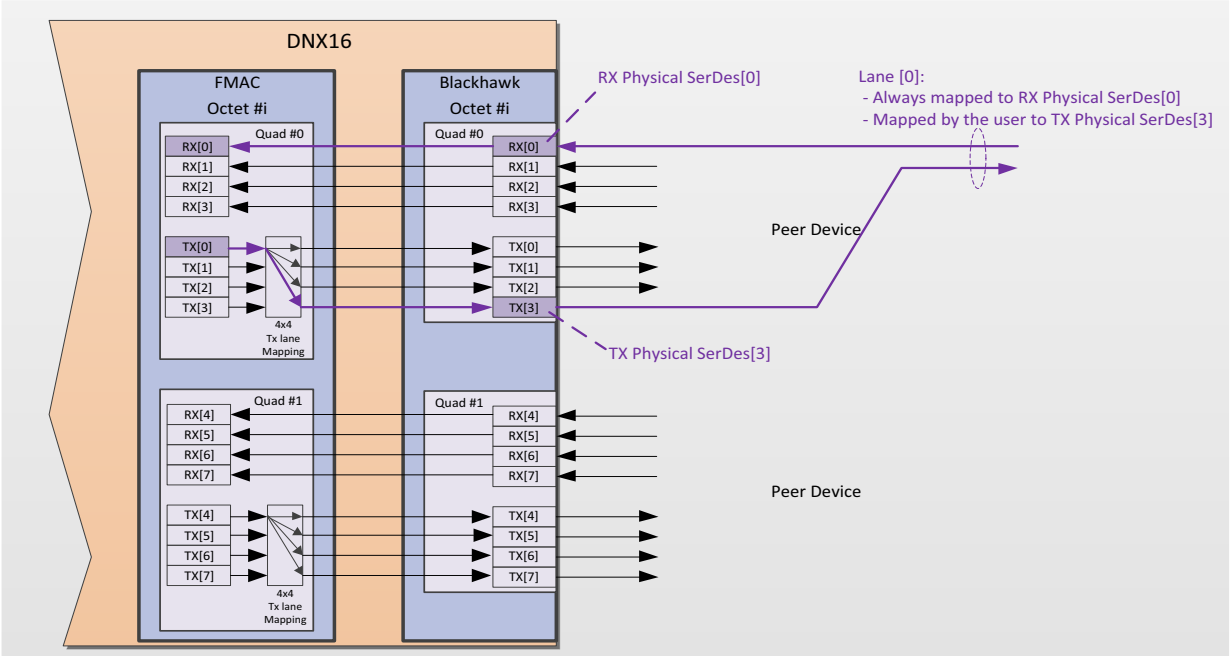


Figure 4 demonstrates the following terms that are used throughout this section:

- RX physical SerDes (on the device ball-out; for example, on the BCM88790, E52 and D52 are FAB_RX_020_P and FAB_RX_020_N).
- TX physical SerDes (on the device ball-out; for example, on the BCM88790, G57 and F57 are FAB_TX_016_P and FAB_TX_016_N).
- Lane, which is a SW entity and includes a pair of RX physical SerDes and TX physical SerDes. Usually, the lane also represents the pairing in the customer board.

NOTE: The index number of the lane is identical to the index number of the RX physical SerDes (in other words, no RX mapping exists).

Figure 4: Lane and RX/TX Physical SerDes



2.3.1.2 Fabric Lane-Mapping Rules

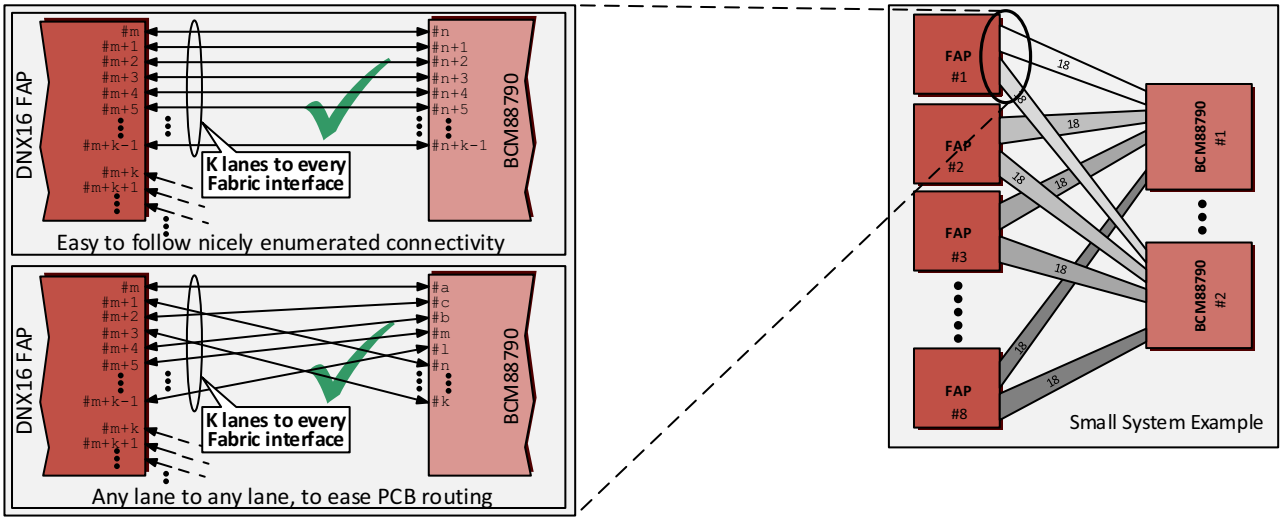
All fabric lanes are identical, and there is no significance to the lane index number with respect to another lane. The important aspect is the number of lanes that are eventually connected between the fabric interfaces of the different devices.

This concept is summarized by the following rule.

Rule 1: Fabric lanes can be swapped freely across the device.

The following figure illustrates Rule 1.

Figure 5: Fabric Lanes Can Be Swapped Freely Across the Device



Note the following disclaimer for Rule 1, when using the BCM88790:

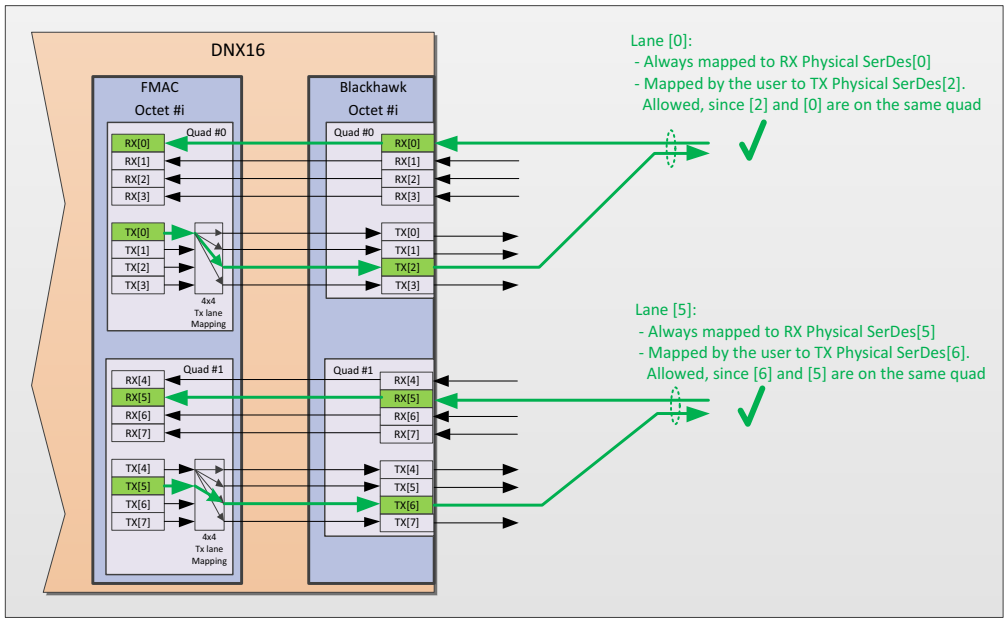
- When links are used in pass-through re-timer mode, the links pairs are predefined and cannot be swapped freely. For more details, refer to the Re-Timer Functionality section and Pass-through Re-timer sub-section in the BCM88790 data sheet (88790-DS1xx).

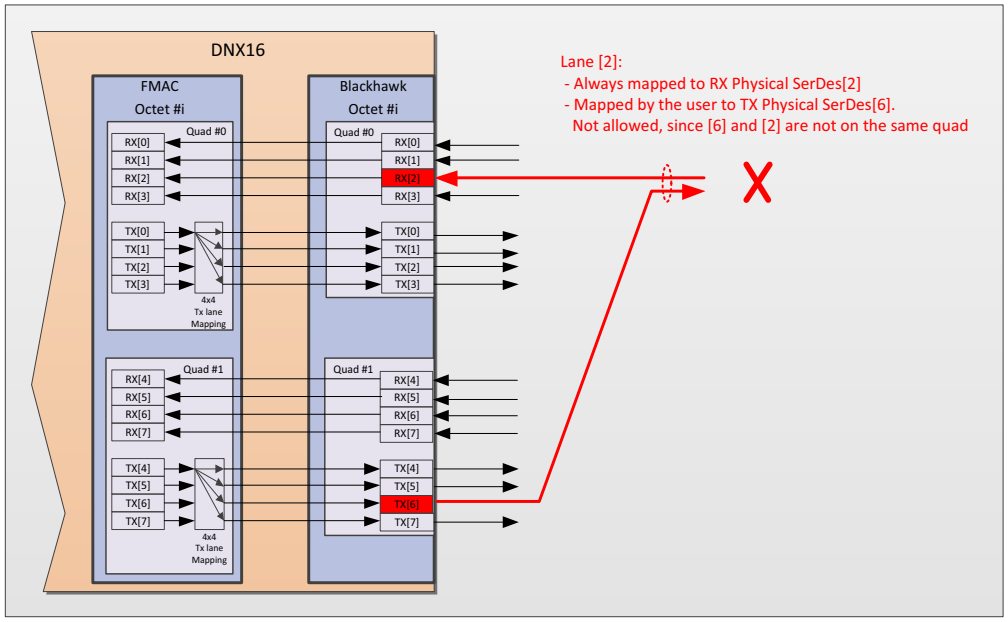
The second fabric lane-mapping rule is explained in the following list and illustrated in the following figure.

Rule 2:

- The RX physical SerDes index is the same as the lane index.
- The TX physical SerDes is assigned to the lane by the TX mapping feature.
- The TX physical SerDes must reside on same quad as the RX physical SerDes.

Figure 6: RX Physical SerDes and TX Physical SerDes Must Reside on the Same Quad





2.3.2 Blackhawk Lane Mapping – NIF Ethernet

2.3.2.1 Overview and Terminology

The following figure demonstrates the NIF Ethernet interface (PM8x50) lane-mapping capabilities.

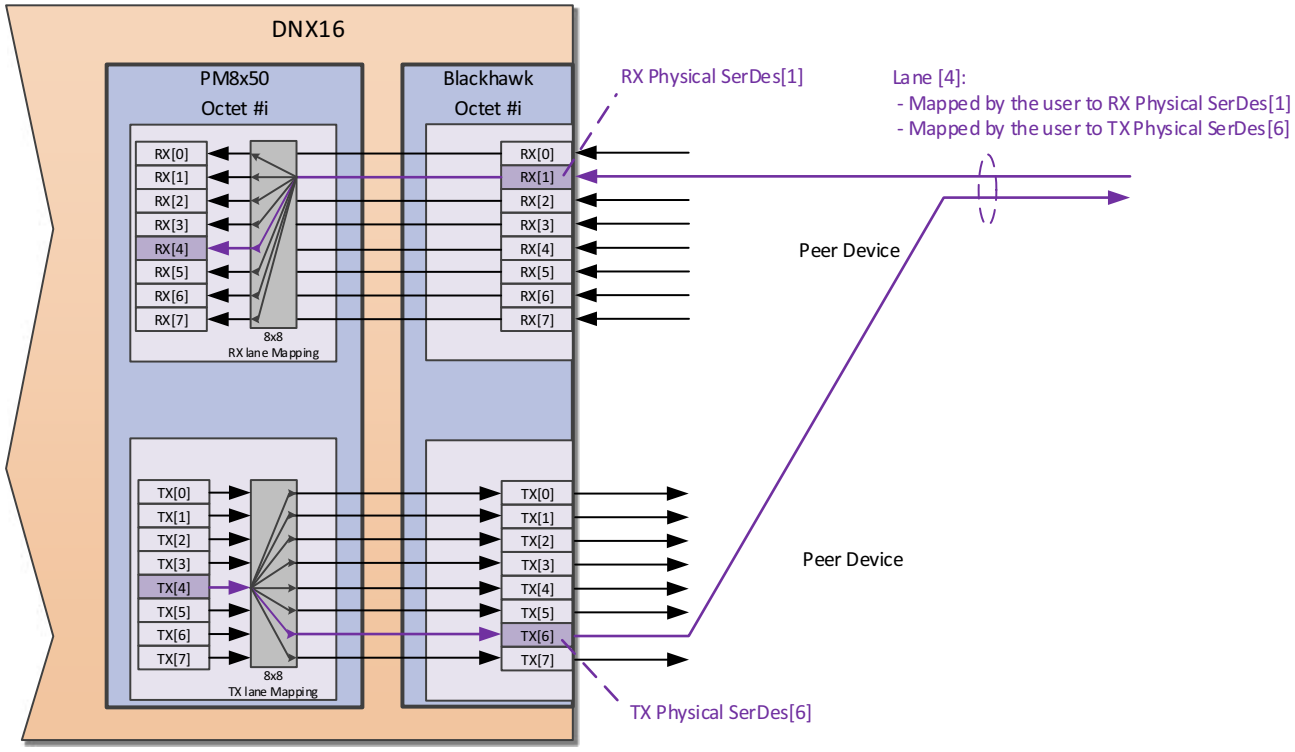
On the Blackhawk SerDes, there is no RX or TX mapping (swapping).

On the PM8x50, both RX and TX mapping (swapping) are supported.

The figure demonstrates the following terms that are used throughout this section:

- RX physical SerDes (on the device ball-out).
- TX physical SerDes (on the device ball-out).
- Lane – A logical SW entity that includes a pair of RX physical SerDes and TX physical SerDes. The logic lane represents the pairing on the customer implementation.

Figure 7: PM8x50 Interface Basic Structure



Mapping:

- The RX physical SerDes is assigned to the lane by the RX mapping feature.
- The TX physical SerDes is assigned to the lane by the TX mapping feature.

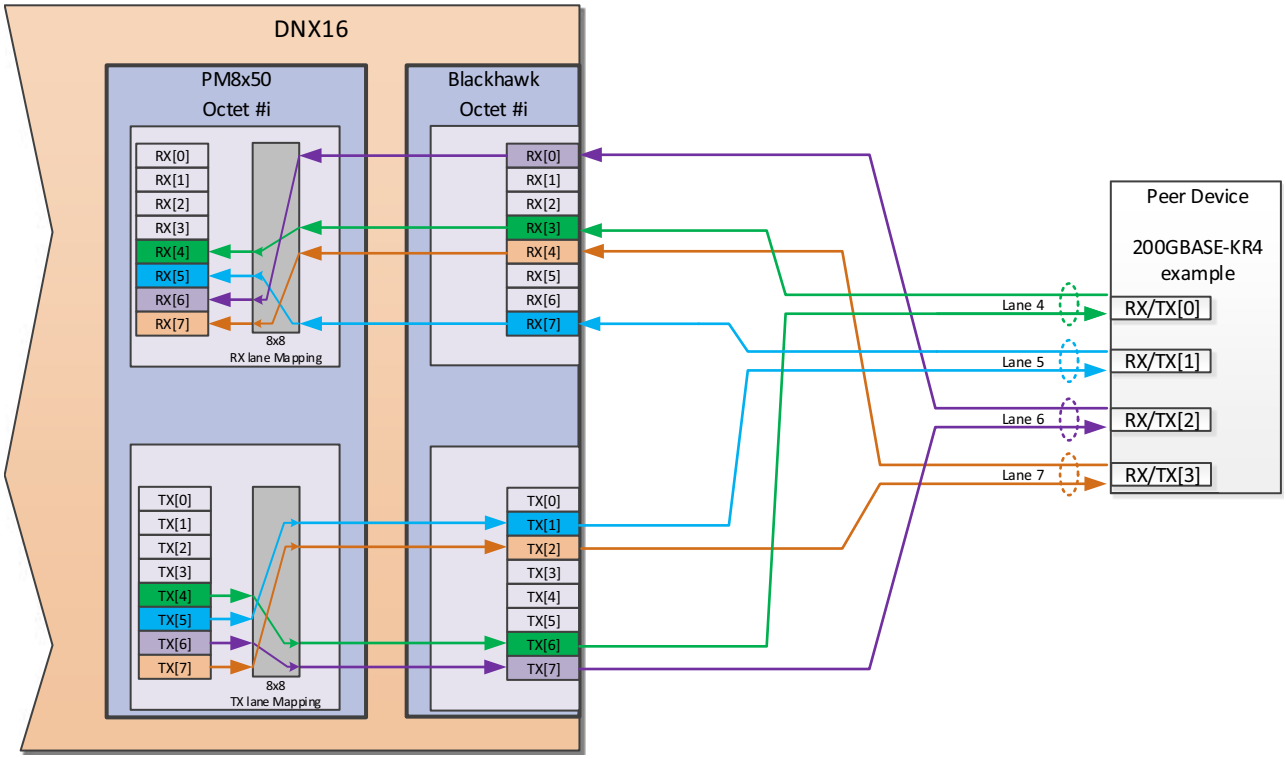
2.3.2.2 NIF Ethernet Lane-Mapping Rules

Rule 1:

- The RX physical SerDes and TX physical SerDes of a lane must reside on same octet.
- All RX physical SerDes and all TX physical SerDes of the port (regardless of whether it is a 1-lane, 2-lane, 4-lane, or 8-lane port) must reside on same octet.

The following figure shows an example of a NIF Ethernet lane swap.

Figure 8: NIF Ethernet Lane Swap Example



NOTE: When mixing Ethernet and ELK on the same octet, the ELK restrictions described in Section 2.3.3, Blackhawk Lane Mapping – BCM88690 ELK-over-ILKN, may limit the Ethernet flexibility.

2.3.3 Blackhawk Lane Mapping – BCM88690 ELK-over-ILKN

This section is applicable to BCM88690 ELK-over-ILKN interfaces.

For more information about ELK interfaces, refer to the External Lookup Engine section in the BCM88690 data sheet (88690-DS1xx).

NOTE: The information in this section applies to ELK-over-ILKN interfaces that use NIF SerDes or Fabric SerDes. For both SerDes types, all rules are the same.

2.3.3.1 Overview and Terminology

On the Blackhawk SerDes, there is no RX or TX mapping (swapping).

On the ILKN block, both RX and TX mapping (swapping) are supported.

The following terms are used throughout this section:

- RX physical SerDes (on the device ball-out).
- TX physical SerDes (on the device ball-out).
- Lane: A logical SW entity that includes a pair of RX physical SerDes and TX physical SerDes. The logic lane represents the pairing on the customer implementation.

Mapping:

- The RX physical SerDes is assigned to the lane by the RX mapping feature.
- The TX physical SerDes is assigned to the lane by the TX mapping feature.

2.3.3.2 ELK-over-ILKN Lane-Mapping Rules

2.3.3.2.1 ELK-over-ILKN Lane-Mapping Rules when RS-FEC Is Not Used

Rule 1: Select the physical RX and TX SerDes that will be used for the ELK.

- **Rule 1a:** When RS-FEC is not used, for each ELK port, up to 16 specific TX/RX physical SerDes pairs can be used. These physical SerDes use two octets.
- **Rule 1b:** When selecting the physical SerDes:
  - ELK interfaces up to 8-lanes should select only the first octet.
  - ELK interfaces up to 16-lanes should fill the first octet and any part of the second octet.
- **Rule 1c:** If physical SerDes RX[i] is assigned to ELK, then physical SerDes TX[i] must be assigned to ELK as well.

Rule 2: Assign RX/TX physical SerDes to a lane.

- The RX physical SerDes and TX physical SerDes of a lane must reside on same octet.

2.3.3.2.2 ELK-over-ILKN Lane-Mapping Rules when RS-FEC Is Used

Rule 1: Select the physical RX and TX SerDes that will be used for the ELK.

- **Rule 1a:** When RS-FEC is used, for each ELK port, up to 12 specific TX/RX physical SerDes pairs can be used. These physical SerDes use one octet and one quad.
- **Rule 1b:** When selecting the physical SerDes:
  - ELK interfaces up to 8 lanes (in other words, 4, 6, or 8 lanes) should select only the first octet.
  - ELK interfaces up to 12 lanes (in other words, 10 or 12 lanes) should fill the first octet and part of the third quad.
- **Rule 1c:** If physical SerDes RX[i] is assigned to ELK, physical SerDes TX[i] must be assigned to ELK as well.
- FEC is calculated on pairs of ILKN lanes (which implies that the ELK-over-ILKN interface must have an even number of lanes). Refer to *Interlaken Reed-Solomon Forward Error Correction Extension Revision 1.0 Dec 2016*.

  Since each two physical RX SerDes, and separately each two TX physical SerDes are attached to one FEC coding block:
  - BCM88690 TX:
    - **Rule 1d:** Outgoing logic lanes [2n] and [2n + 1] (two adjacent lanes that are connected to the same FEC block) should use two adjacent physical TX SerDes [2j, 2j + 1].
    - **Rule 1e:** The two physical TX SerDes [2j, 2j + 1] should be used in this order only.

      Using [2j + 1, 2j] for [2n, 2n + 1] is not supported by the BCM88690 TX.

      This limitation exists because the ELK devices do not support swapping on their RX FEC pairs.

      For example, Lanes 4 and 5 (two adjacent lanes that are connected to the same FEC block) can be connected to physical TX SerDes [2, 3] but cannot be swapped to [3,2].

ATTENTION: When working with the BCM16K KBP (also known as OP2), the BCM16K device should *not* use lane-swap mode because all lane mapping is done by the BCM88690 device.

  - BCM88690 RX:
    - **Rule 1f:** Incoming logic lanes [2m] and [2m + 1] (two adjacent lanes that are connected to the same FEC block) should use two adjacent physical RX SerDes [2i, 2i + 1] or [2i + 1, 2i].
    - **Rule 1g:** Both orders, [2i, 2i + 1] or [2i + 1, 2i], are supported by the BCM88690 RX. The order is auto-detected by the RX FEC.

      For example: Lanes 6 and 7 (two adjacent lanes that are connected to the same FEC block) can be connected to physical RX SerDes [0,1] or [1,0].

Rule 2: Assign the physical RX or TX SerDes to a lane.

- The physical RX SerDes and physical TX SerDes of a lane must reside on same octet.

2.3.4 Blackhawk Lane Mapping – BCM88800 ILKN

The BCM88800 device supports up to 12 ILKN interfaces, with flexible SerDes allocation per ILKN interface.

To define which SerDes can be mapped to which ILKN interface, and under what conditions, refer to the following information in the BCM88800 data sheet (88800-DS1xx):

- Table 1, BCM88800 NIF SerDes Interface Mapping
- Section 3.1.2.2, Interlaken over Fabric SerDes

NOTE: The information in this section applies to ILKN interfaces that use NIF SerDes or Fabric SerDes. For both SerDes types, all rules are the same.

2.3.4.1 Overview and Terminology

Both RX and TX mapping (swapping) are supported within the ILKN block.

The following terms are used throughout this section:

- RX physical SerDes (on the device ball-out).
- TX physical SerDes (on the device ball-out).
- Device lane: A logical software (SW) entity that includes a pair of SerDes made up of one RX physical SerDes and one TX physical SerDes. The BCM88800 includes 128 device lanes for NIF and 48 device lanes for fabric. Device lanes are used for both ILKN and ETH definitions.
- ILKN lane: Each ILKN interface may include up to 24 lanes. The ILKN lane number is defined by the board and the link partner. The ILKN lane should be associated by SW to a device lane (and by this association, to the RX physical SerDes and the TX physical SerDes).

The mapping occurs as follows:

- The RX physical SerDes is assigned to a device lane by the RX mapping feature.
- The TX physical SerDes is assigned to a device lane by the TX mapping feature.
- Device lanes (where each lane represents an RX SerDes and a TX SerDes) are assigned to ILKN interfaces and are ordered to represent the ILKN lane order.

2.3.4.2 ILKN Lane-Mapping Rules

The rules in this section are always applicable, regardless of whether or not FEC is used. If FEC is used, the rules in Section 2.3.4.3, FEC Rules, also apply.

The number of lanes must meet requirements according to interface speed and ILKN core mode (one interface or two interfaces).

The SerDes that are used must meet the allowed options as defined in the BCM88800 data sheet (88800-DS1xx) in Table 1: BCM88800 NIF SerDes Interface Mapping and Section 3.1.2.2 Interlaken over Fabric SerDes.

Rule 1:

If physical SerDes RX[i] is assigned to an ILKN interface, then physical SerDes TX[i] must be assigned to the same ILKN interface as well.

Rule 2:

The RX physical SerDes and TX physical SerDes of a device lane (and therefore, the same ILKN lane) must reside on the same PM (PM50, PM25) or the same Blackhawk for fabric, if fabric is used.

2.3.4.3 FEC Rules

Interfaces using ILKN cores 6 and 7 can use FEC.

FEC is calculated for every two adjacent lanes. (For more information, refer to the *Interlaken Reed-Solomon Forward Error Correction Extension Revision 1.0 Dec 2016*.)

The FEC calculation implies the following:

- An ILKN interface that uses FEC must have an even number of lanes.
- Each two TX physical SerDes, [2j, 2j + 1] are attached to one FEC coding block.
- Each two RX physical SerDes, [2i, 2i + 1] are attached to one FEC coding block.

Rule 3:

BCM88800 TX:

- Outgoing ILKN lanes [2n] and [2n + 1] should use two adjacent physical TX SerDes [2j, 2j + 1].
- The two physical TX SerDes [2j, 2j + 1] should be used in this order only.
- Using TX SerDes [2j + 1, 2j] for ILKN lanes [2n, 2n + 1] is not supported by the BCM88800 TX.

  Example: ILKN lanes 4 and 5 (two adjacent lanes that are connected to the same FEC block) can be connected to physical TX SerDes [2,3] but cannot be swapped to TX SerDes [3,2].

Rule 4:

BCM88800 RX:

- Incoming ILKN lanes [2m] and [2m + 1] should use two adjacent physical RX SerDes [2i, 2i + 1] or [2i + 1, 2i].
- The two orders, [2i, 2i + 1] or [2i + 1, 2i], are supported by the BCM88800 RX because the order is auto-detected by the RX FEC.

  Example: ILKN lanes 6 and 7 (two adjacent lanes that are connected to the same FEC block) can be connected to physical RX SerDes [0,1] or RX SerDes [1,0].

ATTENTION: When working with the BCM16K KBP (also known as OP2), the BCM16K device should not use lane-swap mode because all lane mapping is done by the BCM88800 device.

2.3.5 Blackhawk Lane Mapping – BCM88480 ILKN

The BCM88480 device supports up to four ILKN interfaces, with flexible SerDes allocation per ILKN interface. Only ILKN interface_0 and ILKN interface_1, which are based on ILKN core_0, can use Blackhawk SerDes, but the rate is up to 28.125-Gb/s NRZ.

To define which SerDes can be mapped to which ILKN interface, and under what conditions, refer to the following information in the BCM88480 data sheet (88480-DS1xx):

- Table 1, BCM88480 NIF SerDes Interface Mapping
- Section 3.1.3, Interlaken over Fabric SerDes

2.3.5.1 Overview and Terminology

Both RX and TX mapping (swapping) are supported within the ILKN block.

The following terms are used throughout this section:

- RX physical SerDes (on the device ball-out).
- TX physical SerDes (on the device ball-out).
- Device lane: A logical SW entity that includes a pair of SerDes made up of one RX physical SerDes and one TX physical SerDes. The BCM88480 includes 52 device lanes for NIF. Device lanes are used for both ILKN and ETH definitions.
- ILKN lane: ILKN interfaces that are connected to ILKN core_0 can include up to 12 lanes. ILKN interfaces that are connected to ILKN core_1 can include up to 24 lanes. The ILKN lane number is defined by the board and the link partner. The ILKN lane should be associated by SW to a device lane (and by this association, to the RX physical SerDes and the TX physical SerDes).

Mapping:

- The RX physical SerDes is assigned to a device lane by the RX mapping feature.
- The TX physical SerDes is assigned to a device lane by the TX mapping feature.
- Device lanes (where each lane represents an RX SerDes and a TX SerDes) are assigned to ILKN interfaces and are ordered to represent the ILKN lane order.

2.3.5.2 ILKN Lane-Mapping Rules

The number of lanes must meet requirements according to interface speed and ILKN core mode (one interface or two interfaces).

Rule 1:

If physical SerDes RX[i] is assigned to an ILKN interface, then physical SerDes TX[i] must be assigned to the same ILKN interface as well.

Rule 2:

The RX physical SerDes and TX physical SerDes of a device lane (and therefore, the same ILKN lane) must reside on the same PM (PM50, PM25).

NOTE: A single ILKN interface cannot use both Blackhawk SerDes and Falcon SerDes.

2.3.5.3 FEC Rules

The BCM88480 ILKN interfaces do not support PAM-4 rates and do not include FEC logic.

2.4 Unused Blackhawk Links

Unused TX and RX pairs of Blackhawk octets can be left open (unconnected).

Unused quads should be connected to an active supply and be powered down using software. The filtering and decoupling requirements for the unused quad supplies are relaxed to reduce board components.

In the BCM88690 and BCM88790, when all Blackhawk cores in a group of certain LCPLL are not in use, this LCPLL reference clock is not required.

In the BCM88800 and BCM88480, keep all reference clocks connected, regardless of whether the SerDes cores are used.

NOTE: Links that are enabled (transmitting) but not terminated properly might interfere with other links. Be sure to disable such unterminated links.

2.5 Blackhawk AC Coupling

2.5.1 Blackhawk RX

Blackhawk SerDes RX serial links are internally AC coupled. These AC couplings are on-die and cannot be bypassed. This internal AC coupling eliminates the need for external AC capacitors for most applications.

External capacitors on the RX are still required if the requirement for the maximum Input Common Mode ($V_{IN\text{-}CM\ Max}$) is not met. For more information, refer to the device data sheet.

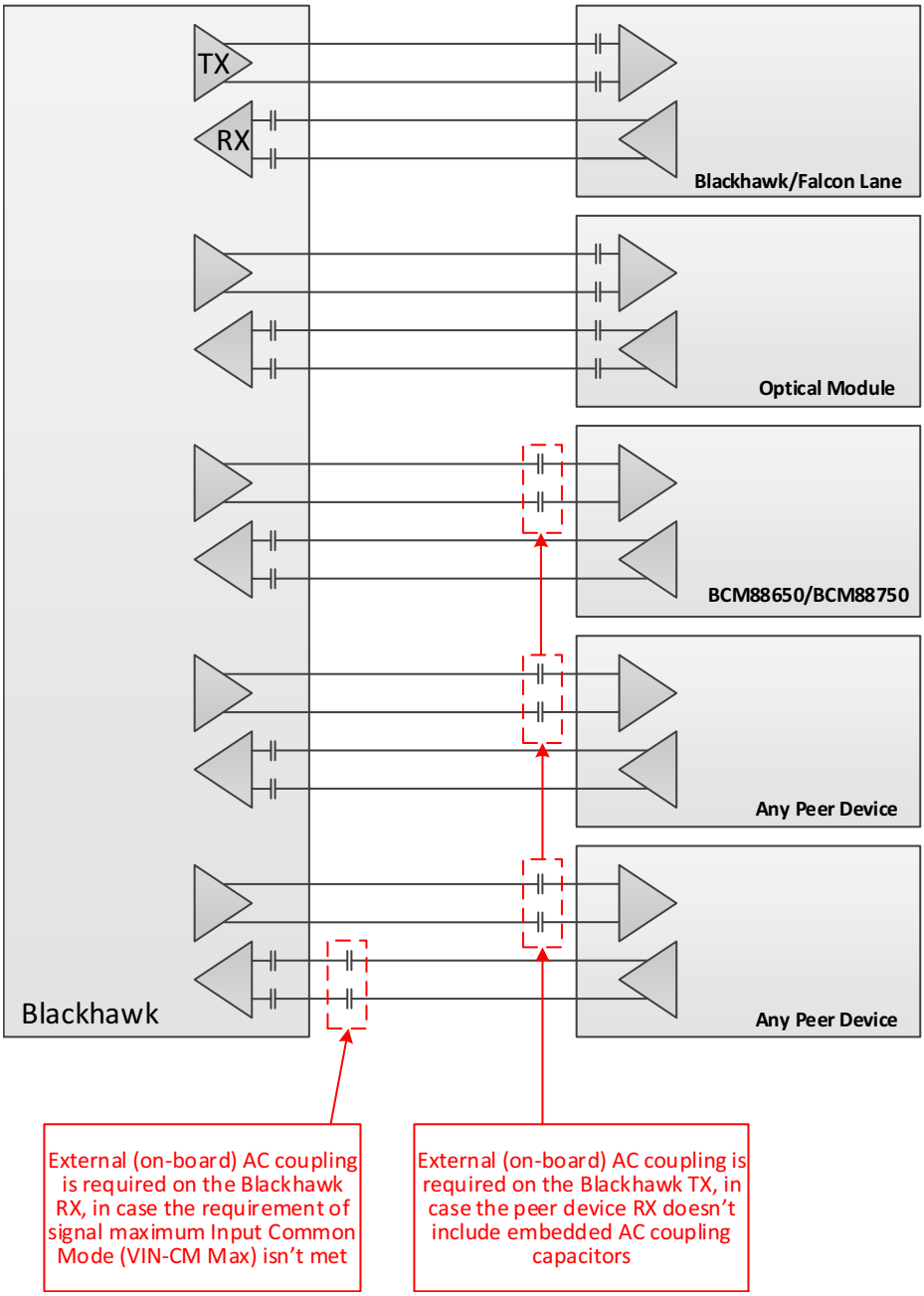
If external capacitors are required, use a small footprint capacitor, such as 0201, to minimize the reflections caused by the structure impedance mismatch.

2.5.2 Blackhawk TX

Any peer device that has an interface to the Blackhawk SerDes must have AC coupling on its receiver. This peer AC coupling can be embedded inside the peer device, as with the Blackhawk/Falcon/Eagle SerDes, or it can be implemented externally on the board level.

NOTE: The requirement for AC coupling means that no DC path is allowed between the Blackhawk TX and the peer device RX.

Figure 9: AC Coupling of Serial Links



2.6 Blackhawk Hot Plug

Hot plugging is the ability to insert a card into a powered chassis or extract a card from a powered chassis. The Blackhawk SerDes supports hot plugging as part of a device on the inserted or extracted card and also as the link partner of a device that is on an inserted or extracted card.

To support the hot plug, the following conditions must be met:

- Link partner TX to Blackhawk RX:
  - The output impedance of the link partner TX driver should be 50Ω ±15Ω to limit the maximum possible current through the Blackhawk RX.
  - The Blackhawk *Absolute maximum RX input* (V_{IN-Abs}) requirement must be kept.
- Blackhawk TX to link partner RX:
  - The interface of the Blackhawk TX to the link partner RX must be AC coupled (that is, no DC path is allowed between the Blackhawk TX and the peer device RX).

The system-level design must guarantee that both devices on the hot-plug scheme have the same reference ground. This requirement can be achieved by using a shorter pin for the signals and a longer pin for the ground.

2.7 Blackhawk Layout Guidelines

This section provides layout guidelines for achieving proper operation of the Blackhawk SerDes.

NOTE: The guidelines detailed in this section are to be considered as good-practice recommendations. They should not replace a detailed simulation for each specific design.

2.7.1 Trace Impedance

On the customer PCB, the Blackhawk RX and TX links should have a target impedance of 95Ω ±10%.

For reference, inside the package:

- The RX trace impedances are 95Ω.
- The TX trace impedances are 90Ω.

2.7.2 Differential Pair Geometry and Stack-Up

NOTE: The guidelines detailed in this section are to be considered as good-practice recommendations. They should not replace a detailed simulation for each specific design.

2.7.2.1 Loosely and Tightly Coupled Differential Pair Structure

In loosely coupled pairs, the intra-pair separation is about 1.5 to 2 times the dielectric height.

In tightly coupled pairs, the intra-pair separation is similar to the dielectric height.

Tightly coupled routing requires less area for routing; however the common-mode impedance cannot be maintained constantly all the way point-to-point and, as usually the trace width is reduced, the fabrication is somehow more complicated.

In general, it is acceptable to select either methodology for Blackhawk routing.

2.7.2.2 Stripline and Reference Planes

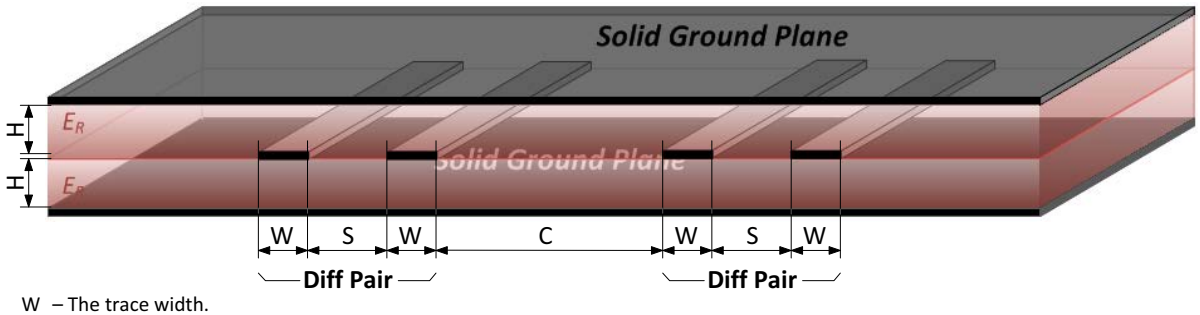
SerDes differential signals must be routed using stripline traces. No routing should be done over the external PCB layers (as in microstrip) other than the short routing to an external transitional via, if used, or to the inner via with a dog-bone breakout.

The two solid reference planes for the stripline structure used for the SerDes routing must be ground. Make sure the distance between the signal traces and the edge of the reference planes is *at least four times* the dielectric height (4 × H).

2.7.2.3 Trace Clearances

Keep a minimum clearance of *at least four times* the dielectric height (4 × H) between the differential pairs and any other signal (other SerDes links or any other non related signals). This amount of clearance minimizes the crosstalk between signals, even if they run in parallel for considerable lengths.

Figure 10: Differential Pair Geometry and Stack-up



W – The trace width.
S – P/N trace separation. About 1½-2×H for loosely coupled pair and similar to the dielectric height for tightly coupled pairs.
H – The PCB dielectric height.
C – Clearance to other traces/pair. Must be ≥ 4×H.

2.7.2.4 Ground Fills and Pours

Ground fills (or ground pours) for shielding the SerDes pairs are not required.

Maintaining a safe 4 × H clearance between pairs is enough to control the crosstalk to a negligible level.

2.7.3 Vias and PTHs

NOTE: The guidelines detailed in this section are to be considered as good-practice recommendations. They should not replace a detailed simulation for each specific design.

The board vias and the board connector's plated-through-holes (PTHs) are a source of severe discontinuity to the signal path. Take measures to reduce their capacitive nature and have their impedance closer to that of the rest of the interconnect. The vias and PTHs are also sources for harming the channel return-loss and crosstalk performances.

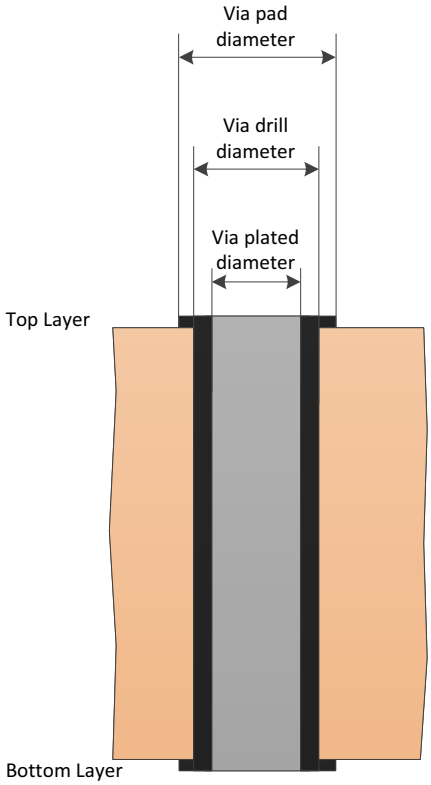
2.7.3.1 Via Drill Diameter

Via drill diameter should be minimized as well as the via pad. Using an 8-mil via and an 18-mil pad decreases the via capacitance and yields better link performance.

The following figure illustrates the differences between the via pad diameter, the drill diameter, and the plated diameter.

For simulations, you should typically use the drilled diameter and not the plated diameter.

Figure 11: Comparison of Via, Drill, and Plated Diameters



2.7.3.2 Keep Two Vias per SerDes Lane

To reduce the number of vias to the absolute possible minimum, SerDes routing must be made in a single layer (aside from the small trace on the top layer, when required). This requirement implies only two vias: one via at the breakout of the BGA, and another via at the peer device BGA or at the PTH of the connector.

2.7.3.3 Remove Pads on Unused Layers

By default, CAD layout tools place pads on the via or PTH on each of the layers. However, the routing is made on only a single layer, so many pads are not used. The presence of the unused pads decreases the electrical qualities of the via by increasing its capacitance and by decreasing its impedance.

All redundant pads must be removed, which most CAD tools support.

Figure 12: Improving Via and PTH Characteristics by Removing Unused Pads



2.7.3.4 Controlling the Via or PTH Stub

Regular vias and PTHs penetrate through the entire extent of the PCB depth. Depending on which layer the signal is routed, a residual portion of the via or PTH length is left open. While low frequency detects this stub as a small negligible lumped capacitance, at the high data rates at which the SerDes runs, this stub has an overwhelming impact on the return loss of the channel and the overall SerDes performance.

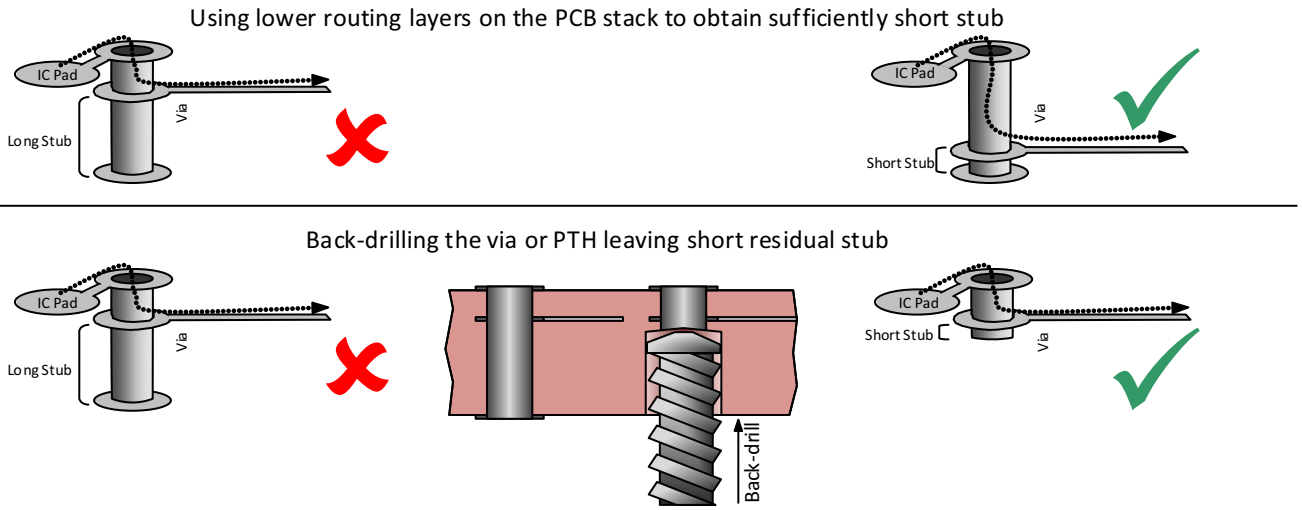
Table 2 lists the maximum allowed PTH stub the design must enforce. For SerDes links, back-drilling is required to carve out the dangling portion of the via or PTH. An approach of combining blind and buried vias is also adequate.

Table 2: Max Via/PTH Stub for SerDes Links

Data Rate	10.3 Gb/s	12.5 Gb/s	25 Gb/s	50 Gb/s
Maximum via or PTH stub	20 mils	15 mils	10 mils	10 mils

NOTE: The values in the table already include the production inaccuracy.

Figure 13: Via or PTH Stub and Back-Drilling



2.7.3.5 Power and GND Via Stub

On the DNX28 devices, the SerDes power and GND plane should be located on a lower layer, close to the bottom layer so the power stub (that cannot be back-drilled because the connection to the capacitor is on the bottom layer) is minimal.

On Blackhawk, the power BGA balls are located relatively far from the SerDes RX/TX balls, so these guidelines do not apply. The SerDes power planes can be located at any layer.

2.7.3.6 Special Anti-Pad Shapes

To further reduce the capacitive nature of the via or PTH carrying the differential signals, it is customary in high-speed design to replace the usual per-via circular anti-pad with an oblong, oval, or other shape that relieves the power or ground plane surrounding both vias.

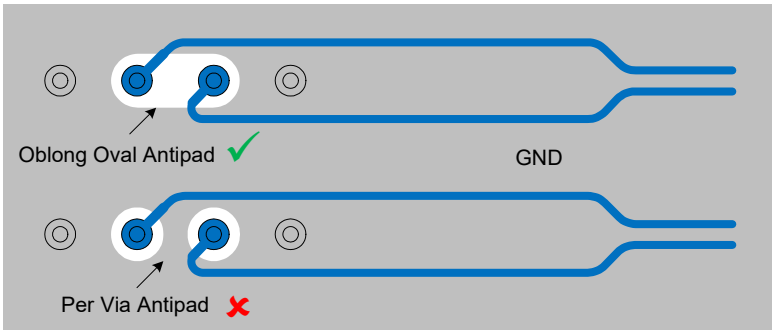
Backplane and optical module connectors provide details on the exact geometry of anti-pads, which should be strictly followed. On the BGA, use a via breakout in an oval or oblong anti-pad shape. Replicate the breakout on all plane layers on which the back-drilling is not performed.

Optimize via structure impedance to target and match the trace impedance. This optimization should be done in simulation per board, and it depends on the board stack-up, anti-pad size, stitching via position, and unused pad removal. The distance between two differential vias can also affect the impedance, although it cannot always be done due to the predefined pitch in the BGA area.

Anti-pad size can be slightly wider in some plane layers, as long as no signals are underneath this wider plane.

When calculating the anti-pad geometry, consider the fabrication miss-registration between PCB layers. Usually a 3-mil guard-band between trace routing to anti-pad clearance is sufficient to overcome fabrication miss-registration.

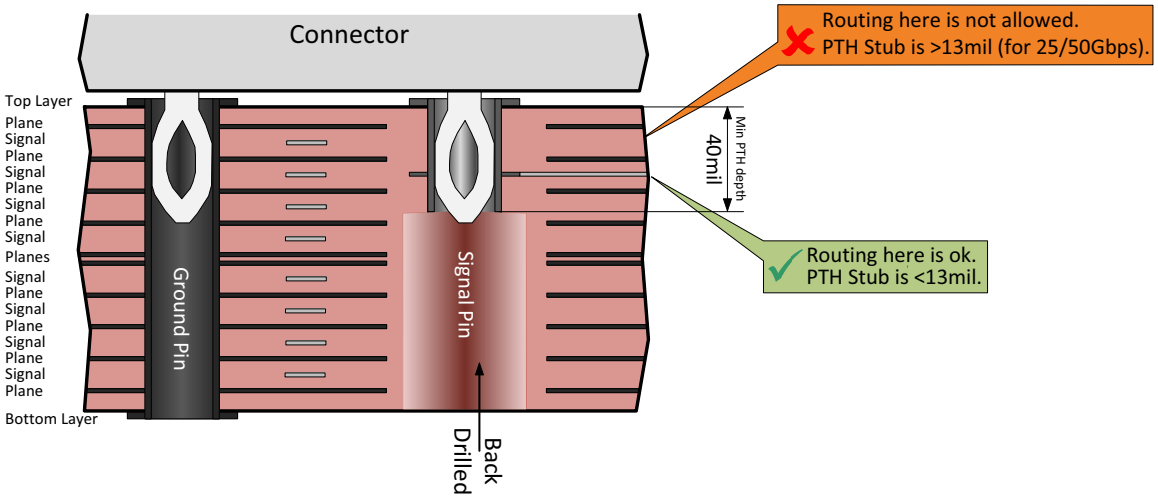
Figure 14: Oblong/Oval Anti-Pad for Vias of Differential Signals



2.7.3.7 Back-Drilling PTH of Press-Fit Connectors

Because of the press-fit nature of the backplane connectors, the PTH must provide a minimal depth. Further back-drilling is not allowed below the minimum PTH depth. Connector manufacturers are aware of the implication of this depth, and they work to shorten the signal pins as much as possible. In the 20-layer PCB shown in the following figure, there are eight routing layers. However, because of the back-drill limit on the connector's pin, the highest routing layer yields a PTH stub that is too long and is therefore inadequate for SerDes routing. In this example, only seven out of the eight routing layers are adequate for SerDes routing at data rates of 25 Gb/s or 50 Gb/s.

Figure 15: Back-Drilling and Routing Constraints on Connectors



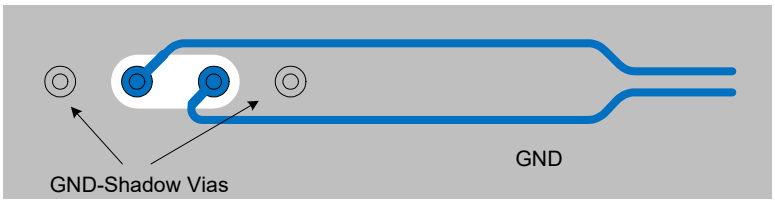
2.7.3.8 GND-Shadow Vias

When a differential-pair is going from layer-to-layer (either via-in-pad or transitional-via), add two GND-shadow vias (stitching-vias) to create a four-via structure.

Optimize the via structure impedance to match the trace impedance or to be close to it. This optimization should be done in simulation per board.

Usually those vias are not back-drilled.

Figure 16: GND-Shadow Vias



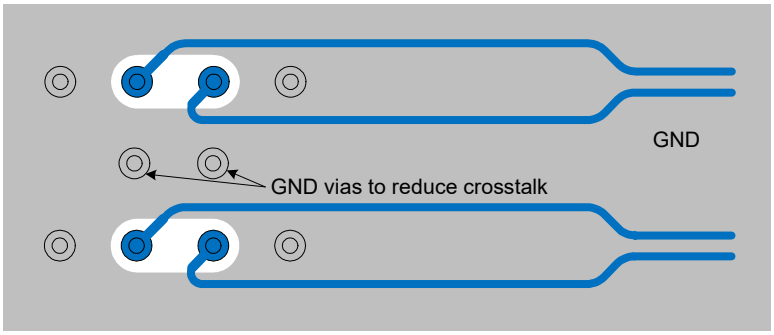
2.7.3.9 Adding GND Vias between Pairs of Differential Signal Vias

To reduce crosstalk, add GND vias between pairs of differential signals vias, as illustrated in the following figure.

Optimize the via structure impedance match the trace impedance or to be close to it. This optimization should be done in simulation per board.

Usually those vias are not back-drilled.

Figure 17: Adding GND Vias Between Pairs of Differential Signals Vias



2.7.3.10 Connector Return-loss and Crosstalk

When selecting the appropriate connector for the design, consider the return-loss and crosstalk parameters. Usually, connectors are sources of crosstalk, and their PTHs are sources for both crosstalk and return-loss.

2.7.4 Maintaining Low Crosstalk

Unlike legacy devices where controlling the insertion-loss was the key parameter and controlling crosstalk was a weaker demand, reducing crosstalk is important in Blackhawk.

Crosstalk reduction is achieved by selecting the appropriate connectors and following the layout guidelines in this document. Make sure the selected connectors are certified to the appropriate rate.

2.7.5 PCB Material

2.7.5.1 Microstrip Plating and Finish

SerDes routing must be made in internal stripline layers where both references are GND planes. Avoid routing in the exterior layers for all high-speed signals, including PCIe and clocks.

Even though high-speed microstrip routing on the exterior layers is minimal to nonexistent, take care with regard to the preferred PCB trace finish and plating process. Avoid using any finish or plating process that involves ferromagnetic materials. For example, the standard gold-plating process involves a thin plating of nickel in between the copper surface and the exterior gold plating. Because this plating is close to the surface of the trace and because of the skin effect, most of the high-speed current is affected by this plating layer. Nickel is a ferromagnetic material with a relative permeability of about 100 μ/μ_0. Skin effect-related losses are proportional to the square root of the relative permeability. Therefore, this nickel-gold-plating example exhibits 10 times the skin effect loss compared to a plain copper trace.

Consult your PCB manufacturer for details about types of available plating and finish processes that are adequate for high-speed applications.

In general, it is recommended to use materials with an average surface roughness of below 2-µm RMS (for example, Megtron-6 HVLP or Isola Tachyon).

On internal layers, pay attention to the PCB manufacturing process of the material (applying mechanical or chemical treatment) to make it less smooth. Control the amount of change.

2.7.5.2 Dielectric Material Adjacent to Routing Layers

When selecting the dielectric material adjacent to routing layers, carefully examine the material's properties, especially the dielectric constant (Dk/Er) and loss tangent (dissipation factor, or Df).

For example, a Megtron-6 core material used on the SVK has a dielectric constant (Dk/Er) of 3.6 and loss tangent/dissipation factor (Df) of 0.004 at 10 GHz.

2.7.5.3 Dielectric Glass Weave

Dielectric materials come with different glass-resin reinforcement styles or weaves. The type of weave can have a dramatic effect on very high-speed signals.

On differential signal routing, if one signal of a P/N pair runs primarily over resin-rich areas (lower dielectric constant) and the other runs over glass-rich areas (higher dielectric constant), the result is skew in the received differential signal. This effect is referred to as a *glass-weave skew* (GWS). Similarly, the weave type may also have an effect on the signal impedance.

Use the following guidelines to avoid issues caused by the weave effect:

- Select a tight (dense) weave material and use mechanically spread (MS) glass. The spreading should be on both horizontal and vertical directions.
- Consider using the following weave types: 1035, 1067, 1078, and 1086.
- Use two-ply mechanically spread glass for each layer instead of one-ply.
- Consider one of the following options:
  - Route the signals in 10-degree to 15-degree angles with respect to the X/Y axis (instead of straight along the X/Y axis). To clarify, any routing between 10 degrees to 15 degrees and 75 degrees to 80 degrees is acceptable; it is not necessary to limit the routing to 10 degrees to 15 degrees.
  - Rotate the design by 10 degrees to 15 degrees with respect to the panel. Before using this option, check with the fabrication vendor since the ability to do this depends on the design size versus panel size.

Another option is to consider is using the *low Dk glass cloth* version of the materials, such as Megtron 6 (N) version (R-5775(N) and R-5670(N)). With these materials, the glass Dk and resin Dk values are very similar.

2.7.6 Blackhawk Skew and Length Matching

2.7.6.1 Pair P/N Skew

2.7.6.1.1 Pair P/N Skew Requirement

For Blackhawk fabric or NIF links at 53.125 Gb/s, the total allowed P-to-N skew should be smaller than 15 mils (2.4 ps, considering a propagation delay of 160 ps/inch and 53.125-Gb/s rate). These 15 mils include the skew across the entire link, end-to-end. For example, consider the following topologies:

- Fabric access processor (FAP) line card ↔ fabric backplane ↔ fabric element (FE) card
- Pizza-box ↔ copper cable ↔ peer system

For best results, divide the 15-mil budget as 2 mils per each card with the DNX16 device on it, and 11 mils for the backplane card or copper cable. Another option is to divide the 15-mil budget as 5 mils per each card (or copper cable).

To summarize, design each card with a skew limit of up to 2 mils (or 5 mils), regardless of whether the card is plugged into a chassis or has copper or fiber-optic module connectors.

When calculating the P/N skew, it is not necessary to include the P/N skew inside the package as well because the previously-described requirement already took the package skew into consideration.

2.7.6.1.2 Pair P/N Skew Sources and Compensation

The common skew sources are as follows:

- BGA breakout
- Away from the BGA and onto the routing towards the end destination, every turn the signal makes adds a small skew. The skew of a right turn is canceled with that of a left turn.
- At the other end of the routing is either a connector (backplane or optical module) or, alternatively, a peer BGA/ASIC. The routing escape of the connector pins often includes a turn in the signals and introduces a skew that is similar to any other turn in the routing of the differential pair.
- Skew related to backplane connectors and their intra-pair P/N skew.

  It is possible to compensate for this skew in the backplane design itself by applying a connectivity that routes the longer pin on the connector of one daughter card to the shorter pin at the connector on the peer card.
- Skew related to a local dielectric constant variation due to the fiberglass weave. See Section 2.7.5.3, Dielectric Glass Weave.

The skew compensation is as follows:

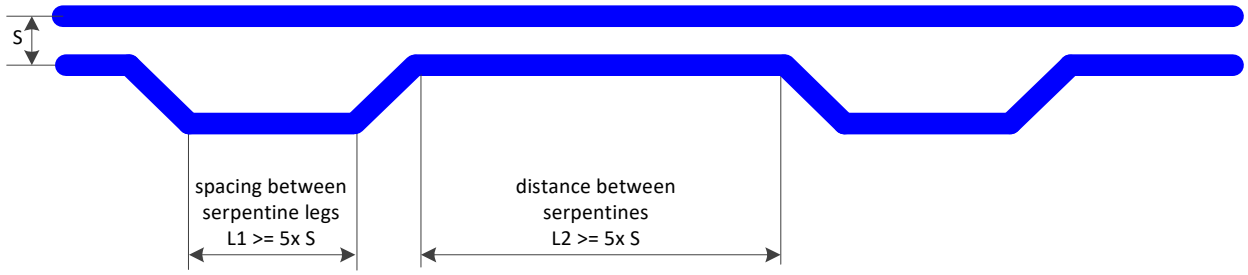
1. Deskewing by swapping the polarity: The Blackhawk SerDes supports swapping the polarity of the P/N for each TX pair and each RX pair (see Section 2.2, Blackhawk P/N Polarity Inversion).
2. Deskewing along the routing path by adding serpentines.

NOTE: Because any turn in the signal will reduce the signal quality, it is not required to immediately deskew any point where skew is generated. For example, if the pair takes a 90-degree turn, there is no need to immediately counter skew it right after the turn. For the relevant data rates and applications, it is acceptable to look at the entire track and deskew it at a certain place.

When using serpentines, keep the following guidelines in mind:

- The spacing (L1) between serpentine legs has to be large enough, at least 5 × the distance between the P and N traces, to minimize the coupling between serpentine legs.
- When another serpentine is required, the distance (L2) between the first serpentine and the second serpentine should also be at least 5 × the distance between the P and N traces.

Figure 18: Serpentine-Legs Spacing Requirements



S – P/N trace separation

By its nature, the serpentine changes the differential impedance of the differential pair.

You can minimize the amount of change by modifying the traces' width in the serpentines. In any case, validate the serpentine structure using SI simulations.

An alternate approach is to use small serpentines that are very close to each other. This approach may provide better impedance response; however, it could introduce a higher skew difference between the electrical length of the signals to the physical/mechanical length of the signals. Taking this difference into consideration makes the layout stage more complicated.

2.7.6.2 Lane (Pair-to-Pair) Skew

This section describes the lane (pair-to-pair) skew considerations on the PCB (not necessarily for the entire system).

2.7.6.2.1 Lane Skew – Fabric

When the fabric lanes are routed over the PCB or backplane, there is no skew requirement, and therefore, no need to perform pair-to-pair length matching (assuming the skew is in the range of up to tens of inches).

When the fabric is over fiber cables, contact Broadcom support to discuss the skew requirement as part of the overall architecture.

2.7.6.2.2 Lane Skew – NIF

For the NIF skew constrains, refer to the relevant IEEE standard. As good practice, limit the lane skew of a NIF port to 1 inch.

2.7.6.2.3 Lane Skew – ILKN

On the same ILKN interface, limit the board lane-to-lane skew to 80 UI. (This is the recommendation for the board design, not a definition of the DNX RX capability, which is according to the standard.)

For 53G, UI = ~18.8 ps.

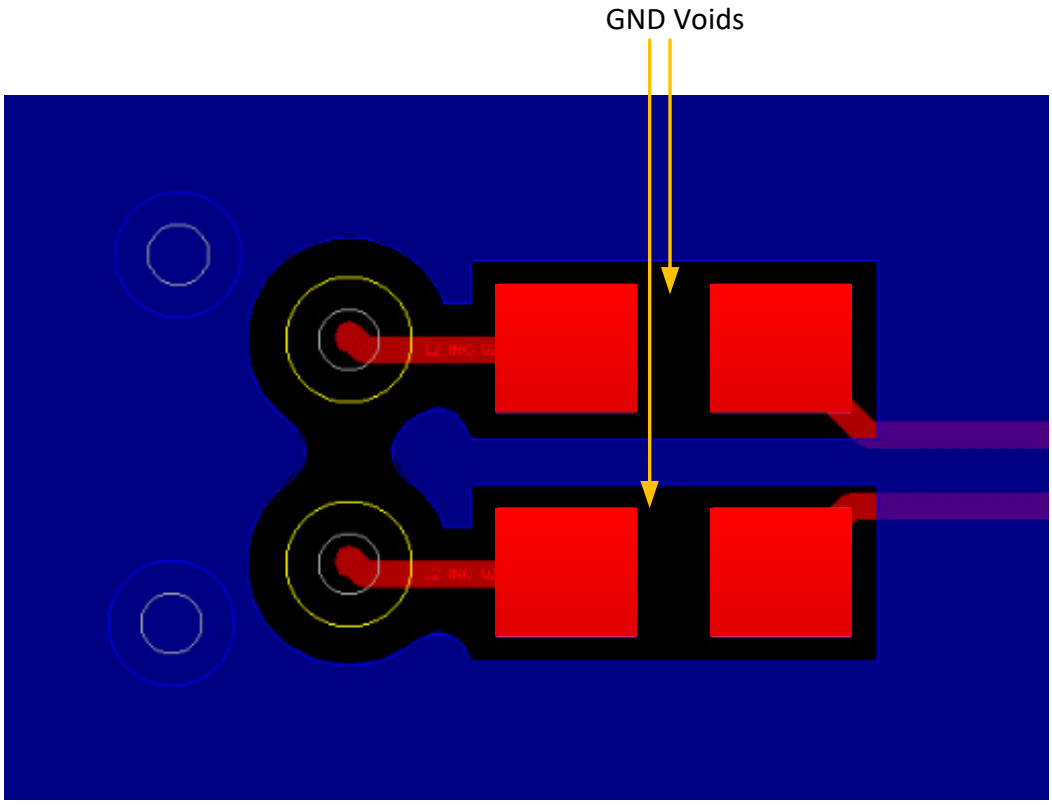
For 25.78G, UI = ~38.8 ps.

2.7.7 AC Coupling Capacitors

As mentioned in Section 2.5, Blackhawk AC Coupling, in most cases, external AC coupling is not required. However, when external AC coupling capacitors are required, pay attention to reducing the impedance discontinuity they cause.

One common approach for reducing the impedance discontinuity (by increasing the structure impedance) is to void the GND reference plane below the capacitors. This void should include at least both pads of the capacitor, as the following figure illustrates.

Figure 19: AC Capacitor GND Voids



Other common approaches are to have one void covering the two capacitors, void more GND planes below the capacitors (and not only the first adjacent plane), or to not route any signals in adjacent layers below the voiding.

Other guidelines are as follows:

- Any via added because of the capacitors should follow all the via layout guidelines described in Section 2.7.3, Vias and PTHs.
- The capacitor package size should be limited to 0201.
- The capacitors placement and routing structure should be symmetrical.

In all cases, optimize the void structure in simulations as well as the capacitor size, capacitor location, capacitor via structure, and so on. The purpose is to avoid reflections.

2.8 External Re-Timer Device Guidelines

Use an external re-timer device to support interconnects that are beyond the specifications of the DNX16 device SerDes. For example, use a re-timer for interconnects that have a higher insertion-loss than supported by the Blackhawk SerDes, or use a re-timer if the system requires a better BER than the Blackhawk SerDes guarantees for this channel.

This section provides information regarding the re-timer selection and is focused on the connectivity to the DNX16 device. Component selection criteria that is not related to connectivity to the DNX16 device is beyond the scope of this document.

In general, no special requirements are necessary when selecting a re-timer device to work with the DNX16 device (as compared to selecting a re-timer to work with other devices); however, there is one exception:

Because the fabric interface FEC is proprietary to the DNX devices, FEC cannot be enabled hop-to-hop but only end-to-end. See Section 2.8.1, Fabric FEC.

NOTE: In this section, the term *re-timer* refers to a device that operates at the bit level only and does not perform any PCS, scrambling, or other logic operation on the data stream (with one possible exception: NIF Ethernet FEC). The re-timer device recovers the information at the digital level (using CDR, DFE, and so on).

This section also applies to selecting MUX devices that operate similarly to the re-timer operation.

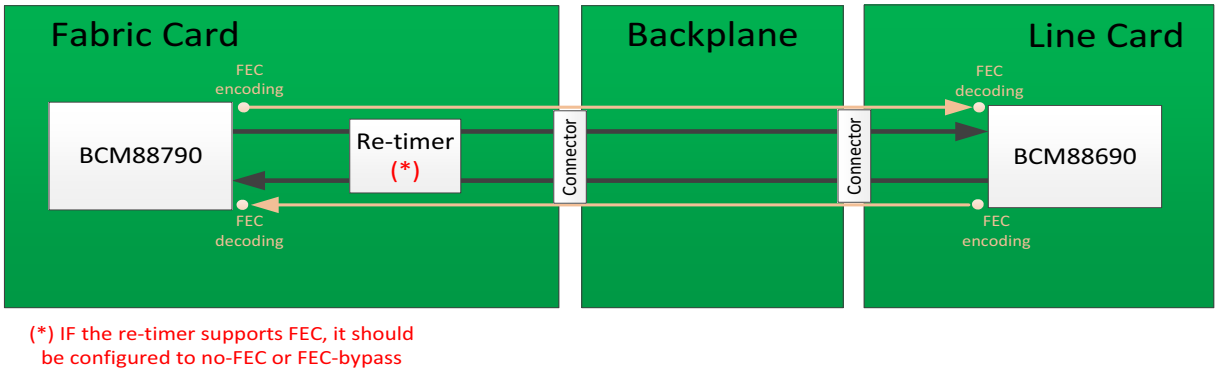
NOTE: As described in the data sheet, the BCM88790 supports internal re-timer functionality. For more information, refer to the Re-Timer Functionality section in the BCM88790 data sheet (88790-DS1xx).

2.8.1 Fabric FEC

The DNX16 fabric FEC is proprietary (for both 25G and 50G) and, therefore, is not supported by the re-timers. The fabric FEC should be operated end-to-end and not hop-by-hop.

If the re-timer does support FEC (usually IEEE-standard FEC), it must be disabled on both sides of the re-timer.

Figure 20: Fabric FEC Is Operated End-to-End



The use of re-timers and end-to-end FEC significantly increases the channel length that can be supported with the fabric interface. The BER performance of a link with a re-timer is equivalent to the performance of the longer portion of this link.

For example, assuming that a fabric SerDes without re-timer can support a channel of ~30 inches.

If a 50-inch channel is split to two parts of 30 inches and 20 inches, the total performance of this link will be approximately as the performance of a 30-inch link.

2.8.2 Link-Training and Force TX FIR Settings

2.8.2.1 Link-Training Connectivity Scheme – General Discussion

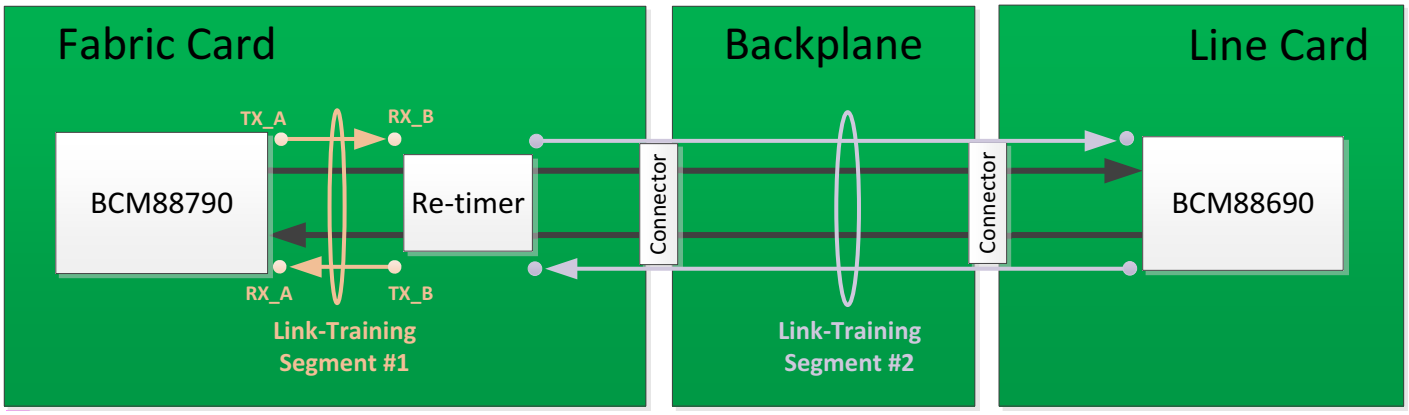
When using link-training (always recommended), link-training is responsible for selecting the TX FIR values, enabling and disabling the PAM-4 pre-coding, and setting the appropriate PAM-4 RX mode (NR/ER).

The link-training operation includes the following characteristics:

- Link-training is operated separately in each segment (in other words, hop-by-hop and not end-to-end). In the following figure, there are two separated segments, and the link-training works separately in each segment.
- Link-training can be operated only on segments in which both RX and TX are connected to the same logical entity on each side. In the following figure, RX_A and TX_A are connected to the same quad on the BCM88790, and RX_B and TX_B are connected to the same re-timer (and following the re-timer rules for lanes-swap and link-training operation). The same is true for Segment #2. This connection strategy is required because, for example, when RX_A requests a TX FIR change on TX_B (its link-partner), RX_A makes the request by sending it over TX_A, which connects to RX_B. The re-timer changes the TX FIR on TX_B according to the requests received on RX_B.

  In this section, such segments are referred to as *bidirectional* segments.

Figure 21: Link-Training Is Operated Separately in Each Segment on Bidirectional Segments



When link-training is disabled, the user manually configures the following settings:

- TX FIR values
- PAM-4 RX mode NR/ER
- PAM-4 pre-coding enabled or disabled
- CDR mode OS/BR

2.8.2.2 Blackhawk Link-Training vs. Force TX FIR

Link-training is strongly recommended because it provides a fast and convenient way to set the TX FIR automatically without needing to follow manual procedures.

When using predefined TX FIR values over PAM-4 for challenging channels (backplane/midplane/copper-cables), there is no expectation that a single set of predefined TX FIR values can overcome board and device process tolerance.

Having a single set of predefined values is possible with very short links (C2M, for example, where link-training is not supported), but it is certainly not expected with backplane/midplane/copper-cables applications. For these applications, using link-training is required.

This information is relevant for PAM-4 only, and there is no change in NRZ requirements as compared with previous generations. For NRZ, it is possible to use the TX FIR sweep method to obtain the values, without using link-training.

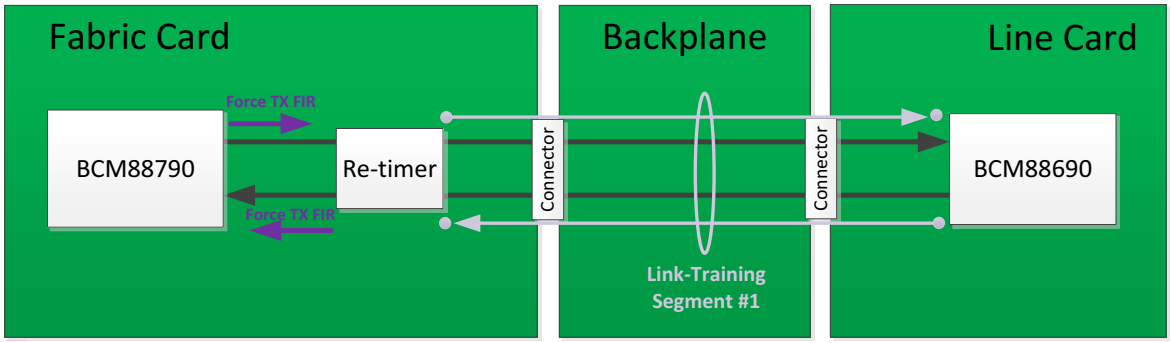
Still, whenever possible, use link-training.

2.8.2.3 Blackhawk PAM-4-Supported Schemes

When combining the requirements described in the previous two sections, the PAM-4 link over the backplane/midplane must support and use link-training.

The local link to the re-timer (on the same board) can use either link-training or force TX FIR. For guidance, refer to the re-timer guidelines. In some cases, the re-timer guidelines are to use link-training only on the line-side and use fixed TX FIR settings on the system-side, as shown in the following figure.

Figure 22: PAM-4 Scheme – Link-Training Is Used on the Backplane, While Force TX FIR Is Used on the Local Link



2.8.2.4 Blackhawk NRZ-Supported Schemes

For NRZ links, it is also possible to select whether to use link-training or force TX FIR for the backplane links.

The local link to the re-timer (on the same board) can use either link-training or force TX FIR. For guidance, refer to the re-timer guidelines. In some cases, the re-timer guidelines using link-training only on the line-side and using fixed TX FIR settings on the system-side, as shown in Figure 23.

When using force TX FIR, the links can be unidirectional.

Figure 23: NRZ Scheme – Link-Training Is Used on the Backplane, While Force TX FIR Is Used on the Local Link

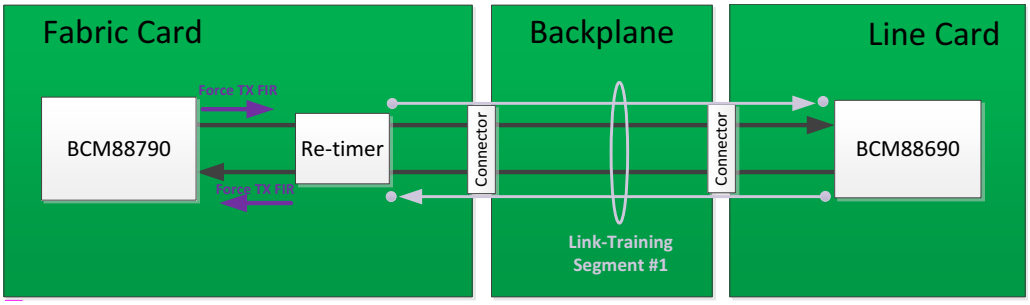


Figure 24: NRZ Scheme – Force TX FIR Is Used on All Links

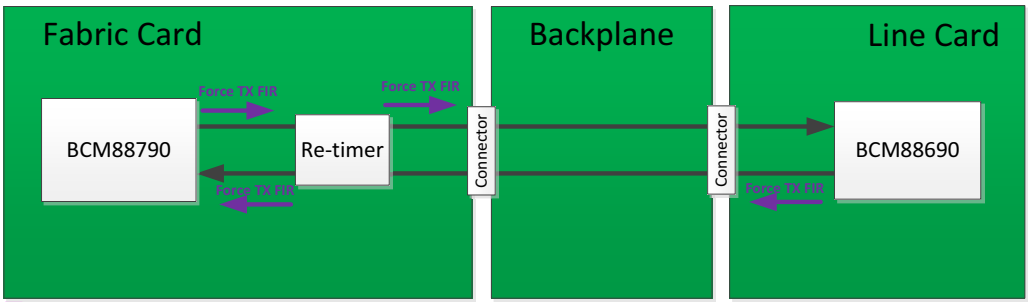
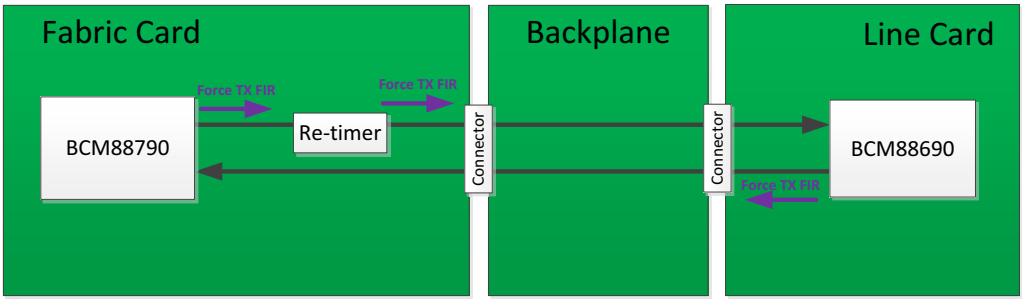


Figure 25: NRZ Scheme – Force TX FIR Is Used on All Links, Unidirectional Links

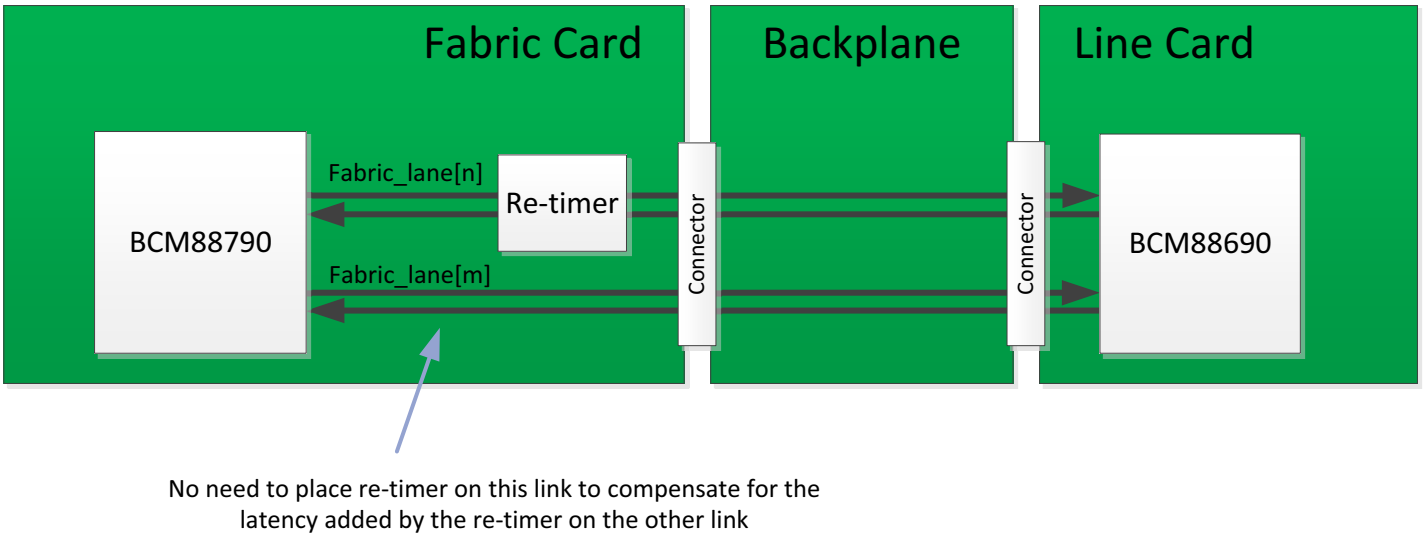


2.8.3 Fabric Latency

When some fabric links have a re-timer in their path and others do not (due to differences in the total length of a link, insertion-loss, and so on), there is no need to compensate for the additional latency introduced by the re-timer device (in other words, to place re-timer devices on all links).

Consider the latency added by the re-timer when calculating the total latency of the channel (and when verifying against the data sheet value for the maximum latency allowed).

Figure 26: Re-Timer Is Not Required on All Links

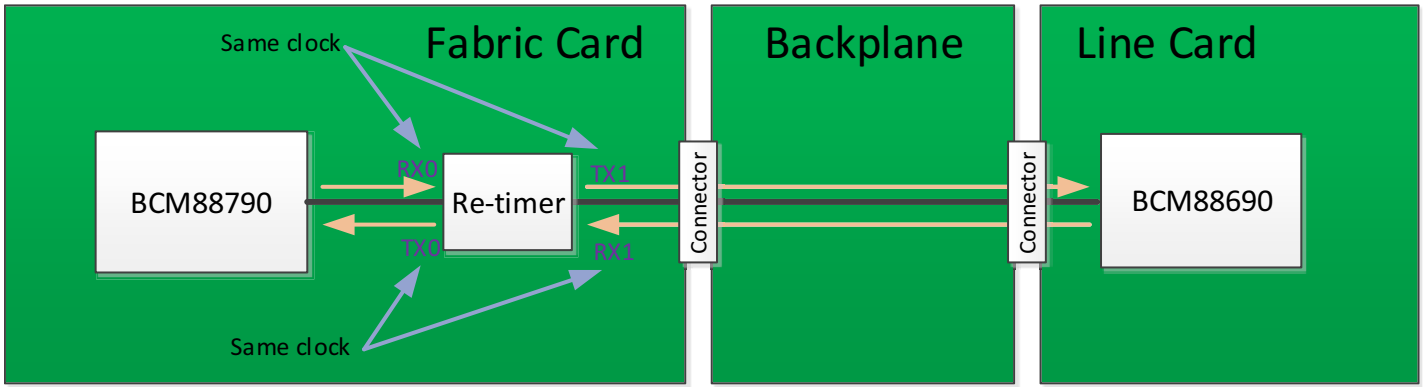


2.8.4 Clock Recovery

A re-timer device recovers the clock on each lane and uses this clock to transmit this lane to the next device. In other words, the re-timer recovers the clock from the [BCM88790 → re-timer] link and uses this clock for the transmission on the [re-timer → BCM88690] link.

As the following figure shows, TX1 is synced to the clock recovered from RX0, and TX0 is synced to the clock recovered from RX1.

Figure 27: Clock Recovery Example



2.8.5 Electrical Interface to the DNX16 Device

Consider the following guidelines:

- For electrical specifications, such as information about differential swing, common mode, AC coupling, and so on, refer to the Electrical Specifications section in the DNX16 device data sheet.
- Follow the electrical specification standard as listed in the DNX16 device data sheet. For example, for a 53-125 Gb/s backplane design, the BCM88790 supports 50GBASE-KR (IEEE 802.3cd CL137). Make sure the physical channel and the re-timer devices also support this standard.
- Follow the channel compliance validation and the IBIS-AMI simulation requirements as described in Section 2.1, Blackhawk Performance.
- Regarding squelch, do not use modules that provide invalid signals toward the Blackhawk, in case of link-down. This issue is noted in Section 2.1.1, Blackhawk Electrical Specification Compliance.

NOTE: For a high-speed SerDes, such as Blackhawk, simulations are required to validate the system design and to provide performance estimations in the form of bit error rate (BER).

2.9 Distributed Disaggregated Chassis Considerations

The distributed disaggregated chassis (DDC) approach replaces the physical chassis with a logical chassis.

The chassis cards are replaced by the following two types of platforms:

- The network cloud packet forwarder (NCP) is the line card pizza box.
- The network cloud fabric (NCF) is the fabric card pizza box.

The DNX fabric connectivity between the platforms is handled by optical or electrical transceivers and cables, which replace the chassis backplane.

This section describes various issues to consider when designing the DNX fabric connectivity of a DDC.

2.9.1 DNX Fabric Links Are Not Ethernet

DNX fabric links use some of the IEEE 802.3 definitions but are not IEEE 802.3 compliant and cannot be considered as Ethernet. DNX fabric links include the following characteristics:

- A fabric link is a single-lane point-to-point connection.
- A link TX SerDes carries data and management/control messages targeted to the link partner RX SerDes.

Because the fabric link is a point-to-point connection, both RX SerDes and TX SerDes should be mapped to the same logical link.

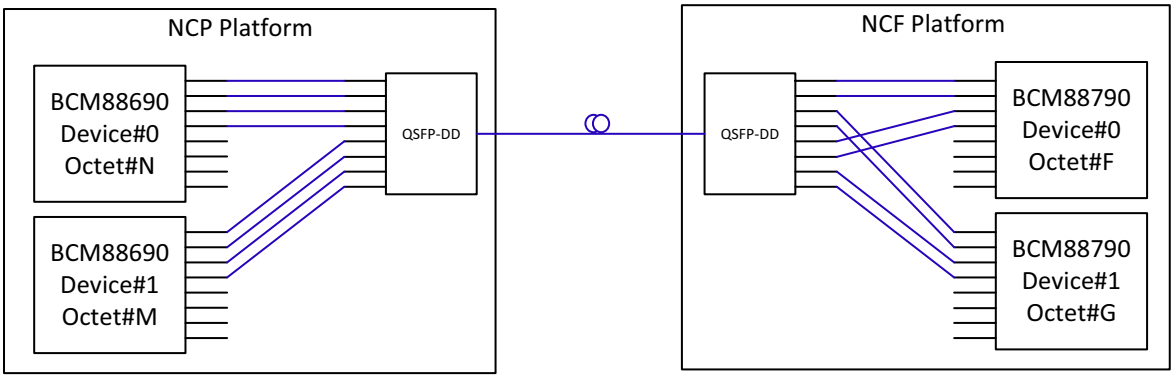
Fabric links are single-lane links, meaning a group of 8 × 50G fabric links cannot be considered as one logical 400G fabric link.

2.9.2 Each Front-Panel Connector Is Connected to More Than One Device

If the NCP or NCF includes more than one device, the typical connectivity guidelines are to connect each front-panel connector (for example, each QSFP-DD module) to all devices present on this platform. The following figure shows an example of this type of connectivity.

In this example, one QSFP-DD module on the NCP platform is connected to two BCM88690 devices. Similarly, one QSFP-DD module on the NCF platform is connected to two BCM88790 devices.

Figure 28: Example: QSFP-DD Connected to More Than One Device in a DDC



This connectivity scheme where a QSFP-DD module is connected to more than one device is also different than the usual Ethernet connectivity in which a QSFP-DD is connected to a 400G Ethernet port of a single device.

2.9.3 Considerations for the Fabric Interconnect on a DDC

In addition to the DNX devices, the fabric interconnect on DDC systems includes the following components (or part of the component):

- Re-timer devices
- Optical transceivers
- Passive copper cables
- Active copper cables (which include CDR, re-timers, or other active devices as part of the cable)

When using standard 400G Ethernet interconnect equipment for DDC fabric connectivity, analyze whether it uses any design assumptions that are based on Ethernet and may not be applicable for DNX fabric.

The following three subsections describe examples of such assumptions.

2.9.3.1 Example 1: Clock Synchronization of the Interconnect Lanes

Verify if the interconnect device assumes that all lanes are synchronized to the same clock. If the device makes this assumption, use the same SerDes reference clocks for all of the DNX devices on each pizza box.

2.9.3.2 Example 2: 400GBASE-DR4 Modules

Some types of interconnects convert or map *n* electrical signals to *n* ÷ 2 optical ports (and vice versa).

An example of this interconnect type is a 400GBASE-DR4 module (IEEE 802.3bs Clause 124), which performs the following conversions:

- TX path: Converts the eight electrical channels (lanes) of 50-Gb/s PAM4 to four optical channels of 100 Gb/s.
- RX path: Converts four optical 100-Gb/s channels to eight electrical 50-Gb/s PAM4 channels (lanes).

When using an interconnect that performs this type of conversion, consider the following issues:

- Is it required to synchronize each pair of 50-Gb/s lanes that are multiplexed into one 100-Gb/s lane to the same SerDes reference clock?

  If this synchronization is required, use the same SerDes reference clocks for all the DNX devices on each pizza box or route each pair to the same DNX device.
- Does the module guarantee that the lane order before the multiplexing is maintained and remains the same after the reverse operation occurs on the peer module? For example, if lanes 6 and 7 are converted to one 100-Gb/s lane and then converted back to two 50-Gb/s lanes, do the modules guarantee that the lane order is still [6, 7], or it is it possible for the order to be arbitrary (per link-up), where the order is either [6, 7] or [7, 6]?

  Some modules guarantee that the pre- and post-conversion lane order remains the same.

  If the lane order changes, the following problems may occur:
  - The RX and TX of a channel are not bundled together (which results in an LLFC mismatch).
  - A channel might be connected to a different FE device than desired (which results in a reachability bitmap mismatch).

  Further discussion about this scenario is beyond the scope of this document. However, as a general guideline, on the NCP platform, route the two channels (lanes) the module converts into one 100-Gb/s channel to the same fabric quad. This routing allows TX lane swapping, if required.

  To summarize, using 400GBASE-DR4 modules (or similar) on the fabric interface of a DDC is not straightforward and requires special attention to the connectivity on the NCP and NCF platforms during the board design and routing phases, along with a thorough understanding of the modules' requirements and behavior.

  If you intend to use modules like these over the interface, contact Broadcom.

2.9.3.3 Example 3: Lane Status

Make sure the interconnect device does not assume that the status of any lane (for example, lane 0) implies the status of all eight lanes.

Chapter 3: Falcon16 SerDes Interface

On the DNX16 devices, the Falcon16 SerDes is used for network interfaces. Falcon16 is a SerDes block that supports four serial links.

NOTE: For supported rates, refer to the data sheet for the specific device.

3.1 Falcon16 Performance

Falcon16 is designed to achieve a BER of 10^{-15} in the following mode: NRZ (Non-Return to Zero) mode without FEC.

Additional information about Falcon16 performance is identical to the Blackhawk performance information in Section 2.1, Blackhawk Performance.

3.2 Falcon16 PLL Frequency and Lane Rate

For information, refer to the Mixed Port Types on PM25 section in the device data sheet (see Related Documents).

3.3 Falcon16 P/N Polarity Inversion

For information, see Section 2.2, Blackhawk P/N Polarity Inversion.

3.4 Falcon16 Lane Mapping

NOTE: Lane mapping (swapping) can be set during device initialization. When setting lane mapping after the device has been initialized, all ports must first be removed from the 4-lane core SerDes. This feature is supported for all DNX16 devices and interfaces.

3.4.1 Falcon16 Lane Mapping – NIF Ethernet

The Falcon16 NIF Ethernet lane-mapping implementation is similar to the Falcon28 implementation (and not to the Blackhawk implementation).

When performing lane mapping, adhere to the following rules:

- The RX physical SerDes and TX physical SerDes of a lane must reside on same quad.
- All RX physical SerDes and all TX physical SerDes of the port (regardless of whether it is a 1-lane, 2-lane, or 4-lane port) must reside on same quad.

3.4.2 Falcon16 Lane Mapping – BCM88800 ILKN

The information about Falcon16 lane mapping for ILKN ports is the same as it is for Blackhawk ILKN ports. See Section 2.3.4, Blackhawk Lane Mapping – BCM88800 ILKN.

3.4.3 Falcon16 Lane Mapping – BCM88480 ILKN

The information about Falcon16 lane mapping for ILKN ports is the same as it is for Blackhawk ILKN ports. See Section 2.3.5, Blackhawk Lane Mapping – BCM88480 ILKN.

3.5 Unused Falcon16 Links

Unused TX and RX pairs of Falcon16 quads can be left open (unconnected).

Unused quads should be connected to an active supply.

In the BCM88800 and BCM88480, keep all reference clocks connected, regardless of whether or not the SerDes cores are used.

NOTE: Links that are enabled (transmitting) but not terminated properly might interfere with other links. Be sure to disable such unterminated links.

3.6 Falcon16 AC Coupling

The information about Falcon16 AC coupling is the same as it is for Blackhawk. See Section 2.5, Blackhawk AC Coupling.

3.7 Falcon16 Hot Plug

The information about Falcon16 hot plugging is the same as it is for Blackhawk. See Section 2.6, Blackhawk Hot Plug.

3.8 Falcon16 Layout Guidelines

The Falcon16 layout guidelines are the same as described in Section 2.7, Blackhawk Layout Guidelines.

In the BCM88290 package, the ball-to-ball pitch is smaller than it is in other DNX16 devices. The pitch was reduced from 1.0 mm to 0.8 mm. To accommodate this ball-to-ball pitch reduction in the BCM88290 device, make BGA breakout adjustments and take signal-integrity impacts into consideration when performing simulations and validations.

Due to the decreased distance between signals (pads, vias, and routing) and signals to power, board designers should be aware that the same breakout rules used with 1.0-mm pitch designs might result in increased crosstalk with the 0.8-mm pitch in the BCM88290 device.

3.9 External Re-Timer Device Guidelines

Guidelines for the Falcon16 external re-timer device are the same as the guidelines for Blackhawk. See Section 2.8, External Re-Timer Device Guidelines.

Chapter 4: Statistics Interface SerDes Limitations

The DNX16 device family includes an option to use Ethernet ports as the STAT_IF (statistics interface). For information about this option, refer to the Statistics Interface section in the System Interfaces chapter in the device data sheet.

STAT_IF and ETHERNET NIF (Network Interface) should not share the same PM50 or a group of PM25.

- BCM88690:

  Each PM50 can be used for STAT_IF.

  Do not share STAT_IF and NIF ETHERNET on the same PM50. (STAT and ILKN can be shared.)

  The options are as follows:

  – Use a single PM50 as a 400G STAT interface.
  – Use a single PM50 for two 200G or four 100G (2 × 50G) STAT interfaces.
  – Use two PM50 for four 100G (4 × 25G) STAT interfaces.
  – Share STAT with ILKN.
- BCM88800:

  Do not share PM50_03 for STAT_IF and NIF ETHERNET. (STAT and ILKN can be shared.)

  If STAT_IF uses any of PM25_04, PM25_05, PM25_06, or PM25_07, other PM25 of this group (PM25_04, PM25_05, PM25_06, or PM25_07) cannot be used for ETH. (They can be used for ILKN.)
- BCM88480:

  If STAT_IF uses any of PM25_05, PM25_06, or PM25_07, other PM25 of this group (PM25_05, PM25_06, or PM25_07) cannot be used for ETH. (They can be used for ILKN.)

Chapter 5: PCIe Interface

5.1 DNX16 PCIe Interface Overview

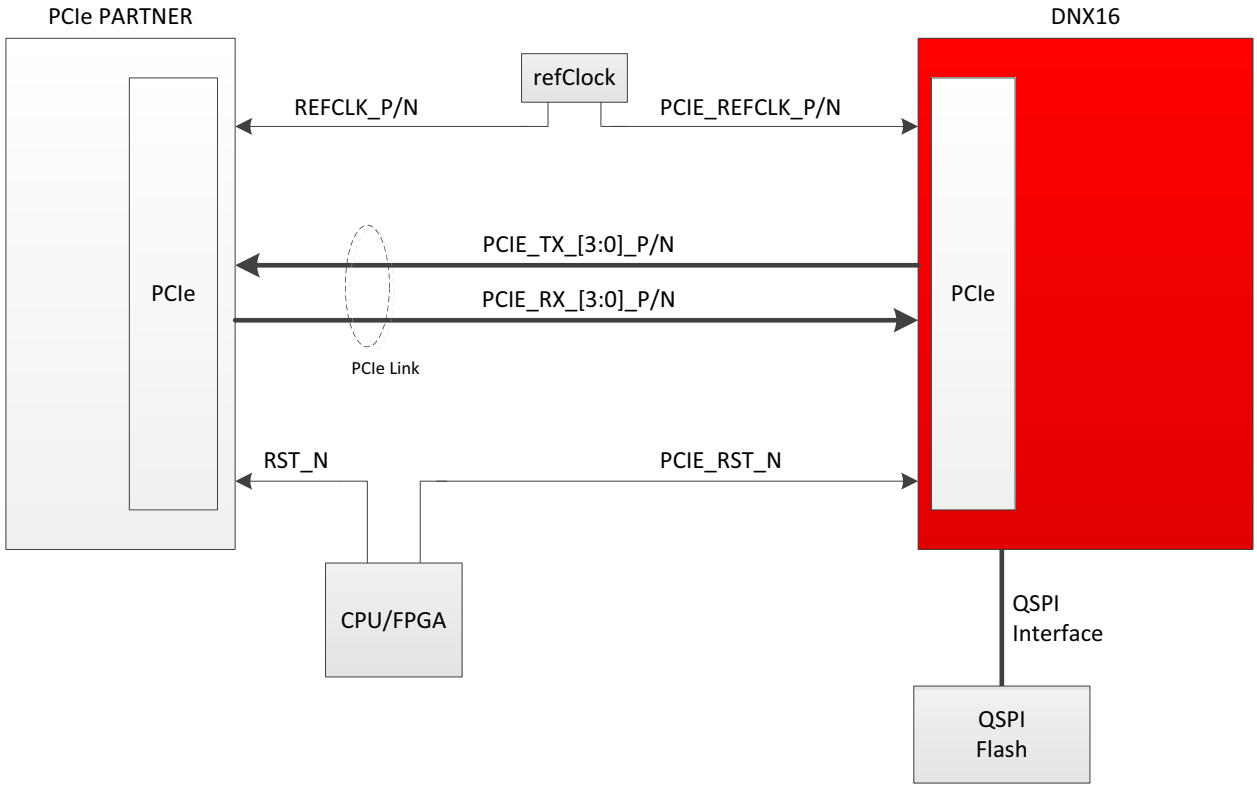
The DNX16 PCI Express (PCIe) is compliant with the *PCI Express Base Specification* Revision 3.0.

The DNX16 PCIe solution supports link widths of x1, x2, and x4 lane endpoints (EPs). Supported link speeds are 2.5 GT/s for PCIe Gen1, 5.0 GT/s for PCIe Gen2, and 8.0 GT/s for PCIe Gen3.

The example in Figure 29 shows a connectivity with the following:

- PCIE_REFCLK_P/N: PCIe common reference clock architecture.
- PCIE_TX_[3:0]_P/N: Four PCIe TX pairs.
- PCIE_RX_[3:0]_P/N: Four PCIe RX pairs.
- PCIE_RST_N: PCIe fundamental reset, used as PCIe cold and warm reset. For assertion and deassertion requirements, refer to the data sheet for the DNX16 device.
- QSPI flash: Stores the PCIe SerDes firmware and operation parameters.

Figure 29: PCIe Connectivity Example



NOTE: Disable active state power management (ASPM) for normal operation. The device does not support this feature. If this feature is enabled, the device may not initialize properly.

5.2 PCIe Link and Lanes

5.2.1 PCIe Link

The supported widths for DNX16 PCIe links are x1, x2, or x4 lanes. Each lane has one TX differential pair and one RX differential pair. These options are supported for PCIe Gen1, Gen2, and Gen3.

5.2.2 PCIe Polarity Swap

DNX16 PCIe supports RX polarity swap only. This operation is done automatically by the device. TX polarity swap is not supported.

5.2.3 Link Forming and Lane Configuration

The following table shows the possible lane connections.

Table 3: PCIe Lane Connection Options

Lane Number	Straight [DNX16 ↔ Partner]
x1	lane-0 ↔ lane-0
x2	lane-0 ↔ lane-0 lane-1 ↔ lane-1
x4	lane-0 ↔ lane-0 lane-1 ↔ lane-1 lane-2 ↔ lane-2 lane-3 ↔ lane-3

5.2.4 Unused Lanes

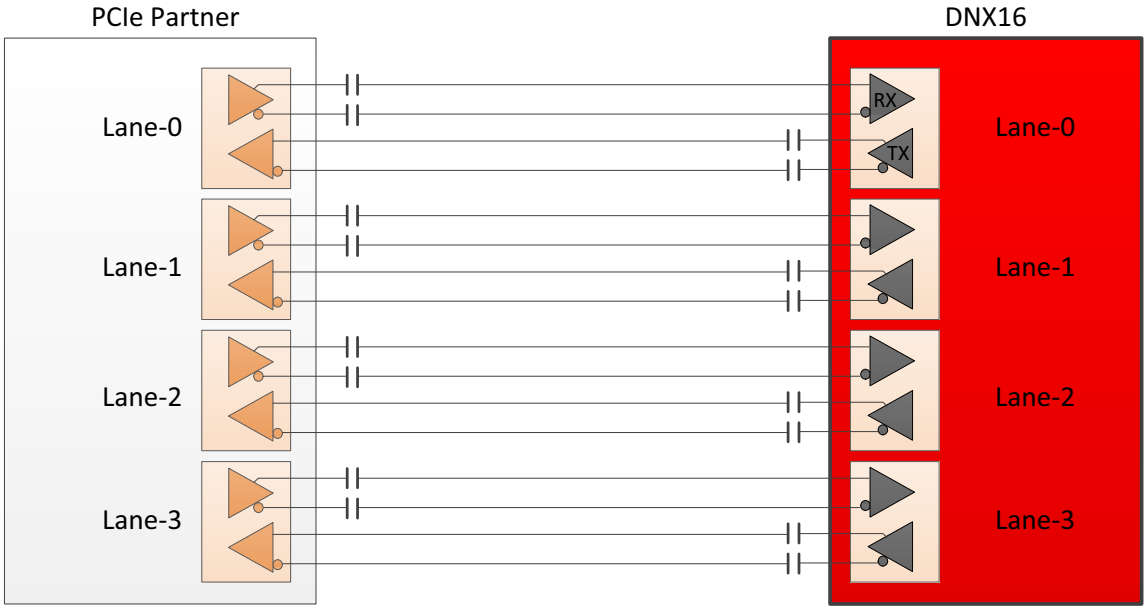
Leave unused RX and TX lanes open.

5.2.5 PCIe AC Coupling

The DNX16 PCIe Gen1, Gen2, and Gen3 requires having external AC-coupling capacitors on both RX and TX pairs.

The following figure shows an example of x4 lanes with AC coupling.

Figure 30: PCIe AC-Coupling Capacitors



NOTE: PCIe Gen1 and Gen2 require an AC-coupling capacitor value between 75 nF and 265 nF. Gen3 requires an AC-coupling capacitor value between 176 nF and 265 nF. Select a capacitor that fits your application. A typical selection is 220 nF.

The PCIe standard states that it is preferable to place the capacitors close to the transmitter. In a system where both devices are always connected, the capacitors can be placed anywhere.

5.3 PCIe QSPI Flash

The QSPI flash holds the PCIe firmware, the PCIe configuration, and other DNX device configuration settings.

Successful loading of the PCIe firmware and configuration from the QSPI flash is mandatory for proper PCIe operation and link-up.

QSPI flash must be connected to each DNX16 device for proper PCIe Gen1, Gen2, and Gen3 operation. This requirement is mandatory.

The PCIe link (including Gen1 and Gen2) cannot be guaranteed without loading the QSPI firmware image. By default, the DNX16 device advertises its capabilities as Gen1, Gen2, and Gen3. There is no external pin to force the DNX16 device to advertise its capabilities as Gen1 and Gen2 only.

Per standard PCIe operation, the link endpoints negotiate their capabilities, and the actual link-rate is set to the highest common rate of both ends.

For PCIe Gen1, Gen2, and Gen3 operation, set the appropriate PUC to QSPI Enable.

The QSPI firmware image is available on the Engineering Support Portal (ESP).

5.3.1 QSPI Controller External Interface (BSPI and MSPI Modes)

The following table describes the different operational modes of the QSPI controller.

Table 4: QSPI Controller Modes

Mode	When Used	Read/Write	Single-Lane or Quad-Lane Mode[a]
BSPI at boot	At device initialization (when the device is taken out of reset) for loading the PCIe firmware, PCIe configuration, and other configurations.	Read only	Single-lane mode. (The setting is hard-wired in the DNX devices).
BSPI after boot	Used by the SDK for QSPI device operations.	Read or write	Single-lane or quad-lane mode is set by the compilation flag. ■ IPROC_BSPI_READ_CMD ■ IPROC_BSPI_QUAD_READ_CMD
MSPI	After the SDK initialization is complete, for flash write or read using SDK or BCM shell commands.	Read or write	Single-lane mode.

a. In single-lane operation, the MISO/MOSI/HOLD/WP signals are unidirectional. In quad-lane operation, the MISO/MOSI/HOLD/WP signals are bidirectional.

The QSPI selection and the board should allow operation of both BSPI and MSPI, as well as both single-lane and quad-lane operation. A key difference between single-lane operation and quad-lane operation is the interface signal direction. In single-lane operation, interface signals are unidirectional. In quad-lane operation, part of the interface signals are bidirectional.

When designing the QSPI interface on the board, consider the quad-lane bidirectional signal operation:

- Buffers that have another master cannot be simple unidirectional buffers. In this case, consider using analog buffers to support bidirectional signals.
- Voltage level translation cannot be done with unidirectional buffers or analog buffers. Use flash devices with the same operational voltage as the Broadcom device.
- Simple serial termination may not be sufficient. Consider termination schemes that are compatible with bidirectional signals.

5.3.2 QSPI Flash Device Selection Guidelines

When selecting a QSPI flash device, use the following guidelines:

- The interface voltage must comply with the value specified in the DNX device data sheet:
  - 3.3V: BCM88790 and BCM88690
  - 1.8V: BCM88800, BCM88480, and BCM88290
- The flash size should be 128 Mb or 256 Mb. Using a smaller flash size might not provide enough space if the firmware size increases in the future.
- The flash must include a HOLD pin to connect to the DNX QSPI controller. The controller does not support a RESET pin.

  If the flash device has a RESET pin, follow the flash device vendor recommendations. Be sure to complete the QSPI flash device reset cycle prior to taking the DNX device out of reset. For maximum flexibility, connect the flash device RESET pin input to the same source that is driving the DNX16 SYS_RESET_N input pin. See the connectivity example in Figure 31. Regardless of the RESET pin, the flash device must include a HOLD pin.
- The default DNX boot uses single lane. The flash should support operation in all three lane modes (single lane, dual lane, and quad lane).
- The default DNX boot uses 3-byte addressing. When programing the flash device by using an external programmer, use 3-byte addressing. The flash should support operating in both addressing modes (3-byte addressing and 4-byte addressing).
- The QSPI controller issues FAST_READ commands. READ (03h) commands are not supported. Therefore, the QSPI device's operating frequency should be based on the FAST_READ command instead of the READ (03h) command.
- The QSPI device must comply with the values specified in the QSPI section of the Electrical Specifications chapter in the DNX device data sheet. For example, ensure the minimum operating speed aligns with the data sheet value.
- To ease system and SDK integration, select a device from Section 5.3.3, QSPI Flash Device Part Numbers for Reference, or select a device that is compatible with a device listed in this section.

5.3.3 QSPI Flash Device Part Numbers for Reference

The tables in this section list QSPI serial flash vendors and part numbers that can be used for reference. These devices are listed here as examples of parts to consider.

Because vendors update their part number as the technology matures, consult the flash vendors for updated part numbers.

Broadcom does not qualify or test QSPI flash devices. It is the responsibility of the customer to qualify the device for each design and to check compatibility with recommended devices. The devices in the following table are listed only for reference and to assist with device selection, but other compliant devices can also be used.

The BCM88690 and BCM88790 support QSPI flash devices with 3.3V I/Os. The following table lists supported QSPI flash devices.

Table 5: 3.3V I/O QSPI Flash (BCM88790 and BCM88690)

Vendor	Family Part Number	I/O Voltage
Micron	MT25QL128/MT25QL256	3.3V
Cypress	S25FL128S/S25FL256S	3.3V
Winbond	25Q256FV	3.3V
Winbond	W25Q256JVFIM	3.3V

The BCM88800, BCM88480, and BCM88290 support QSPI flash devices with 1.8V I/Os. The following table lists supported QSPI flash devices.

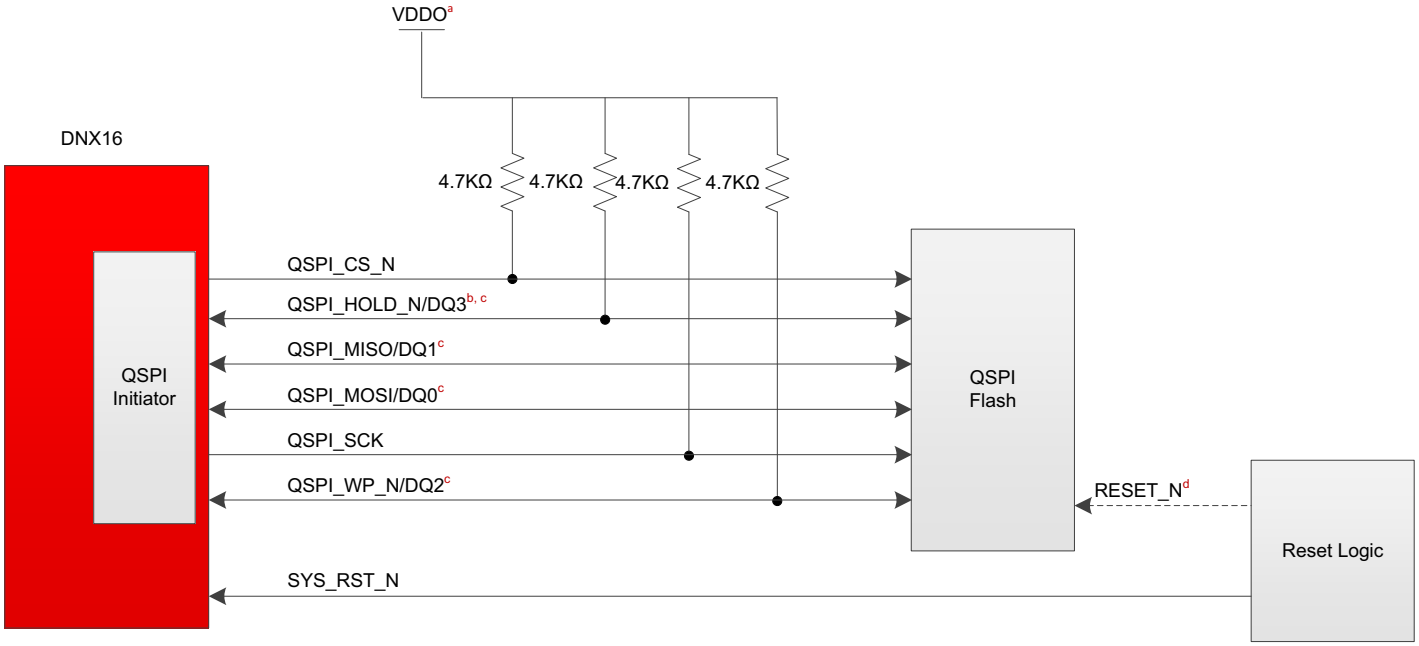
Table 6: 1.8V I/O QSPI Flash (BCM88800, BCM88480, and BCM88290)

Vendor	Family Part Number	I/O Voltage
Micron	MT25QU128/MT25QU256	1.8V
Winbond	W25Q128JWFIQ	1.8V

5.3.4 QSPI Interface Connectivity

The following figure shows the connection of the QSPI flash to a DNX16 device.

Figure 31: QSPI Interface Connection Example



a. 3.3V or 1.8V, according to the device data sheet.
b. QSPI_HOLD_N is a required connection between the DNX device and the QSPI flash device. Used as bidirectional in quad-lane operation.
c. Different modes of QSPI interface operation are possible. In some modes, the DQ[3:0] are inputs/outputs.
d. An optional connection. If the QSPI flash device supports a RESET_N pin, connect it to the HOST/FPGA that drives DNX SYS_RST_N. The QSPI device should be ready for operation before DNX07 device SYS_RST_N deassertion.

NOTE: CS_N and SCK are always derived by the device. If other devices with a QSPI initiator must be connected to the QSPI flash bus, use buffers to disconnect the DNX16 QSPI interface from the QSPI flash during programming.

5.3.5 DNX16 QSPI Flash Programming

As stated in Section 5.3, PCIe QSPI Flash, using the QSPI flash is mandatory, and establishing a Gen1, Gen2, and Gen3 link cannot be guaranteed if the QSPI flash is not programmed properly.

NOTE: After programming the QSPI flash, a device reset is required for loading and running the latest programmed image.

This section provides detailed information about the different options that are available for programming the flash.

At the system level, other options might be possible based on the specific system design, but the principles covered in this section apply to those options as well.

Program in 3-byte addressing mode only. Programming in 4-byte addressing mode is not supported.

There are three use cases when programming the flash:

1. **First flash programming**: This occurs in production, for example.
2. **Field upgrade**: The flash is already programmed, a valid PCIe link can be established, and the SDK is loaded successfully.
3. **Field upgrade recovery**: In the field, the flash image is corrupted or missing, and therefore a valid PCIe link cannot be established.

NOTE: In some systems, it is common to have the flash store two sets of images so that if one image is corrupted, there is a valid backup image to load.

The QSPI implementation of the DNX16 devices *does not* support such operation, and as result, if the QSPI image is corrupted, the flash must be reprogrammed by the system without relying on the PCIe link between the host and the DNX16 device.

The implementation of two images can be made at the system level, for example, by using two QSPI flash devices. The CPU host controls which flash device is active and connected to the DNX16 device. In this implementation, the DNX16 device sees only one QSPI flash at any given time.

NOTE: When an external device or programmer tool is used for programming the flash, the DNX16 device should be kept in reset during the programming process.

The following table describes the different options for programming the flash and whether the option is valid with each of the three use-cases.

Table 7: Flash Programming Options

No.	Programming Option	First Flash Programming	Field Upgrade	Field Upgrade Recovery
1	Out of the board, using external flash programmer before assembly. After programming the FW file to the QSPI flash, pad the remaining area with 0xFF.	✓	X	X
2	On-board, by the DNX16 device, using the BCM shell utility. (Running after the PCIe link is established and the SDK loading is complete)	X Cannot guarantee any PCIe link without flash.	✓	X Cannot guarantee any PCIe link without flash.
3	On-board, using external programmer connected to the Flash by header[a].	✓	X Possible but not realistic because technician access to the board is required.	X Possible but not realistic because technician access to the board is required.
4	On-board, using an external device present on-board[a]. (For example FPGA, I^2C to I/O device, CPU).	✓	✓	✓
5	On-board, by the DNX16 device, using the I^2C interface between the host and the DNX16 device.	✓	✓	✓
6	On-board, using boundary scan (JTAG).	✓ Broadcom provides only the device BSDL file.	X	X

a. When an external device or programmer tool is used for programming the flash, the DNX16 device should be kept in reset during the programming process. Even if the DNX16 device is kept in reset, the QSPI interface outputs are still driven. To avoid contention on the bus:

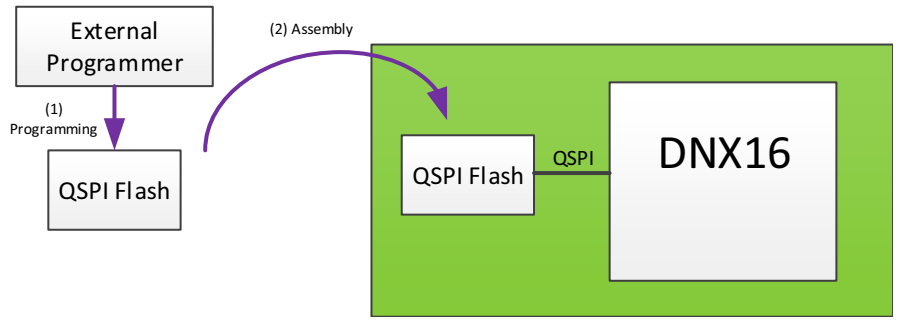
Option 1: Buffers are implemented to isolate the DNX16 device from the flash while the other entity programs the flash.

Option 2: The DNX16 is kept in power-down while the other entity programs the flash.

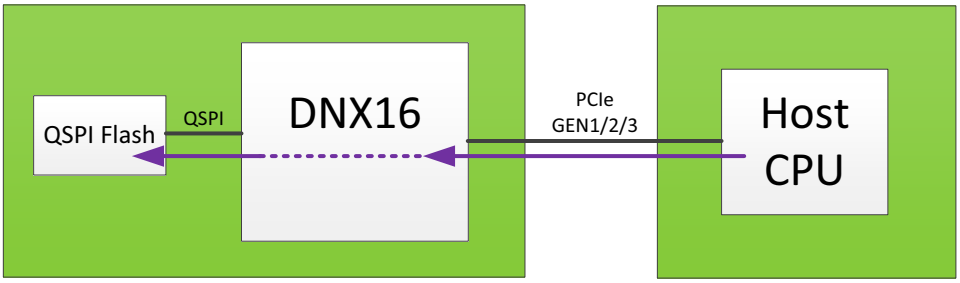
The following figures illustrate the conceptual connectivity and programming path in each option.

NOTE: The number for each item in the following list corresponds to the option number in Table 7.

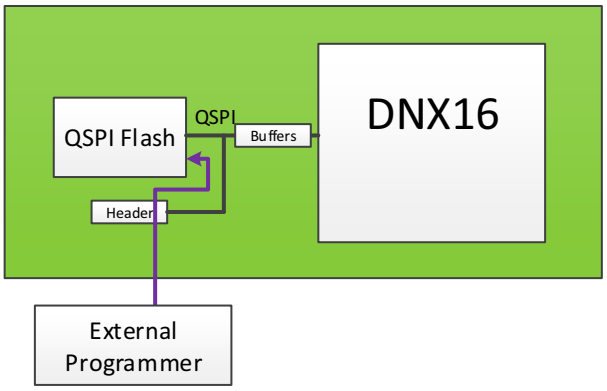
1. Out of the board, using external flash programmer before assembly. After programming the FW file to the QSPI flash, pad the remaining area with 0xFF.



2. On-board, by the DNX16 device, using BCM shell utility.

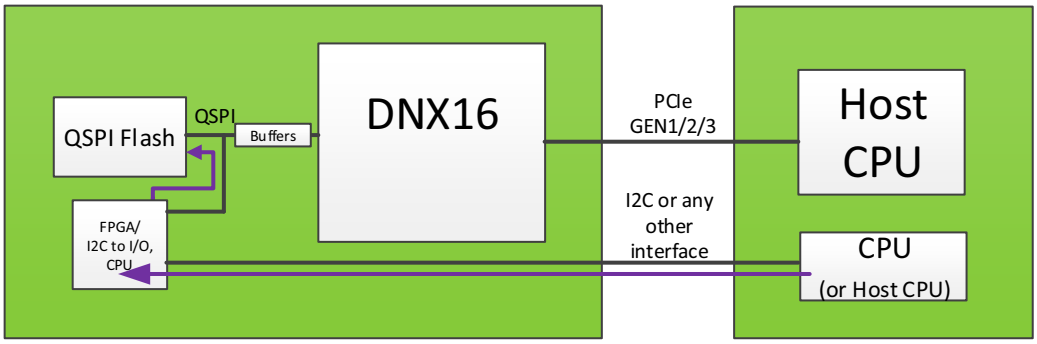


3. On-board, using external programmer connected to the flash by header.



NOTE: Buffers that have another initiator cannot be simple unidirectional buffers. In this case, consider using analog buffers to support bidirectional signals.

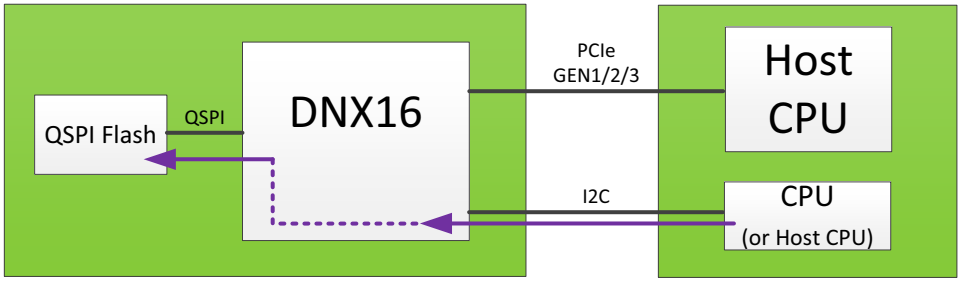
4. On-board, using an external device present on-board (for example, FPGA, I^2C to I/O device, CPU).



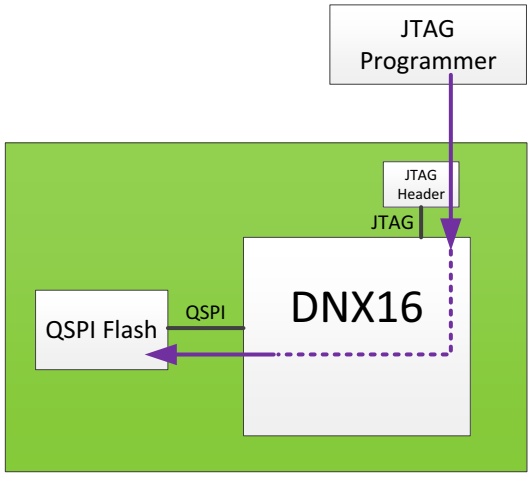
NOTE: Buffers that have another initiator cannot be simple unidirectional buffers. In this case, consider using analog buffers to support bidirectional signals.

5. On-board, by the DNX16 device, using the I^2C interface between the host and the DNX16 device.

NOTE: Not available on SDK 6.5.13.



6. On-board, using boundary scan (JTAG).



5.4 PCIe Layout Guidelines

This section provides layout guidelines for achieving proper operation of the PCIe links.

PCIe signal routing should follow the guidelines in Section 2.7, Blackhawk Layout Guidelines. The information that follows is intended to supplement the guidelines described previously.

NOTE: The guidelines detailed in this section should not replace a detailed simulation for each specific design.

5.4.1 Trace Impedance

The PCIe RX and TX links should have a target impedance of 100Ω ±10% on the customer PCB.

5.4.2 Skew and Length Match

The total allowed P to N skew should be smaller than 5 mils.

5.5 Calibration Resistors

DNX16 PCIe SerDes calibration resistors are shared with Blackhawk SerDes. For more details, see Section 10.1, RESCAL, and refer to the data sheet for the DNX16 device.

5.6 PCIe Modeling and Simulations

To enable PCIe simulations, the following models are available on the Engineering Support Portal (ESP):

- PCIe IBIS-AMI model
- PCIe device package model (s32p)

For more information about these files, see Table 51, IBIS-AMI Models, and Table 52, Package Models.

NOTE:

- The PCIe SerDes is not supported by the Broadcom LinkEye tool.
- Replace the package provided with the IBIS-AMI model with a device-specific package available on the ESP.
- The insertion-loss values available in the package model are applicable for operation at 20°C. When the device is operating at 110°C, add ~5% to those values.

Chapter 6: High Bandwidth Memory

The BCM88690 device package has two integrated High Bandwidth Memory (HBM) DRAMs. The BCM88800 integrates one HBM DRAM. The HBMs are manufactured by industry-leading memory vendors.

For power information, see Section 9.5, HBM Power Supply. For thermal information, see Section 12.3, BCM88690 and BCM88800 HBM Thermal Design Considerations.

6.1 HBM DA Test Port

The JESD235 JEDEC standard for the HBM defines a Direct Access (DA) test port used for vendor-specific memory test implementations.

Customers should leave the DA test port pin open (not connected).

NOTE: In the past, Broadcom recommended routing the DA test port to a connector. This recommendation is no longer valid.

6.2 X-Ray Inspection Considerations

Excessive X-ray inspection may cause performance degradation of the HBM.

To mitigate performance degradation issues, use the following guidelines:

- Manage the cumulative doses to an amount that is less than 30 mGy.
- Scrap the products after X-ray inspection of the HBM.

For additional information, contact Samsung.

Chapter 7: GDDR6 Interfaces (BCM88480 and BCM88290)

7.1 Introduction

The BCM88480 and BCM88290 device use GDDR6 as external DRAM. The BCM88480 has two GDDR6 interfaces (DDR0 and DDR1), and the BCM88290 has one interface (DDR1).

Each DRAM interface is connected to a single GDDR6 device and has two channels: Channel-A and Channel-B. Each channel operates independently (GDDR6 two channel mode). The following figures show the connectivity between the GDDR6 interfaces and the BCM88480 (Figure 32) or BCM88290 (Figure 33) device, power supplies, and reference clock.

For more information about power, see Chapter 9, Power Aspects and Section 9.10, GDDR6-Related Power Supply.

Figure 32: BCM88480 GDDR6 DRAM High-Level Blocks

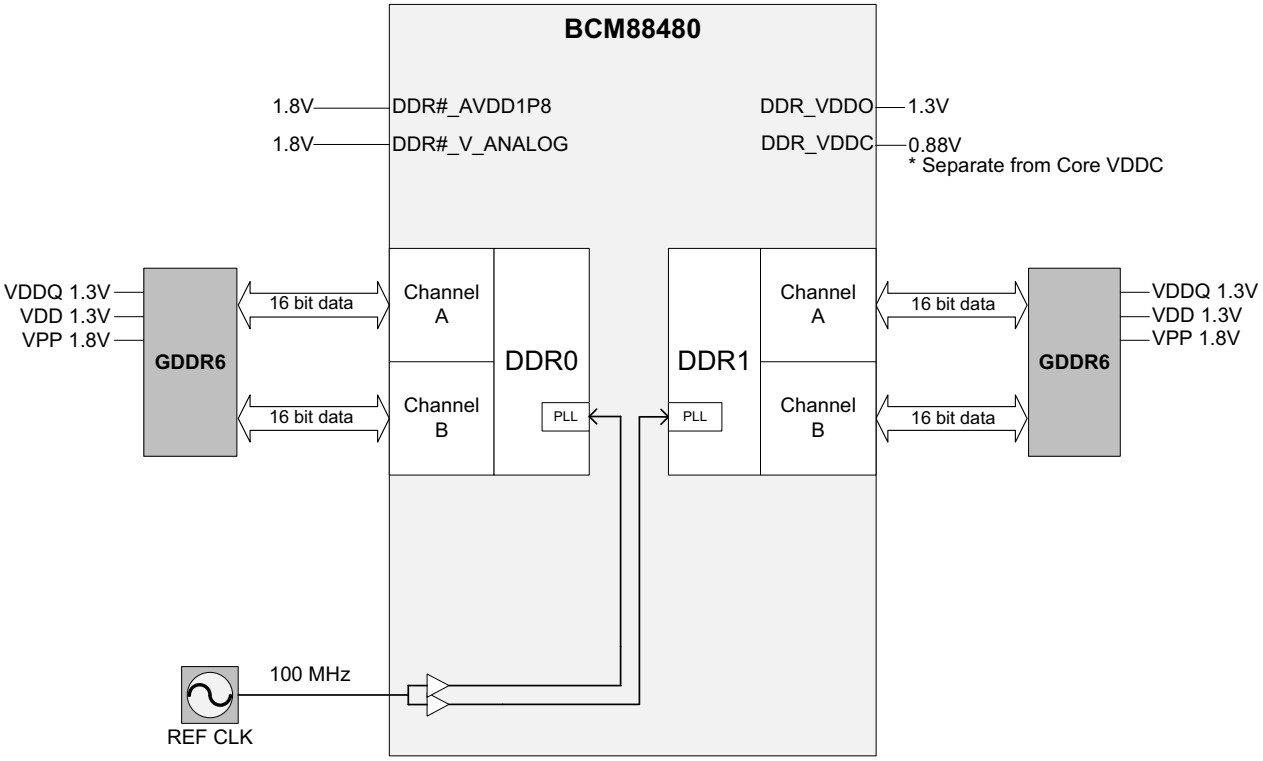
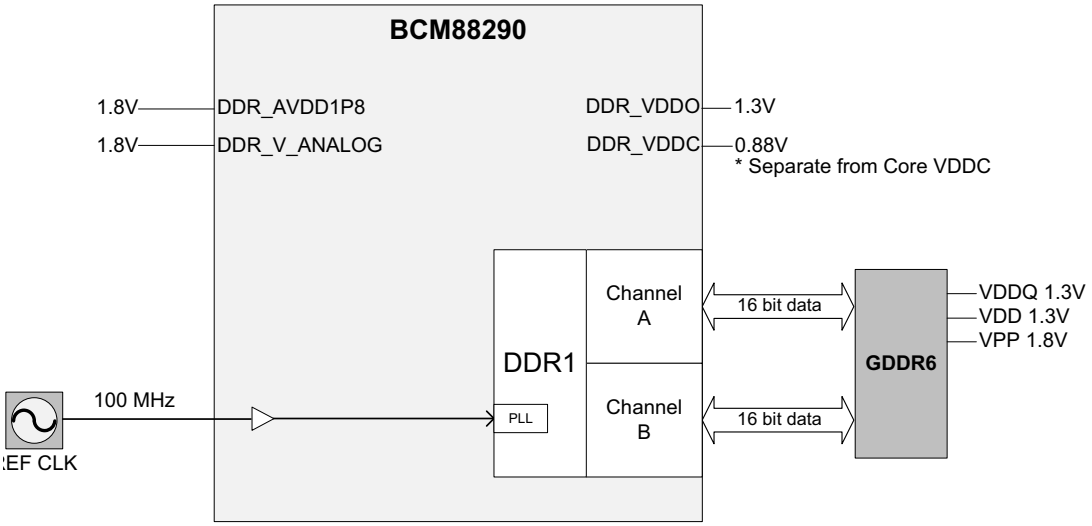


Figure 33: BCM88290 GDDR6 DRAM High-Level Blocks



NOTE: If the BCM88480 GDDR6 DDR0 or DDR1 interface or the BCM88290 DDR1 interface is not used and is not connected to a GDDR6 device, the following connectivity requirements apply:

- Connect all BCM88480 or BCM88290 GDDR6 interface-related power signals, analog signals, and DDR_REFCLK_P/N.
- Leave open all BCM88480 or BCM88290 controller signals related to the GDDR6 interface, including WCK, DQ, DBI, EDC, CA, CABI, CKE, CK, RESET, and so on.

7.2 Logical Connectivity

The following figure shows the logical connectivity and recommend placement (not to scale) of the BCM88480 and BCM88290 relative to the GDDR6 devices. In the figure, Channel-A is indicated by blue text, and Channel-B indicated by black text. B0 represents byte-0 in the channel, B1 represent byte-1 in the channel.

NOTE: The BCM88480/BCM88290 and GDDR6 devices must be placed on the top layer of the board. The figure is not to scale.

For information about power, see Chapter 9, Power Aspects, and Section 9.10, GDDR6-Related Power Supply.

NOTE: Follow the layout and connectivity shown in the following figure. For additional details, see Section 7.10, BCM88480 GDDR6 Connectivity and Swapping, and Section 7.11, BCM88290 GDDR6 Connectivity and Swapping.

Figure 34: Recommended Placement for BCM88480 and DRAM Devices (Top View)

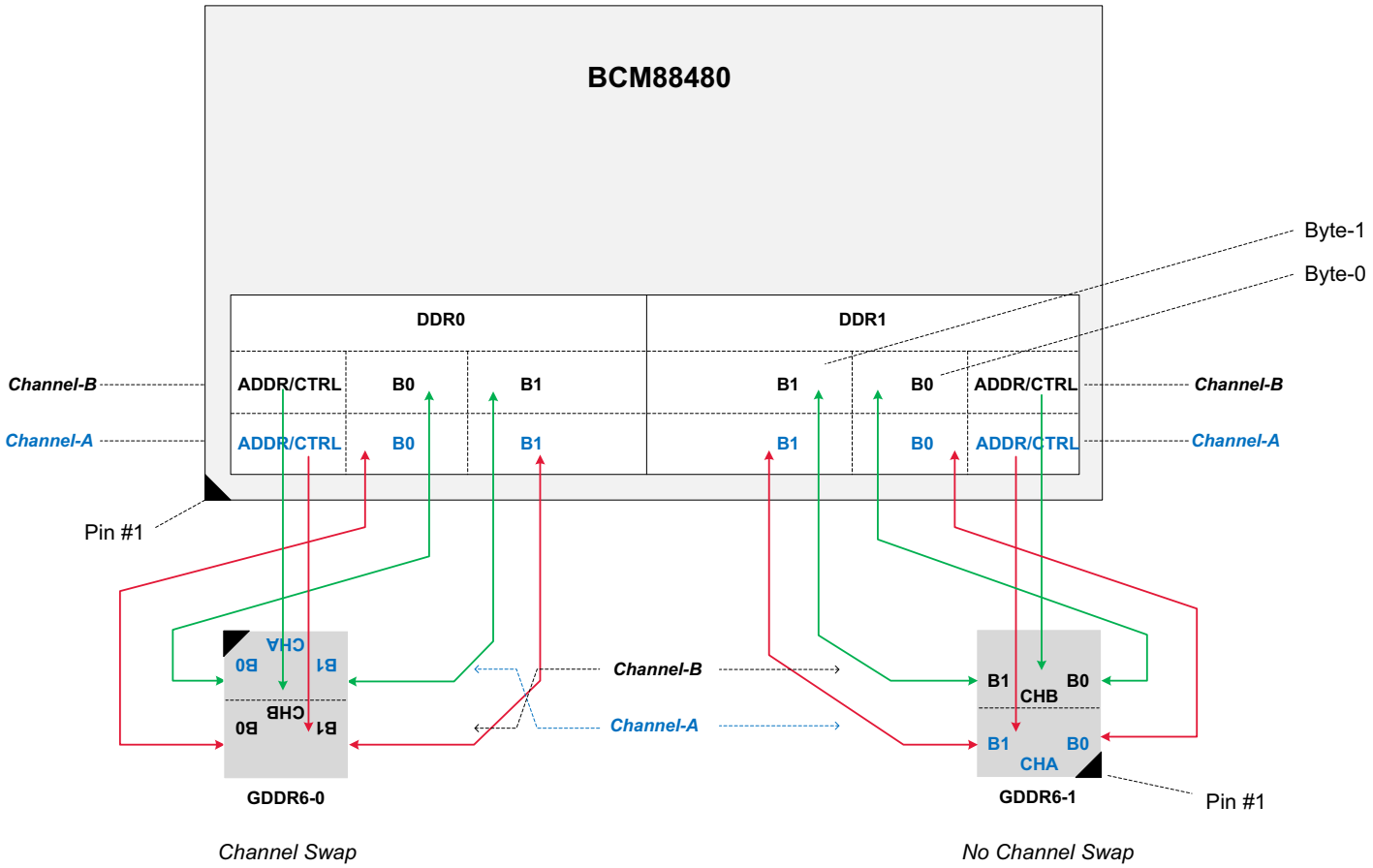
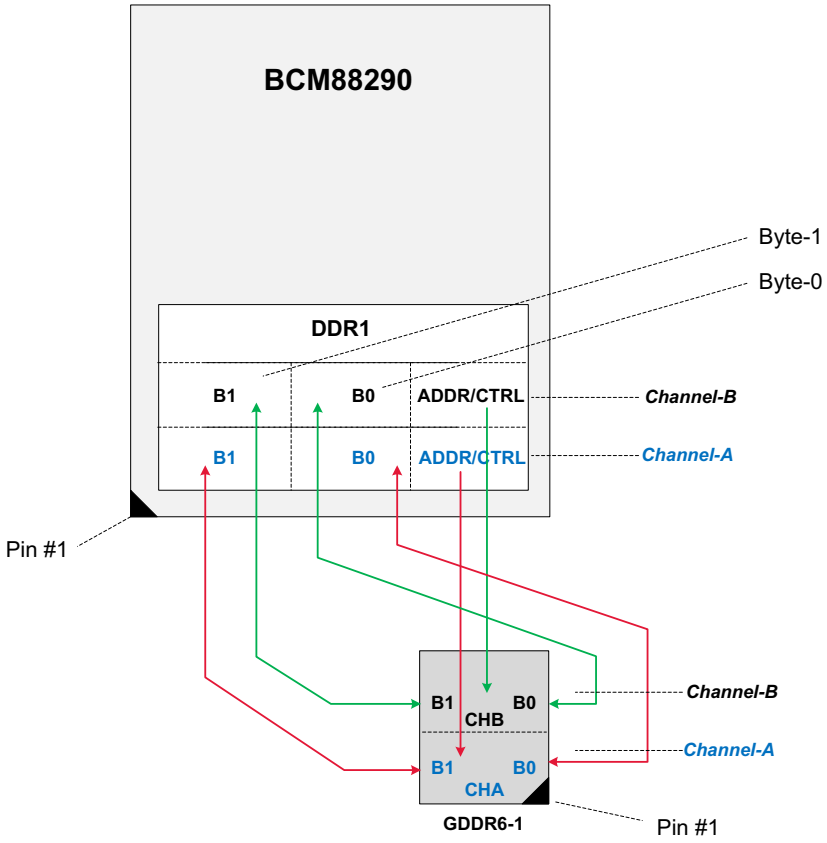


Figure 35: Recommended Placement for BCM88290 and DRAM Device (Top View)



No Channel Swap

7.3 GDDR6 Device Selection

The following table lists suggested GDDR6 DRAM part numbers and vendors. The devices in the table are listed for reference and to assist with device selection, but other compliant devices can also be used.

When attaching two GDDR6 devices to the BCM88480 on the same PCB, both GDDR6 devices must have the same part number.

The BCM88480 GDDR6 interface operates at 6.8 Gb/s. The BCM88290 GDDR6 interface operates at 6.0 Gb/s.

Broadcom does not qualify or test DRAM devices. It is the responsibility of the customer to qualify the DRAM device for each design and to perform system testing over different temperature and voltage conditions under real traffic.

For devices not listed in the table, contact Broadcom support. In the Broadcom support case, attach the vendor's GDDR6 data sheet.

Some devices support both 1.25V and 1.35V, as the following table shows. However, the required operation is 1.3V. For information about operation at 1.3V, contact the DRAM vendor. The GDDR6 *Max Bit Rate*, as specified in the following table, is not the maximum bit rate of the DNX GDDR6 PHY. The operation of the DNX GDDR6 PHY is set to the default bit rate in all cases. Any selection of the GDDR6 device should be 8 Gb/s or above.

NOTE:

- The BCM88480 and BCM88290 support 8-Gb and 16-Gb GDDR6 devices. Customers who want to use 24-Gb or 32-Gb GDDR6 devices should contact Broadcom. The size of memory used is independent of the GDDR6 device in use. Follow the data sheet specifications for the maximum memory used by each channel.
- The GDDR6 interface should operate at 1.3V only (not 1.25V or 1.35V). Refer to the memory vendor documentation for more information.
- On 8-Gb and 16-Gb DRAM devices, there is no CA10 pin. Externally terminate the BCM88480/BCM88290 CA10 pin on-board. For more information about CA10, see Section 7.4.1.2, CA[10].

Table 8: Reference Vendors and Part Numbers for GDDR6

Vendor	Part Number	GDDR6 Device Density (Gb)	WCK	SDK Status
Samsung	K4Z80325BC-HC14[a, b] K4Z80325BC-HC16[a, b]	8	Per Byte	Supported
Samsung	K4ZAF325BC-SC16	16[c]	Per Byte	Supported from SDK 6.5.25
Samsung	K4ZAF325BM-HC14[b] K4ZAF325BM-HC16[b]	16[c]	Per Byte	Supported from SDK 6.5.25
Micron	MT61M256M32JE-10N:A[b, d] MT61M256M32JE-12N:A[b, d]	8	Per Word[e]	Supported
Micron	MT61M512M32KPA-12N:C[b] MT61M512M32KPA-14N:C[b]	16[c]	Per Byte	Supported from SDK 6.5.24
Micron	MT61M512M32KPA-12AAT:E	16[c]	Per Byte	Supported from SDK 6.5.31

a. Do not use K4Z80325BC-HC12.

b. This device is targeted to be declared end-of-life (EOL). Contact the memory vendor for the current status and recommendations.

c. Buffering uses only the lower half of the GDDR6 16Gb device (lower 8Gb), while the upper half is not used. BIST at power up tests only the lower half of the GDDR6 device.

d. A temperature read from the device cannot be performed by the BCM88480/BCM88290. For details, contact the memory vendor.

e. Boards with a Micron 8G device should include all WCK routing to support a Micron 16G device. The Micron 8G device uses two WCK signals per device, while the Micron 16G device uses four WCK signals per device. For additional information, see Section 7.4.1.1, WCK.

NOTE: DRAM vendors update the part number as the technology matures. For updated part numbers and availability, consult the DRAM vendors. For StrataDNX applications, devices with better longevity and reliability than consumer devices are preferred. Some DRAM brands may use a networking mark or provide another indication that a device is intended for networking applications rather than consumer applications.

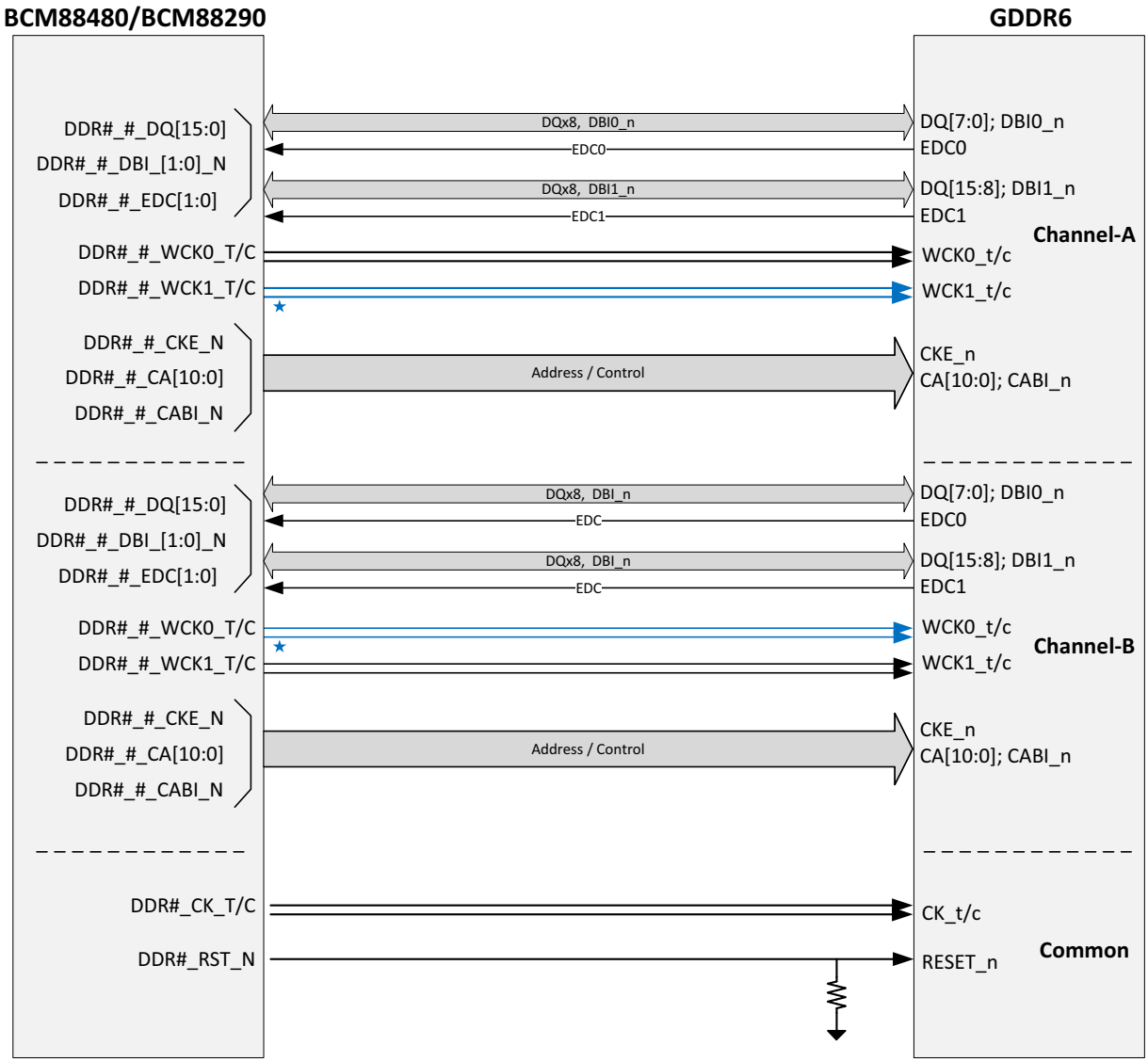
7.4 GDDR6 DRAM Interface Signal Description

The following table defines the signals of a single interface. The signal naming in the table is based on the GDDR6 specification. The GDDR6 signal names in the BCM88480 and BCM88290 signal naming may vary slightly. See Figure 36.

Table 9: GDDR6 DRAM Interface Signal Description

Signals	Description
Per Channel	
DQ[7:0], DBI0_n, EDC0 DQ[15:8], DBI1_n, EDC1	2 bytes, with each byte (8xDQ) followed by DBIn and EDC signals. DQ and DBIn signals are bidirectional, while the EDC signal is driven by the GDDR6 device only. Data-byte signals are sampled on the rising and falling WCK edge.
WCK0_t, WCK0_c WCK1_t, WCK1_c	Two pairs of differential clocks, each associated with one data byte: {DQ, DBI_N and EDC}. See also Section 7.4.1, GDDR6 Part Number Dependent Signals.
CKE_n CA[10:0] CABI_n	Command and address lines are sampled on the rising and falling CK edge. See also Section 7.4.1.2, CA[10].
Common Per GDDR6 Device	
CK_t, CK_c	Differential clock inputs for address and command. See also Section 7.4.1.1, WCK.
RESET_n (DDR#_RST_N)	Reset signal. See also Section 7.5.6, GDDR6 RESET_n.

Figure 36: High-Level Interface Connectivity to One GDDR6 Device



7.4.1 GDDR6 Part Number Dependent Signals

The GDDR6 WCK and CA[10] signals are not the same across all parts.

7.4.1.1 WCK

The following table shows the use of WCK per-word or per-byte based on GDDR6 device support. For devices that support WCK per-word, the GDDR6 Channel-A WCK1_t/c and Channel-B WCK0_t/c are not used.

For an example of GDDR6 devices and their WCK support, see Table 8.

Route all four WCK signals to each GDDR6 device. Any unused WCK signals will drive a constant value.

Table 10: GDDR6 WCK Per-Byte or Per-Word

Channel	Clock	Per Byte	Per Word
A	WCK0	Used	Used
A	WCK1	Used	Not used
B	WCK0	Used	Not used
B	WCK1	Used	Used

7.4.1.2 CA[10]

CA[10] should be terminated in all cases. This termination is required to reduce potential crosstalk and other SI issues. BCM88480/BCM88290 CA[10] toggles even when it is not in use, and no option is available to disable the toggling.

NOTE: The BRCM88480 and BCM88290 support 8-Gb and 16-Gb GDDR6 devices. Customers who want to use 24-Gb or 32-Gb devices should contact Broadcom.

The following table shows the termination schemes possible for CA[10].

Table 11: CA[10] Use Cases and Termination Schema

Use Case	External Termination Required	Notes
8/16-Gb GDDR6, BCM88480/BCM88290 CA[10] is not routed to the DRAM	Yes	See Figure 38
8/16-Gb GDDR6, BCM88480/BCM88290 CA[10] is routed to the DRAM (the board is ready for larger devices)	Yes	See Figure 38
24/32-Gb GDDR6	No	See Figure 39

Figure 37: 8/16-Gb GDDR6, BCM88480/BCM88290 CA[10] Is Not Routed to the DRAM (External Termination)

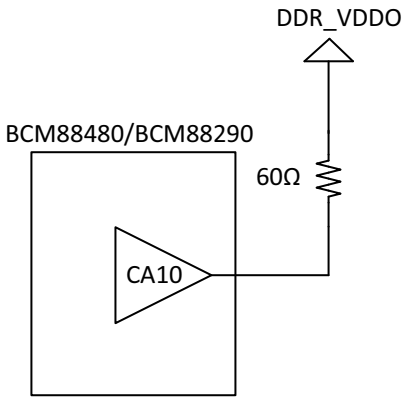


Figure 38: 8/16-Gb GDDR6, BCM88480 CA[10] Is Routed to the DRAM (External Termination)

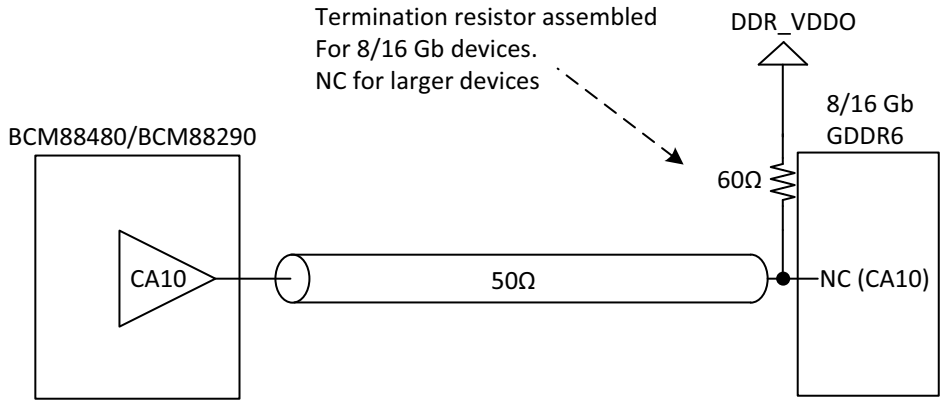
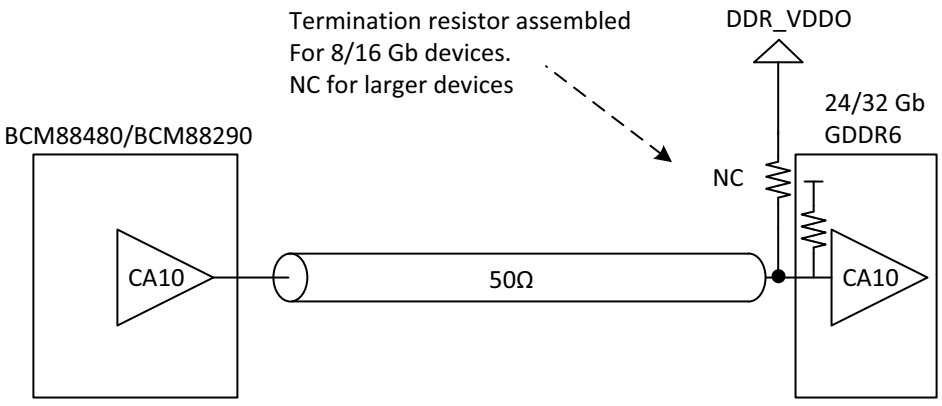


Figure 39: 24/32-Gb GDDR6 (Internal Termination)



7.5 Special Signals

This section contains information about pins that require special handling.

7.5.1 BCM88480 VREF_MON Pins

The BCM88480 DDR#_VREF_MON pins are analog outputs. For debugging, enable scope probing on these signals.

For routing requirements, see Section 7.6.4, Analog Signals.

NOTE: In the BCM88290 device, these pins are not present.

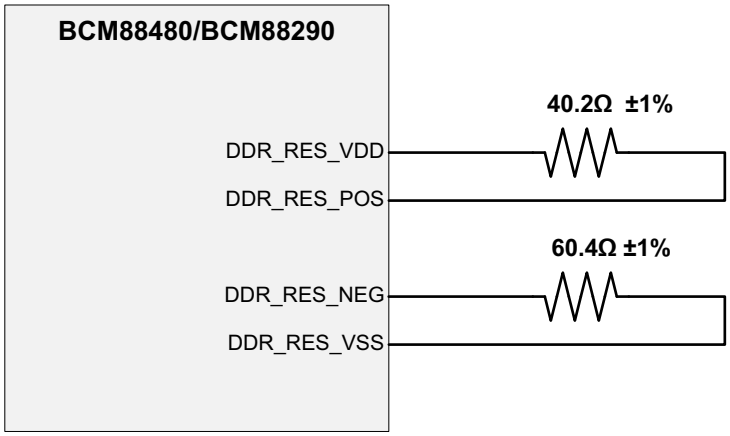
7.5.2 BCM88480/BCM88290 Calibration Pins

The BCM88480/BCM88290 DRAM interfaces require external calibration resistors, as the following figure shows. Resistors are shared between the DDR interfaces.

The placement of calibration resistors under the BGA should take precedence over other considerations.

For routing requirements, see Section 7.6.4, Analog Signals.

Figure 40: BCM88480/BCM88290 Resistance Calibration Pins Example



7.5.3 GDDR6 Device VREFC Pin

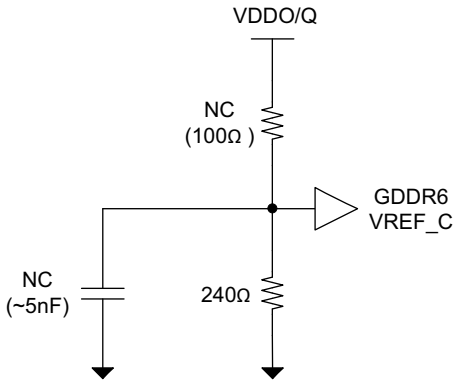
GDDR6 devices support two options for VREFC: internal and external (0.7 × VDDQ). The BCM88480/BCM88290 solution uses the internal VREFC in the GDDR6 devices.

Circuitry for internal and external VRECF is as follows:

- Circuitry for internal VREFC operation: 240Ω resistor to GND.
- Circuitry for external VREFC: As shown in the following figure.

For routing requirements see Section 7.6.4, Analog Signals. Each GDDR6 must have its own external VREFC circuit (in other words, do not share the VREFC between two devices).

Figure 41: GDDR6 Device VREFC Reference Voltage



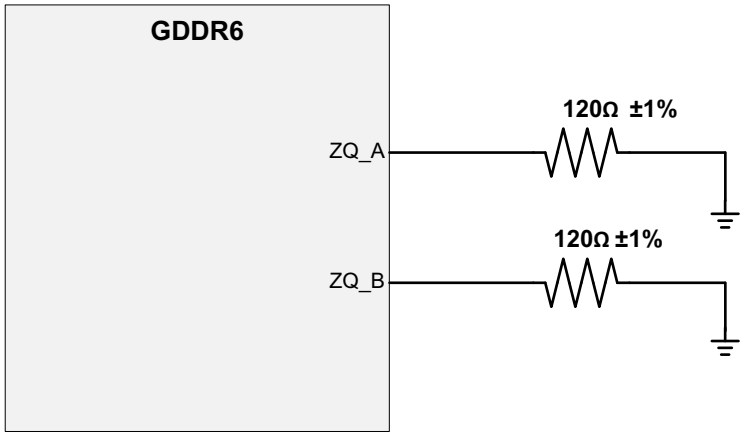
7.5.4 GDDR6 Device ZQ Calibration Pins

GDDR6 device calibration requires a 120Ω ±1% resistor to be connected between the ZQ pin and ground.

The placement of calibration resistors under the BGA should be take precedence over other considerations.

For routing requirements, see Section 7.6.4, Analog Signals.

Figure 42: GDDR6 Device Resistance Calibration Pins Example



7.5.5 GDDR6 Device JTAG

Standard GDDR6 devices have JTAG signals {TDI, TDO, TMS, TCK}. There is no TRST signal on the GDDR6 JTAG.

NOTE: DRAM JTAG interface I/Os operate at the GDDR6 VDDQ voltage (1.3V), while the BCM88480/BCM88290 JTAG interface I/Os operate at 1.8V. If the DRAM and BCM88480/BCM88290 JTAG interface I/Os are used on the same chain, proper handling is required due to the difference in I/O voltages and to avoid damaging other devices.

7.5.6 GDDR6 RESET_n

The GDDR6 RESET_n signal (connected to BCM88480/BCM88290 DDR#_RST_N) should be pulled low with a weak pull-down. A 10-kΩ resistor to ground is recommended.

7.6 PCB Routing Guidelines

7.6.1 Trace Impedance

Ensure trace impedance is matched through the complete channel, including vias, trace necks, or any other topology that may impact the impedance.

Table 12: GDDR6 Interface Trace Impedance

Signal	Trace Type	Trace Impedance (Ω)	Variation
Data group	Single-ended stripline	46	±10%
Address and control group	Single-ended stripline	50[a]	±10%
CK and WCK	Differential stripline	100	±10%

a. The driver is optimized to 46Ω; however, CA is a reduced rate, so 50Ω can be used to ease the breakout routing.

7.6.2 Reference Planes

For reference planes, use the following guidelines:

- Route all DRAM signals using stripline traces.
- VDDO and GND must be the reference planes for all signals {DQ, DBI, EDC, WCK, CA, CKE, CABI, CK} through the entire route.
- Do not route anything over the external PCB layers other than *dog-bone* connectivity from the device pads to the via.
- Keep a distance of at least three times the width of the signal from the edge of signal to the edges of reference planes.

7.6.3 CK and WCK

CK and WCK traces must be as clean as possible. Guidelines are as follows:

- Make the route as short as possible from the BCM88480/BCM88290 device to the GDDR6 device. Avoid serpentines.
- Route as differential pairs.

7.6.4 Analog Signals

The BCM88480/BCM88290 DDR analog signals and GDDR6 analog signals should be routed as a first priority. The following table lists the analog signals.

Table 13: BCM88480/BCM88290 DDR# and GDDR6 Analog Signals

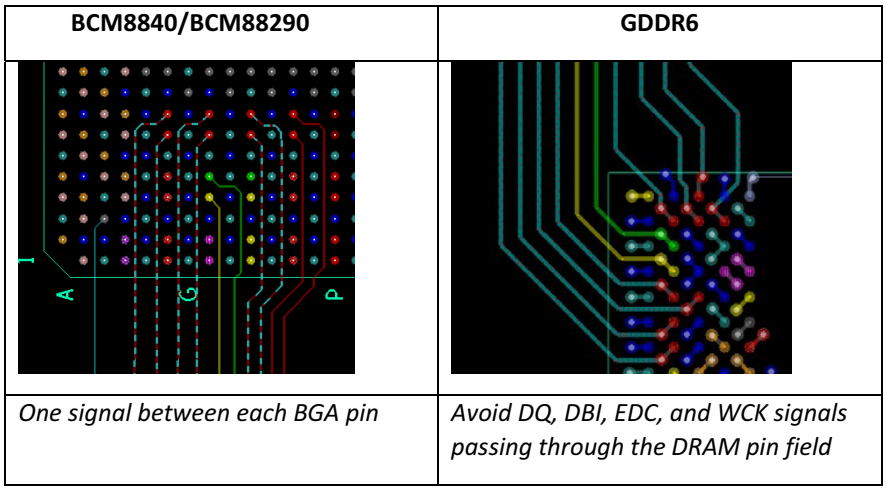
Device	Signal	Layout Requirement
BCM88480	VREF monitor: ■ DDR0_VREF_MON ■ DDR1_VREF_MON	Use wide traces that are three times the width of the data signal trace. Wide traces minimize trace resistance and inductance. Keep a clearance of four times the dielectric height of these traces and other signals.
BCM88480/BCM88290	Calibration pins: ■ DDR_RES_VOD ■ DDR_RES_POS ■ DDR_RES_NEG ■ DDR_RES_VSS	Use a very short trace. Place a resistor beneath the device.
GDDR6	VREF voltage: ■ VREFC	Use wide traces that are three times the width of the data signal trace. Wide traces minimize trace resistance and inductance. Keep a clearance of four times the dielectric height of these traces and other signals.
	Calibration pins: ■ ZQ_A ■ ZQ_B	Use a very short trace. Place a resistor beneath the device.

7.6.5 Layer Partitioning

Use the following guidelines to route signals on a specific PCB layer:

- Route a whole data byte {DQ, DBI, and EDC} on the same routing layer. This guideline applies to both WCK per-byte and WCK per-word.
- WCK can be routed on a different layer than the data-related signals {DQ, DBI, and EDC}.
- Route data-related signals {DQ, DBI, and EDC} and WCK signals on the top three signal layers.
- In the BGA pin escape region, route only one signal between each of the BGA pins. See the BCM88480 example in Figure 43.
- Avoid DQ/DBI/EDC/WCK signals passing through the DRAM pin field. See the GDDR6 example in Figure 43.

Figure 43: Escape from BCM88480/BCM88290 and GDDR6



7.6.6 Signal Vias

Signal vias have the following requirements:

- Back drill all signal vias (or use blind vias instead).
- All signals use two vias.
- Every signal via should have a return ground via. Most signals have a return ground via based on the package footprint. For signals that do not have a return ground via, be sure to add one.
- Keep the same via structure for all data bytes.
- Keep the same via structure for all control and address signals.

7.6.7 Power Vias and Vias Related to Decoupling

The following guidelines are required for the decoupling capacitor vias and the power vias for the BCM88480/BCM88290 and GDDR6:

- Ensure that every power via has an adjacent return ground via.
- Add stitching vias in random patterns to reduce plane impedance and inductance when placed on different layers (the same plane on more than one PCB layer).
- Make sure inductance is low (use wide traces, increase the number of vias, and so on).

For power via guidelines related to the voltage rails, see Table 14.

7.6.8 Power Planes

When designing the board stackup, keep any GDDR6 and BCM88480/BCM88290 DDR power planes with GND planes above and below.

The following power-shape planes should have a solid GND plane above and below:

- GDDR6 VDDQ is connected to BCM88480/BCM88290 DDR_VDDO and can be shared with other GDDR6 VDDQ.
- GDDR6 VDD can be shared with other GDDR6 VDD.
- BCM88480 DDR0_AVDD1P8 and DDR1_AVDD1P8 can be shared.
- BCM88480 DDR0_V_ANALOG and DDR1_V_ANALOG can be shared.

Table 14: DRAM and BRCM DDR# Power Planes

GDDR6 DRAM Power Planes	BCM88480/BCM88290 DDR# Power Planes
VDDQ (connected to BCM88480/BCM88290 DDR_VDDO) VDD VPP	DDR_VDDO (connected with GDDR6 VDDQ) DDR_VDDC DDR#_AVDD1P8 DDR#_V_ANALOG

7.6.9 Trace Clearances

Measure trace clearance from edge to edge. That is, measure from the edge of the signal trace to the edge of the closest trace on the same layer.

Table 15: BCM88480/BCM88290 DDR# Interface Trace Clearance

Group/Signal	Clearance	Note
Data group (does not include EDC)	> 3W	More than three times the width of the trace
EDC	> 4W	More than four times the width of the trace
Address group	> 3W	More than three times the width of the trace
WCK	> 30 mils	More than 30 mils
CK	> 30 mils	More than 30 mils
Reset	> 3W	More than three times the width of the trace

7.7 GDDR6 Signal Length and Skew Matching

Meet the following length matching rules. The signal length is in mils (1 mil = 1/1000 inch) and includes the following measurements:

- The trace route on all layers
- The length of two vias from the top layer to routing layers
- The BCM88480/BCM88290 package length

Designs with a socket for the BCM88480/BCM88290 may be required to place GDDR6 devices further from the BCM88480/BCM88290 device than designs without a socket. The BCM88480/BCM88290 SVK is an example of this placement. Boards that do not use a BCM88480 socket can place the GDDR6 devices closer to the BCM88480/BCM88290, which reduces the total route.

The following general rules apply to each channel design using a GDDR6 layout:

- The maximum length for all signals is 4000 mils (not including RESET_n, which has no requirement).
- For each differential pair WCK and CK, the skew between each #_t to #_c should be no greater than ±5 mils.
- The DQ group skew for 2 bytes should be less than 1000 mils.
- The address and control group skew should be less than 250 mils.
- The address and control group to CK skew should be less than 1000 mils.

Table 16: GDDR6 General Length Rules Summary

No.	Parameter Description	Related Signals	Skew Requirement	Length Requirement
1	Differential CK skew	CK_t, CK_c	< 5 mils	< 4000 mils
2	Differential WCK0 skew	WCK0_t, WCK0_c	< 5 mils	< 4000 mils
3	Differential WCK1 skew	WCK1_t, WCK1_c	< 5 mils	< 4000 mils
4	Group skew for both bytes together (2B within channel)	DQ[7:0], DBI0_n, EDC0 DQ[15:8], DBI1_n, EDC1	< 1000 mils	< 4000 mils
5	Address and control group skew (within channel)	CA[10:0], CKE_n, CABI_n	< 250 mils	< 4000 mils
6	Address and control group to CK skew (within channel)	CK_t, CK_c, CA[10:0], CKE_n, CABI_n	< 1000 mils	< 4000 mils

NOTE: Signal names in the table are the GDDR6 device pin names. If the design includes pin swapping, the requirements still apply and are based on the GDDR6 pin and not on the swapped BCM88480/BCM88290 pin to which it connects.

7.8 Channel Swapping

This section describes the SOC property to use when the BCM88480/BCM88290 DDR channels are not connected to the same channels on the GDDR6 device, which is shown in Figure 34, Recommended Placement for BCM88480 and DRAM Devices (Top View) and defined in Section 7.10, BCM88480 GDDR6 Connectivity and Swapping.

The GDDR6 channel swap SOC property format is ext_ram_channel_swap_en_dramX=Y, where the variables are as follows:

- dramX:
  - dram0 for BCM88480 DDR0
  - dram1 for BCM88480 DDR1 and BCM88290 DDR1
- Y:
  - 0 for no channel swap
  - 1 for channel swap

To follow the connectivity shown in Figure 34, define the following SOC properties:

- ext_ram_channel_swap_en_dram0=1. This SOC defines a channel swap for BCM88480 DDR0 to GDDR6-0.
- ext_ram_channel_swap_en_dram1=0. This SOC defines no channel swap for BCM8840/BCM88290 DDR1 to GDDR6-1.

NOTE: No other swapping is permitted. If a design requires a different setting, it must be approved by Broadcom customer support.

7.9 DQ Swapping

This section describes the SOC property to use when swapping GDDR6 data bits.

The DQ swap SOC property format is ext_ram_dq_swap_dramX_byteY_bitZ=M, where the variables are as follows:

- dramX indicates the DDR number:
  - dram0 for BCM88480 DDR0
  - dram1 for BCM88480 DDR1 and BCM88290 DDR1
- byteY is the byte number of the related BCM88480 DDR0 or DDR1 (BCM88290 DDR1) and can be one of the following:
  - byte0
  - byte1
  - byte2
  - byte3

  Byte is after channel swap, if any.
- bitZ is the bit number in byteY and can be one of the following:
  - bit0
  - bit1
  - bit2
  - bit3
  - bit4
  - bit5
  - bit6
  - bit7
- M is the GDDR6 bit number connected to BCM88480/BCM88290 byteY. M can be 0, 1, 2, 3, 4, 5, 6, or 7.

7.9.1 BCM88480 DQ Swap SOC Example

According to the recommended connectivity and the information in Table 17, BCM88480 DDR0 Interface Connectivity and Table 18, BCM88480 DDR1 Interface Connectivity, use the following DQ swap SOC properties:

- ext_ram_dq_swap_dram0_byte2_bit0.BCM8848X=7
- ext_ram_dq_swap_dram0_byte2_bit1.BCM8848X=6
- ext_ram_dq_swap_dram0_byte2_bit2.BCM8848X=5
- ext_ram_dq_swap_dram0_byte2_bit3.BCM8848X=4
- ext_ram_dq_swap_dram0_byte2_bit4.BCM8848X=2
- ext_ram_dq_swap_dram0_byte2_bit5.BCM8848X=1
- ext_ram_dq_swap_dram0_byte2_bit6.BCM8848X=3
- ext_ram_dq_swap_dram0_byte2_bit7.BCM8848X=0
- ext_ram_dq_swap_dram0_byte3_bit0.BCM8848X=0
- ext_ram_dq_swap_dram0_byte3_bit1.BCM8848X=1
- ext_ram_dq_swap_dram0_byte3_bit2.BCM8848X=2
- ext_ram_dq_swap_dram0_byte3_bit3.BCM8848X=3
- ext_ram_dq_swap_dram0_byte3_bit4.BCM8848X=7
- ext_ram_dq_swap_dram0_byte3_bit5.BCM8848X=6
- ext_ram_dq_swap_dram0_byte3_bit6.BCM8848X=5
- ext_ram_dq_swap_dram0_byte3_bit7.BCM8848X=4
- ext_ram_dq_swap_dram0_byte0_bit0.BCM8848X=0
- ext_ram_dq_swap_dram0_byte0_bit1.BCM8848X=1
- ext_ram_dq_swap_dram0_byte0_bit2.BCM8848X=2
- ext_ram_dq_swap_dram0_byte0_bit3.BCM8848X=3
- ext_ram_dq_swap_dram0_byte0_bit4.BCM8848X=5
- ext_ram_dq_swap_dram0_byte0_bit5.BCM8848X=6
- ext_ram_dq_swap_dram0_byte0_bit6.BCM8848X=4
- ext_ram_dq_swap_dram0_byte0_bit7.BCM8848X=7
- ext_ram_dq_swap_dram0_byte1_bit0.BCM8848X=7
- ext_ram_dq_swap_dram0_byte1_bit1.BCM8848X=6
- ext_ram_dq_swap_dram0_byte1_bit2.BCM8848X=4
- ext_ram_dq_swap_dram0_byte1_bit3.BCM8848X=5
- ext_ram_dq_swap_dram0_byte1_bit4.BCM8848X=2
- ext_ram_dq_swap_dram0_byte1_bit5.BCM8848X=3
- ext_ram_dq_swap_dram0_byte1_bit6.BCM8848X=0
- ext_ram_dq_swap_dram0_byte1_bit7.BCM8848X=1
- ext_ram_dq_swap_dram1_byte0_bit0.BCM8848X=7
- ext_ram_dq_swap_dram1_byte0_bit1.BCM8848X=6
- ext_ram_dq_swap_dram1_byte0_bit2.BCM8848X=5
- ext_ram_dq_swap_dram1_byte0_bit3.BCM8848X=4
- ext_ram_dq_swap_dram1_byte0_bit4.BCM8848X=2
- ext_ram_dq_swap_dram1_byte0_bit5.BCM8848X=0
- ext_ram_dq_swap_dram1_byte0_bit6.BCM8848X=3
- ext_ram_dq_swap_dram1_byte0_bit7.BCM8848X=1
- ext_ram_dq_swap_dram1_byte1_bit0.BCM8848X=0
- ext_ram_dq_swap_dram1_byte1_bit1.BCM8848X=1

- ext_ram_dq_swap_dram1_byte1_bit2.BCM8848X=2
- ext_ram_dq_swap_dram1_byte1_bit3.BCM8848X=3
- ext_ram_dq_swap_dram1_byte1_bit4.BCM8848X=7
- ext_ram_dq_swap_dram1_byte1_bit5.BCM8848X=6
- ext_ram_dq_swap_dram1_byte1_bit6.BCM8848X=5
- ext_ram_dq_swap_dram1_byte1_bit7.BCM8848X=4
- ext_ram_dq_swap_dram1_byte2_bit0.BCM8848X=1
- ext_ram_dq_swap_dram1_byte2_bit1.BCM8848X=0
- ext_ram_dq_swap_dram1_byte2_bit2.BCM8848X=3
- ext_ram_dq_swap_dram1_byte2_bit3.BCM8848X=2
- ext_ram_dq_swap_dram1_byte2_bit4.BCM8848X=5
- ext_ram_dq_swap_dram1_byte2_bit5.BCM8848X=6
- ext_ram_dq_swap_dram1_byte2_bit6.BCM8848X=4
- ext_ram_dq_swap_dram1_byte2_bit7.BCM8848X=7
- ext_ram_dq_swap_dram1_byte3_bit0.BCM8848X=7
- ext_ram_dq_swap_dram1_byte3_bit1.BCM8848X=6
- ext_ram_dq_swap_dram1_byte3_bit2.BCM8848X=4
- ext_ram_dq_swap_dram1_byte3_bit3.BCM8848X=5
- ext_ram_dq_swap_dram1_byte3_bit4.BCM8848X=3
- ext_ram_dq_swap_dram1_byte3_bit5.BCM8848X=2
- ext_ram_dq_swap_dram1_byte3_bit6.BCM8848X=1
- ext_ram_dq_swap_dram1_byte3_bit7.BCM8848X=0

7.9.2 BCM88290 DQ Swap SOC Example

According to the recommended connectivity and the information in Table 19, BCM88290 DDR1 Interface Connectivity, use the following DQ swap SOC properties:

- ext_ram_dq_swap_dram1_byte0_bit0=7
- ext_ram_dq_swap_dram1_byte0_bit1=6
- ext_ram_dq_swap_dram1_byte0_bit2=5
- ext_ram_dq_swap_dram1_byte0_bit3=4
- ext_ram_dq_swap_dram1_byte0_bit4=1
- ext_ram_dq_swap_dram1_byte0_bit5=2
- ext_ram_dq_swap_dram1_byte0_bit6=0
- ext_ram_dq_swap_dram1_byte0_bit7=3
- ext_ram_dq_swap_dram1_byte1_bit0=0
- ext_ram_dq_swap_dram1_byte1_bit1=1
- ext_ram_dq_swap_dram1_byte1_bit2=2
- ext_ram_dq_swap_dram1_byte1_bit3=3
- ext_ram_dq_swap_dram1_byte1_bit4=7
- ext_ram_dq_swap_dram1_byte1_bit5=6
- ext_ram_dq_swap_dram1_byte1_bit6=5
- ext_ram_dq_swap_dram1_byte1_bit7=4
- ext_ram_dq_swap_dram1_byte2_bit0=5
- ext_ram_dq_swap_dram1_byte2_bit1=4
- ext_ram_dq_swap_dram1_byte2_bit2=6
- ext_ram_dq_swap_dram1_byte2_bit3=7
- ext_ram_dq_swap_dram1_byte2_bit4=3
- ext_ram_dq_swap_dram1_byte2_bit5=1
- ext_ram_dq_swap_dram1_byte2_bit6=2
- ext_ram_dq_swap_dram1_byte2_bit7=0
- ext_ram_dq_swap_dram1_byte3_bit0=4
- ext_ram_dq_swap_dram1_byte3_bit1=5
- ext_ram_dq_swap_dram1_byte3_bit2=7
- ext_ram_dq_swap_dram1_byte3_bit3=6
- ext_ram_dq_swap_dram1_byte3_bit4=3
- ext_ram_dq_swap_dram1_byte3_bit5=2
- ext_ram_dq_swap_dram1_byte3_bit6=1
- ext_ram_dq_swap_dram1_byte3_bit7=0

7.10 BCM88480 GDDR6 Connectivity and Swapping

Table 17 and Table 18 show the logical DRAM interface connectivity requirements between the BCM88480 device and the GDDR6 device.

NOTE:

- Follow the connectivity in the following two tables without deviation.
- Follow the latest pinout spreadsheet available on the ESP. The following tables are aligned with BCM88480_PinList_v1.0.zip.

The DDR# swapping and bit mapping summary is as follows:

- Channel swapping is used only on the BCM88480 DDR0 interface; it is not used on the BCM88480 DDR1 interface.
- Byte swapping is *not* used.
- DQ (per byte) bit mapping is used on all bytes.
- CA bit mapping is *not* used.

Table 17: BCM88480 DDR0 Interface Connectivity

Item	BCM88480 Pin Name	BCM88480 Pin Number	GDDR6 Pin Name	GDDR6 Pin Number
1	DDR0_A_CA00	A10	CA00_B	L3
2	DDR0_A_CA01	A4	CA01_B	M11
3	DDR0_A_CA02	B6	CA02_B	M4
4	DDR0_A_CA03	D4	CA03_B	L12
5	DDR0_A_CA04	A7	CA04_B	L5
6	DDR0_A_CA05	B3	CA05_B	L10
7	DDR0_A_CA06	D7	CA06_B	K12
8	DDR0_A_CA07	C5	CA07_B	K11
9	DDR0_A_CA08	B9	CA08_B	K4
10	DDR0_A_CA09	D10	CA09_B	K3
11	DDR0_A_CA10	G9	CA10_B	M5
12	DDR0_A_CABI_N	C8	CABI_n_B	K5
13	DDR0_A_CKE_N	B1	CKE_n_B	M10
14	DDR0_A_DBI0_N	K4	DBI0_n_B	R2
15	DDR0_A_DBI1_N	V5	DBI1_n_B	R13
16	DDR0_A_DQ07	F2	DQ00_B	U4
17	DDR0_A_DQ05	F1	DQ01_B	V3
18	DDR0_A_DQ04	F4	DQ02_B	U3
19	DDR0_A_DQ06	F5	DQ03_B	U2
20	DDR0_A_DQ03	M4	DQ04_B	P3
21	DDR0_A_DQ02	M5	DQ05_B	P2
22	DDR0_A_DQ01	M1	DQ06_B	N2
23	DDR0_A_DQ00	M2	DQ07_B	M2
24	DDR0_A_DQ08	Y2	DQ08_B	U11
25	DDR0_A_DQ09	Y1	DQ09_B	V12
26	DDR0_A_DQ10	Y4	DQ10_B	U12

Table 17: BCM88480 DDR0 Interface Connectivity (Continued)

Item	BCM88480 Pin Name	BCM88480 Pin Number	GDDR6 Pin Name	GDDR6 Pin Number
27	DDR0_A_DQ11	Y5	DQ11_B	U13
28	DDR0_A_DQ15	P4	DQ12_B	P12
29	DDR0_A_DQ14	P5	DQ13_B	P13
30	DDR0_A_DQ13	P1	DQ14_B	N13
31	DDR0_A_DQ12	P2	DQ15_B	M13
32	DDR0_A_EDC0	K5	EDC0_B	T2
33	DDR0_A_EDC1	V4	EDC1_B	T13
34	DDR0_A_WCK0_C	K1	WCK0_c_B	R5
35	DDR0_A_WCK0_T	K2	WCK0_t_B	R4
36	DDR0_A_WCK1_C	T2	WCK1_c_B	R10
37	DDR0_A_WCK1_T	T1	WCK1_t_B	R11
38	DDR0_B_CA00	A11	CA00_A	H3
39	DDR0_B_CA01	A5	CA01_A	G11
40	DDR0_B_CA02	B10	CA02_A	G4
41	DDR0_B_CA03	D5	CA03_A	H12
42	DDR0_B_CA04	A8	CA04_A	H5
43	DDR0_B_CA05	B4	CA05_A	H10
44	DDR0_B_CA06	D8	CA06_A	J12
45	DDR0_B_CA07	C6	CA07_A	J11
46	DDR0_B_CA08	C9	CA08_A	J4
47	DDR0_B_CA09	C11	CA09_A	J3
48	DDR0_B_CA10	F10	CA10_A	G5
49	DDR0_B_CABI_N	B7	CABI_n_A	J5
50	DDR0_B_CKE_N	A2	CKE_n_A	G10
51	DDR0_B_DBI0_N	H4	DBI0_n_A	D2
52	DDR0_B_DBI1_N	T5	DBI1_n_A	D13
53	DDR0_B_DQ00	M8	DQ00_A	B4
54	DDR0_B_DQ01	M7	DQ01_A	A3
55	DDR0_B_DQ02	K8	DQ02_A	B3
56	DDR0_B_DQ03	K7	DQ03_A	B2
57	DDR0_B_DQ06	H8	DQ04_A	E3
58	DDR0_B_DQ04	H7	DQ05_A	E2
59	DDR0_B_DQ05	F7	DQ06_A	F2
60	DDR0_B_DQ07	F8	DQ07_A	G2
61	DDR0_B_DQ14	P8	DQ08_A	B11
62	DDR0_B_DQ15	P7	DQ09_A	A12
63	DDR0_B_DQ12	T8	DQ10_A	B12
64	DDR0_B_DQ13	T7	DQ11_A	B13
65	DDR0_B_DQ10	V8	DQ12_A	E12
66	DDR0_B_DQ11	V7	DQ13_A	E13

Table 17: BCM88480 DDR0 Interface Connectivity (Continued)

Item	BCM88480 Pin Name	BCM88480 Pin Number	GDDR6 Pin Name	GDDR6 Pin Number
67	DDR0_B_DQ09	Y7	DQ14_A	F13
68	DDR0_B_DQ08	Y8	DQ15_A	G13
69	DDR0_B_EDC0	H5	EDC0_A	C2
70	DDR0_B_EDC1	T4	EDC1_A	C13
71	DDR0_B_WCK0_C	H1	WCK0_c_A	D5
72	DDR0_B_WCK0_T	H2	WCK0_t_A	D4
73	DDR0_B_WCK1_C	V1	WCK1_c_A	D10
74	DDR0_B_WCK1_T	V2	WCK1_t_A	D11
75	DDR0_CK_C	D2	CK_c	K10
76	DDR0_CK_T	D1	CK_t	J10
77	DDR0_RST_N	C3	RESET_n	J1

Table 18: BCM88480 DDR1 Interface Connectivity

Item	BCM88480 Pin Name	BCM88480 Pin Number	GDDR6 Pin Name	GDDR6 Pin Number
1	DDR1_A_CA00	BB10	CA00_A	H3
2	DDR1_A_CA01	BB4	CA01_A	G11
3	DDR1_A_CA02	BA6	CA02_A	G4
4	DDR1_A_CA03	AW4	CA03_A	H12
5	DDR1_A_CA04	BB7	CA04_A	H5
6	DDR1_A_CA05	BA3	CA05_A	H10
7	DDR1_A_CA06	AW7	CA06_A	J12
8	DDR1_A_CA07	AY5	CA07_A	J11
9	DDR1_A_CA08	BA9	CA08_A	J4
10	DDR1_A_CA09	AW10	CA09_A	J3
11	DDR1_A_CA10	AT9	CA10_A	G5
12	DDR1_A_CABI_N	AY8	CABI_n_A	J5
13	DDR1_A_CKE_N	BA1	CKE_n_A	G10
14	DDR1_A_DBI0_N	AN4	DBI0_n_A	D2
15	DDR1_A_DBI1_N	AE5	DBI1_n_A	D13
16	DDR1_A_DQ05	AU2	DQ00_A	B4
17	DDR1_A_DQ07	AU1	DQ01_A	A3
18	DDR1_A_DQ04	AU4	DQ02_A	B3
19	DDR1_A_DQ06	AU5	DQ03_A	B2
20	DDR1_A_DQ03	AL4	DQ04_A	E3
21	DDR1_A_DQ02	AL5	DQ05_A	E2
22	DDR1_A_DQ01	AL1	DQ06_A	F2
23	DDR1_A_DQ00	AL2	DQ07_A	G2
24	DDR1_A_DQ08	AC2	DQ08_A	B11
25	DDR1_A_DQ09	AC1	DQ09_A	A12

Table 18: BCM88480 DDR1 Interface Connectivity (Continued)

Item	BCM88480 Pin Name	BCM88480 Pin Number	GDDR6 Pin Name	GDDR6 Pin Number
26	DDR1_A_DQ10	AC4	DQ10_A	B12
27	DDR1_A_DQ11	AC5	DQ11_A	B13
28	DDR1_A_DQ15	AJ4	DQ12_A	E12
29	DDR1_A_DQ14	AJ5	DQ13_A	E13
30	DDR1_A_DQ13	AJ1	DQ14_A	F13
31	DDR1_A_DQ12	AJ2	DQ15_A	G13
32	DDR1_A_EDC0	AN5	EDC0_A	C2
33	DDR1_A_EDC1	AE4	EDC1_A	C13
34	DDR1_A_WCK0_C	AN1	WCK0_c_A	D5
35	DDR1_A_WCK0_T	AN2	WCK0_t_A	D4
36	DDR1_A_WCK1_C	AG1	WCK1_c_A	D10
37	DDR1_A_WCK1_T	AG2	WCK1_t_A	D11
38	DDR1_B_CA00	BB11	CA00_B	L3
39	DDR1_B_CA01	BB5	CA01_B	M11
40	DDR1_B_CA02	BA10	CA02_B	M4
41	DDR1_B_CA03	AW5	CA03_B	L12
42	DDR1_B_CA04	BB8	CA04_B	L5
43	DDR1_B_CA05	BA4	CA05_B	L10
44	DDR1_B_CA06	AW8	CA06_B	K12
45	DDR1_B_CA07	AY6	CA07_B	K11
46	DDR1_B_CA08	AY9	CA08_B	K4
47	DDR1_B_CA09	AY11	CA09_B	K3
48	DDR1_B_CA10	AU10	CA10_B	M5
49	DDR1_B_CABI_N	BA7	CABI_n_B	K5
50	DDR1_B_CKE_N	BB2	CKE_n_B	M10
51	DDR1_B_DBI0_N	AR4	DBI0_n_B	R2
52	DDR1_B_DBI1_N	AG5	DBI1_n_B	R13
53	DDR1_B_DQ01	AL8	DQ00_B	U4
54	DDR1_B_DQ00	AL7	DQ01_B	V3
55	DDR1_B_DQ03	AN8	DQ02_B	U3
56	DDR1_B_DQ02	AN7	DQ03_B	U2
57	DDR1_B_DQ06	AR8	DQ04_B	P3
58	DDR1_B_DQ04	AR7	DQ05_B	P2
59	DDR1_B_DQ05	AU7	DQ06_B	N2
60	DDR1_B_DQ07	AU8	DQ07_B	M2
61	DDR1_B_DQ15	AJ8	DQ08_B	U11
62	DDR1_B_DQ14	AJ7	DQ09_B	V12
63	DDR1_B_DQ13	AG8	DQ10_B	U12
64	DDR1_B_DQ12	AG7	DQ11_B	U13
65	DDR1_B_DQ10	AE8	DQ12_B	P12

Table 18: BCM88480 DDR1 Interface Connectivity (Continued)

Item	BCM88480 Pin Name	BCM88480 Pin Number	GDDR6 Pin Name	GDDR6 Pin Number
66	DDR1_B_DQ11	AE7	DQ13_B	P13
67	DDR1_B_DQ09	AC7	DQ14_B	N13
68	DDR1_B_DQ08	AC8	DQ15_B	M13
69	DDR1_B_EDC0	AR5	EDC0_B	T2
70	DDR1_B_EDC1	AG4	EDC1_B	T13
71	DDR1_B_WCK0_C	AR2	WCK0_c_B	R5
72	DDR1_B_WCK0_T	AR1	WCK0_t_B	R4
73	DDR1_B_WCK1_C	AE1	WCK1_c_B	R10
74	DDR1_B_WCK1_T	AE2	WCK1_t_B	R11
75	DDR1_CK_C	AW2	CK_c	K10
76	DDR1_CK_T	AW1	CK_t	J10
77	DDR1_RST_N	AY3	RESET_n	J1

7.11 BCM88290 GDDR6 Connectivity and Swapping

The following table shows the logical DRAM interface connectivity requirements between the BCM88290 device and the GDDR6 device.

NOTE:

- Follow the connectivity in the following table without deviation.
- Follow the latest pinout spreadsheet available on docSAFE. The following table is aligned with BCM88290_PinList_v1p1.zip.

The DDR# swapping and bit mapping summary is as follows:

- Channel swapping is *not* used
- Byte swapping is *not* used.
- DQ (per byte) bit mapping is used on all bytes.
- CA bit mapping is *not* used.

Table 19: BCM88290 DDR1 Interface Connectivity

Item	BCM88290 Pin Name	BCM88290 Pin Number	GDDR6 Pin Name	GDDR6 Pin Number
1	DDR1_CK_C	AM2	CK_c	K10
2	DDR1_CK_T	AM1	CK_t	J10
3	DDR1_RST_N	AN3	RESET_n	J1
4	DDR1_A_CKE_N	AP1	CKE_n_A	G10
5	DDR1_A_CA00	AR10	CA00_A	H3
6	DDR1_A_CA01	AR4	CA01_A	G11
7	DDR1_A_CA02	AP6	CA02_A	G4
8	DDR1_A_CA03	AM4	CA03_A	H12
9	DDR1_A_CA04	AR7	CA04_A	H5
10	DDR1_A_CA05	AP3	CA05_A	H10
11	DDR1_A_CA06	AM7	CA06_A	J12

Table 19: BCM88290 DDR1 Interface Connectivity (Continued)

Item	BCM88290 Pin Name	BCM88290 Pin Number	GDDR6 Pin Name	GDDR6 Pin Number
12	DDR1_A_CA07	AN5	CA07_A	J11
13	DDR1_A_CA08	AP9	CA08_A	J4
14	DDR1_A_CA09	AM10	CA09_A	J3
15	DDR1_A_CA10	AN12	CA10_A	G5
16	DDR1_A_CABI_N	AN8	CABI_n_A	J5
17	DDR1_A_DBI0_N	AH4	DBI0_n_A	D2
18	DDR1_A_DQ00	AD2	DQ07_A	G2
19	DDR1_A_DQ01	AD1	DQ06_A	F2
20	DDR1_A_DQ02	AD5	DQ05_A	E2
21	DDR1_A_DQ03	AD4	DQ04_A	E3
22	DDR1_A_DQ04	AK4	DQ01_A	A3
23	DDR1_A_DQ05	AK2	DQ02_A	B3
24	DDR1_A_DQ06	AK5	DQ00_A	B4
25	DDR1_A_DQ07	AK1	DQ03_A	B2
26	DDR1_A_EDC0	AH5	EDC0_A	C2
27	DDR1_A_WCK0_C	AF5	WCK0_c_A	D5
28	DDR1_A_WCK0_T	AF4	WCK0_t_A	D4
29	DDR1_A_DBI1_N	Y4	DBI1_n_A	D13
30	DDR1_A_DQ08	T2	DQ08_A	B11
31	DDR1_A_DQ09	T1	DQ09_A	A12
32	DDR1_A_DQ10	T4	DQ10_A	B12
33	DDR1_A_DQ11	T5	DQ11_A	B13
34	DDR1_A_DQ12	AB2	DQ15_A	G13
35	DDR1_A_DQ13	AB1	DQ14_A	F13
36	DDR1_A_DQ14	AB5	DQ13_A	E13
37	DDR1_A_DQ15	AB4	DQ12_A	E12
38	DDR1_A_EDC1	Y5	EDC1_A	C13
39	DDR1_A_WCK1_C	Y1	WCK1_c_A	D10
40	DDR1_A_WCK1_T	Y2	WCK1_t_A	D11
41	DDR1_B_CKE_N	AR2	CKE_n_B	M10
42	DDR1_B_CA00	AR11	CA00_B	L3
43	DDR1_B_CA01	AR5	CA01_B	M11
44	DDR1_B_CA02	AP10	CA02_B	M4
45	DDR1_B_CA03	AM5	CA03_B	L12
46	DDR1_B_CA04	AR8	CA04_B	L5
47	DDR1_B_CA05	AP4	CA05_B	L10
48	DDR1_B_CA06	AM8	CA06_B	K12
49	DDR1_B_CA07	AN6	CA07_B	K11
50	DDR1_B_CA08	AN9	CA08_B	K4
51	DDR1_B_CA09	AN11	CA09_B	K3

Table 19: BCM88290 DDR1 Interface Connectivity (Continued)

Item	BCM88290 Pin Name	BCM88290 Pin Number	GDDR6 Pin Name	GDDR6 Pin Number
52	DDR1_B_CA10	AM11	CA10_B	M5
53	DDR1_B_CABI_N	AP7	CABI_n_B	K5
54	DDR1_B_DBI0_N	AF7	DBI0_n_B	R2
55	DDR1_B_DQ00	AF1	DQ05_B	P2
56	DDR1_B_DQ01	AF2	DQ04_B	P3
57	DDR1_B_DQ02	AH1	DQ06_B	N2
58	DDR1_B_DQ03	AH2	DQ07_B	M2
59	DDR1_B_DQ04	AD7	DQ03_B	U2
60	DDR1_B_DQ05	AB7	DQ01_B	V3
61	DDR1_B_DQ06	AD8	DQ02_B	U3
62	DDR1_B_DQ07	AB8	DQ00_B	U4
63	DDR1_B_EDC0	AF8	EDC0_B	T2
64	DDR1_B_WCK0_C	AK8	WCK0_c_B	R5
65	DDR1_B_WCK0_T	AK7	WCK0_t_B	R4
66	DDR1_B_DBI1_N	V5	DBI1_n_B	R13
67	DDR1_B_DQ08	T8	DQ12_B	P12
68	DDR1_B_DQ09	T7	DQ13_B	P13
69	DDR1_B_DQ10	P2	DQ15_B	M13
70	DDR1_B_DQ11	P1	DQ14_B	N13
71	DDR1_B_DQ12	V7	DQ11_B	U13
72	DDR1_B_DQ13	V8	DQ10_B	U12
73	DDR1_B_DQ14	Y7	DQ09_B	V12
74	DDR1_B_DQ15	Y8	DQ08_B	U11
75	DDR1_B_EDC1	V4	EDC1_B	T13
76	DDR1_B_WCK1_C	V1	WCK1_c_B	R10
77	DDR1_B_WCK1_T	V2	WCK1_t_B	R11

Chapter 8: Clock and PLL

8.1 Differential Clock Driving

To drive the various differential clocks in each DNX16 device, use the clock schemes shown in the following tables:

- BCM88790: Table 20
- BCM88690: Table 21
- BCM88800: Table 22
- BCM88480: Table 23 and Table 24
- BCM88290: Table 25 and Table 26

Table 20: BCM88790 Clock Schemes

No.	Signal	Standard	Frequency (MHz)	CML Internal Termination (Figure 44)	CML External Termination (Figure 45)
1	FAB_LCPLL[#]_REFCLK_P/N[a]	CML	156.25	✓	—
2	C_PLL_REFCLK_P/N	CML	25	✓	—
3	U_PLL_REFCLK_P/N	CML	25	✓	—
4	PCIE_REFCLK_P/N	CML	100	—	✓

a. See Section 8.2.3, BCM88790 Re-Timer Mode – Special Requirements.

Table 21: BCM88690 Clock Schemes

No.	Signal	Standard	Frequency (MHz)	CML Internal Termination (Figure 44)	CML External Termination (Figure 45)
1	FAB_LCPLL[#]_REFCLK_P/N	CML	156.25	✓	—
2	NIF_LCPLL[#]_REFCLK_P/N	CML	156.25	✓	—
3	C_PLL_REFCLK_P/N	CML	25	✓	—
4	U_PLL_REFCLK_P/N	CML	25	✓	—
5	TS_PLL_REFCLK_P/N	CML	25	✓	—
6	DRAM_PHY[#]_REFCLK_P/N[a]	CML	100	✓	—
7	PCIE_REFCLK_P/N	CML	100	—	✓

a. The clock must be connected even if the HBM is not used or the device does not include an HBM.

Table 22: BCM88800 Clock Schemes

No.	Signal	Standard	Frequency (MHz)	CML Internal Termination (Figure 44)	CML External Termination (Figure 45)
1	FAB_PLL_REFCLK_P/N[a]	CML	156.25	✓	—
2	FAB_[1:0]_REFCLK_P/N[a, b]	CML	156.25	✓	—
3	NIF_[4:0]_REFCLK_P/N[a, c]	CML	156.25	✓	—
4	C_PLL_REFCLK_P/N	CML	25	✓	—

Table 22: BCM88800 Clock Schemes (Continued)

No.	Signal	Standard	Frequency (MHz)	CML Internal Termination (Figure 44)	CML External Termination (Figure 45)
5	U_PLL_REFCLK_P/N	CML	25	✓	—
6	TS_PLL_REFCLK_P/N	CML	25	✓	—
7	DRAM_PHY_REFCLK_P/N[d]	CML	100	✓	—
8	PCIE_REFCLK_P/N	CML	100	—	✓

a. The clock must be connected even if the interface is not used.

b. The two reference clocks must be sourced from the same clock source. There is no requirement for phase alignment.

c. The five reference clocks must be sourced from the same clock source. There is no requirement for phase alignment.

d. The clock must be connected even if the HBM is not used or the device does not include an HBM.

Table 23: BCM88480 CML Clock Schemes

No.	Signal	Standard	Frequency (MHz)	CML Internal Termination (Figure 44)	CML External Termination (Figure 45)
1	NIF_PLL_REFCLK_P/N	CML	156.25	✓	—
2	NIF_[3:0]_REFCLK_P/N[a, b]	CML	156.25	✓	—
3	C_PLL_REFCLK_P/N	CML	25	✓	—
4	U_PLL_REFCLK_P/N	CML	25	✓	—
5	TS_PLL_REFCLK_P/N	CML	25	✓	—
6	FLEXE_PLL_REFCLK_P/N[c, d]	CML	155.52	✓	—
7	PCIE_REFCLK_P/N	CML	100	—	✓

a. The clock must be connected even if the interface is not used.

b. The four reference clocks must be sourced from the same clock source. There is no requirement for phase alignment.

c. See Section 8.3, BCM88480 FlexE Reference Clocks.

d. When FlexE is not in use or not used in a distributed system, the clock must still be connected. In this case, the clock can be 156.25 MHz or 155.52 MHz.

Table 24: BCM88480 Non-CML Clocks

No	Signal	Standard	Frequency (MHz)	Termination
1	DDR_REFCLK_P/N[a]	LVDS	100	See Figure 46 and Figure 47
2	FLEXE_8K_REFCLK[b, c]	CMOS 1.8V	8 kHz	N/A

a. The clock must be connected even if the interface is not used.

b. See Section 8.3, BCM88480 FlexE Reference Clocks.

c. When FlexE is not in use or not used in a distributed system, this clock is not required (unlike FLEXE_PLL_REFCLK_P/N).

Table 25: BCM88290 CML Clock Schemes

No.	Signal	Standard	Frequency (MHz)	CML Internal Termination (Figure 44)	CML External Termination (Figure 45)
1	NIF_PLL_REFCLK_P/N	CML	156.25	✓	—
2	NIF_[3:2]_REFCLK_P/N[a, b]	CML	156.25	✓	—
3	C_PLL_REFCLK_P/N	CML	25	✓	—
4	U_PLL_REFCLK_P/N	CML	25	✓	—
5	TS_PLL_REFCLK_P/N	CML	25	✓	—
6	FLEXE_PLL_REFCLK_P/N[a]	CML	156.25	✓	—
7	PCIE_REFCLK_P/N	CML	100	—	✓

a. The clock must be connected even if the interface is not used.

b. The two reference clocks must be sourced from the same clock source. There is no requirement for phase alignment.

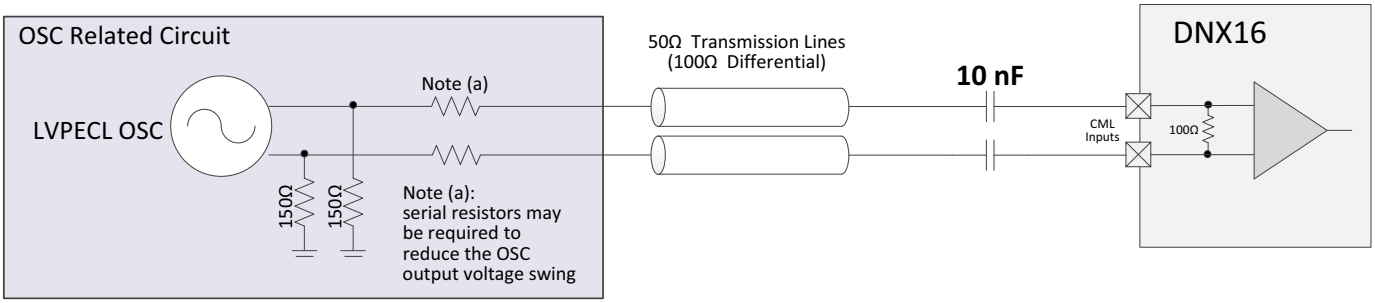
Table 26: BCM88290 Non-CML Clocks

No.	Signal	Standard	Frequency (MHz)	Termination
1	DDR1_REFCLK_P/N[a]	LVDS	100	See Figure 46 and Figure 47

a. The clock must be connected even if the interface is not used.

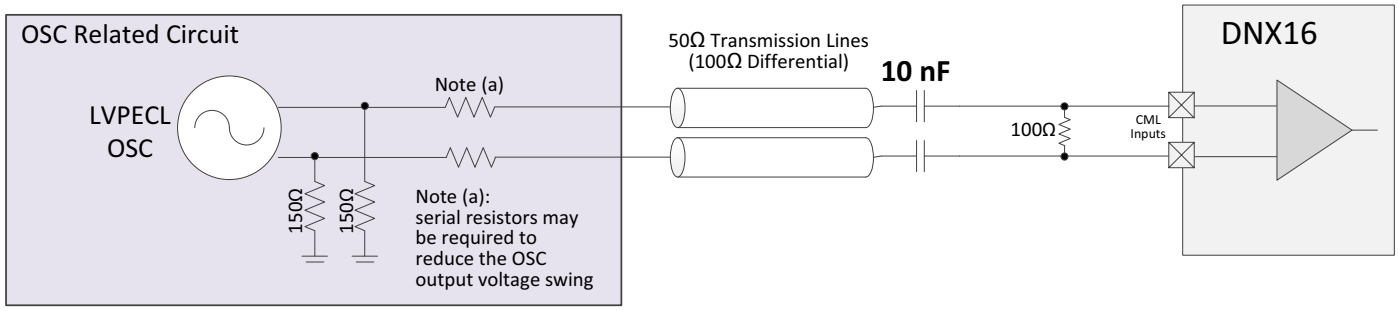
The following figure demonstrates the required clock scheme when the termination is internal. As an example, an LVPECL oscillator is used.

Figure 44: Differential Clock Scheme – CML Internal Termination



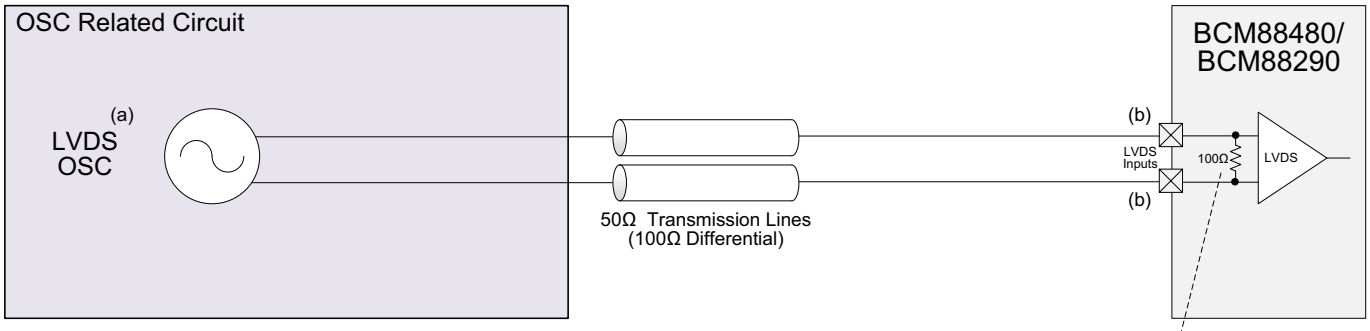
The following figure demonstrates the required clock scheme when the termination is external. As an example, an LVPECL oscillator is used.

Figure 45: Differential Clock Scheme – CML External Termination



The following two figures demonstrate the required clock scheme for the BCM88480 DDR_REFCLK_P/N. Figure 46 shows an example of a DC-coupled LVDS with no components between the driver and receiver. Figure 47 shows an example of an AC-coupled LVPECL oscillator and optional biasing circuit.

Figure 46: Differential Clock Schema for BCM88480/BCM88290 DDR_REFCLK_P/N LVDS Input – DC Coupled



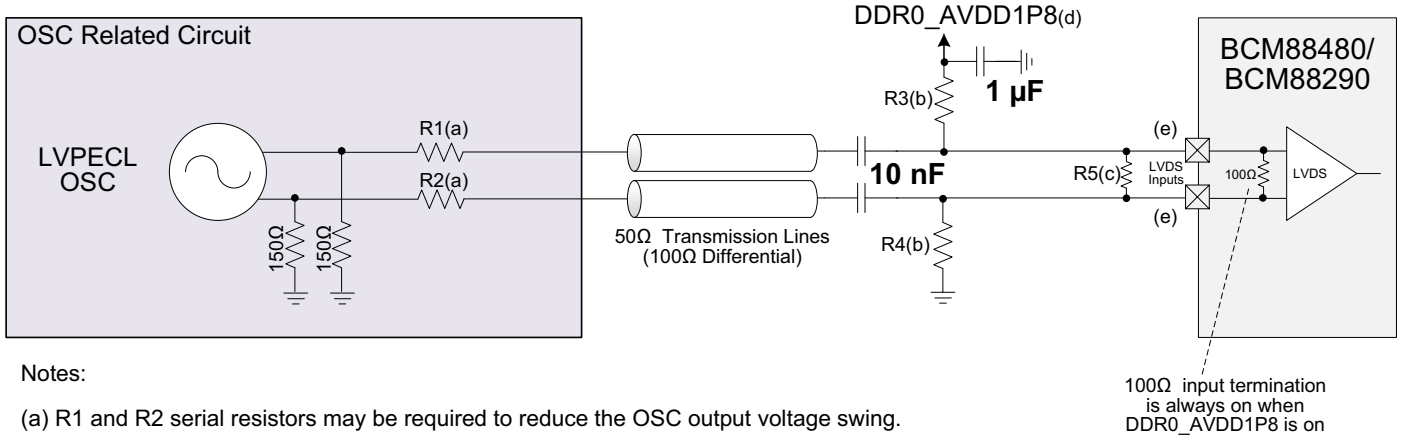
Notes:

(a) 1.8V LVDS Oscillator/Driver

(b) LVDS receiver pads cannot have a signal or clock when DDR0_AVDD1P8 is not present (powered up). Refer to the BCM88480 or BCM88290 data sheet.

NOTE: When DDR0_AVDD1P8 is not present, no signals or clocks should be on the LVDS inputs.

Figure 47: Differential Clock Schema for BCM88480/BCM88290 DDR_REFCLK_P/N LVDS Input – AC Coupled



Notes:

(a) R1 and R2 serial resistors may be required to reduce the OSC output voltage swing.

(b) R3 and R4 are set to a typical offset voltage of 0.875V. R3 = 3.7 kΩ, R4 = 3.5 kΩ.

(c) R5 is set NC, located under BGA.

(d) DDR0_AVDD1P8 or sourced from DDR0_AVDD1P8.

(e) LVDS receiver pads cannot have a signal or clock when DDR0_AVDD1P8 is not present (powered up). Refer to the BCM88480 or BCM88290 data sheet.

NOTE: When DDR0_AVDD1P8 is not present, no signals or clocks should be on the LVDS inputs.

8.2 FAB/NIF REFCLK

The fabric and NIF REFCLK connectivity and termination is described in Section 8.1, Differential Clock Driving. This section provides additional information that is specific to those two types of reference clocks.

The fabric and NIF REFCLK signals for each DNX16 device are as follows:

- BCM88790: FAB_LCPLL[#]_REFCLK_P/N
- BCM88690: FAB_LCPLL[#]_REFCLK_P/N and NIF_LCPLL[#]_REFCLK_P/N
- BCM88800: FAB_PLL_REFCLK_P/N, FAB_[#]_REFCLK_P/N, and NIF_[#]_REFCLK_P/N
- BCM88480: NIF_PLL_REFCLK_P/N and NIF_[3:0]_REFCLK_P/N

8.2.1 FAB/NIF REFCLK Routing

The input jitter requirements for fabric and NIF REFCLK are very tight (refer to the device data sheet), so take extra care with the clock signal routing. Route those signals as high-speed SerDes signals (perform back-drilling, reduce the number of vias, control impedance, optimize the via-structure and anti-pad, and so on).

8.2.2 Suggested Part Numbers

Fabric and NIF REFCLK have strict PPM and jitter requirements, as described in the data sheet for each device.

Use the tables in this section as references for clock source selection. The data sheet specifications for the parts listed in the following tables were found suitable to be used with the DNX16 devices (at the time those part numbers were added to this document). It is expected that customers review and validate the DNX16 device requirements with the clock source vendors.

Table 27: PLL-Based Clock Generator

No.	Vendor	P/N
1	IDT	8V49NS1412 8V49NS0412

Table 28: Jitter Attenuators

No.	Vendor	P/N
1	IDT	8V19N474 8V19N478

Table 29: Synchronization Timing DPLL

No.	Vendor	P/N
1	IDT	8A35027C

8.2.3 BCM88790 Re-Timer Mode – Special Requirements

As described in the BCM88790 data sheet, the BCM88790 supports re-timer functionality. This section provides additional design guidelines when using this feature.

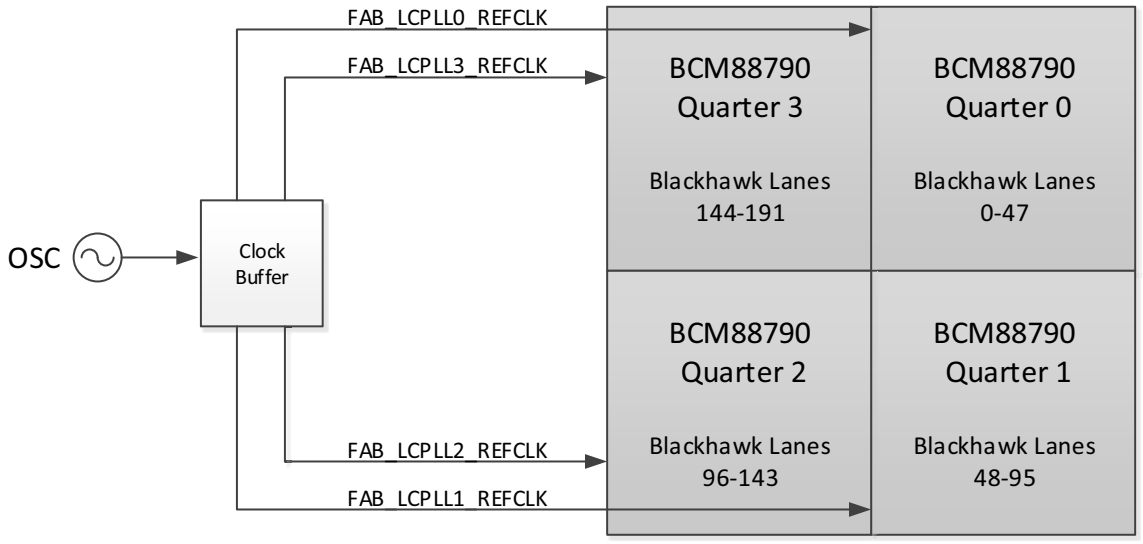
For more details, refer to the Re-Timer Functionality section in the BCM88790 data sheet (88790-DS1xx).

When the re-timer feature is used, the BCM88790 FAB_LCPLL[3:0]_REFCLK must share the same clock reference source. In other words, FAB_LCPLL[3:0]_REFCLK cannot use separate clock sources even if the sources have the same frequency. There is no requirement for phase alignment.

NOTE: Using the same clock source is usually the default connectivity in most applications.

NOTE: The requirement for using the same reference clock source applies to the BCM88790 device but does not apply to the fabric link partner. For example, it is not required on the BCM88690, even if the BCM88690 is connected to a BCM88790 that is using the re-timer functionality.

Figure 48: BCM88790 FAB_LCPLL[3:0]_REFCLK Connected to the Same Clock Source



8.3 BCM88480 FlexE Reference Clocks

The FlexE core has the following two reference clocks:

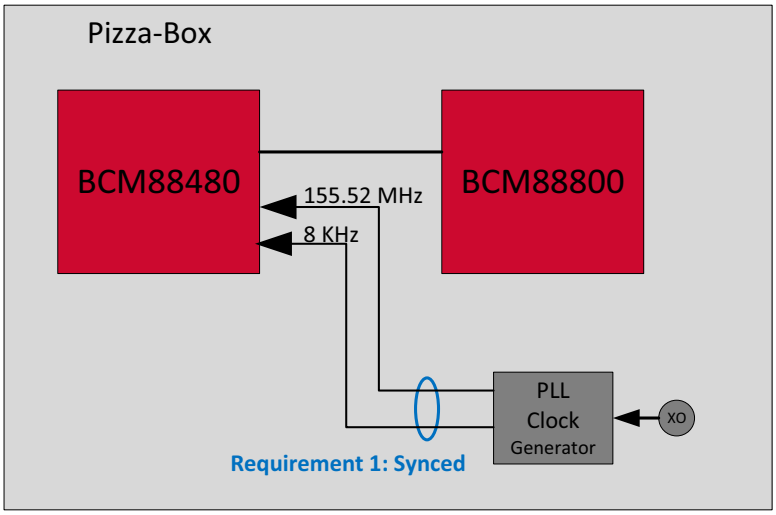
- FLEXE_8K_REFCLK (8 kHz)
- FLEXE_PLL_REFCLK_P/N (155.52 MHz)

As described in the device data sheet (88480-DS1xx), when using the FlexE in a distributed system, the FlexE core reference clocks have the following two requirements at the system level:

- Requirement 1: The 8-kHz and 155.52-MHz clocks of all BCM88480 devices on the same node must be synced (that is, sourced from the same source). It is not required to sync these clocks with any other clock on board (TSPLL, IEEE 1588 clocks, BroadSync® solution, or SerDes TX clocks).
- Requirement 2: The maximum skew between the 8-kHz clocks of all BCM88480 devices on the same node must be less than 50 ns.

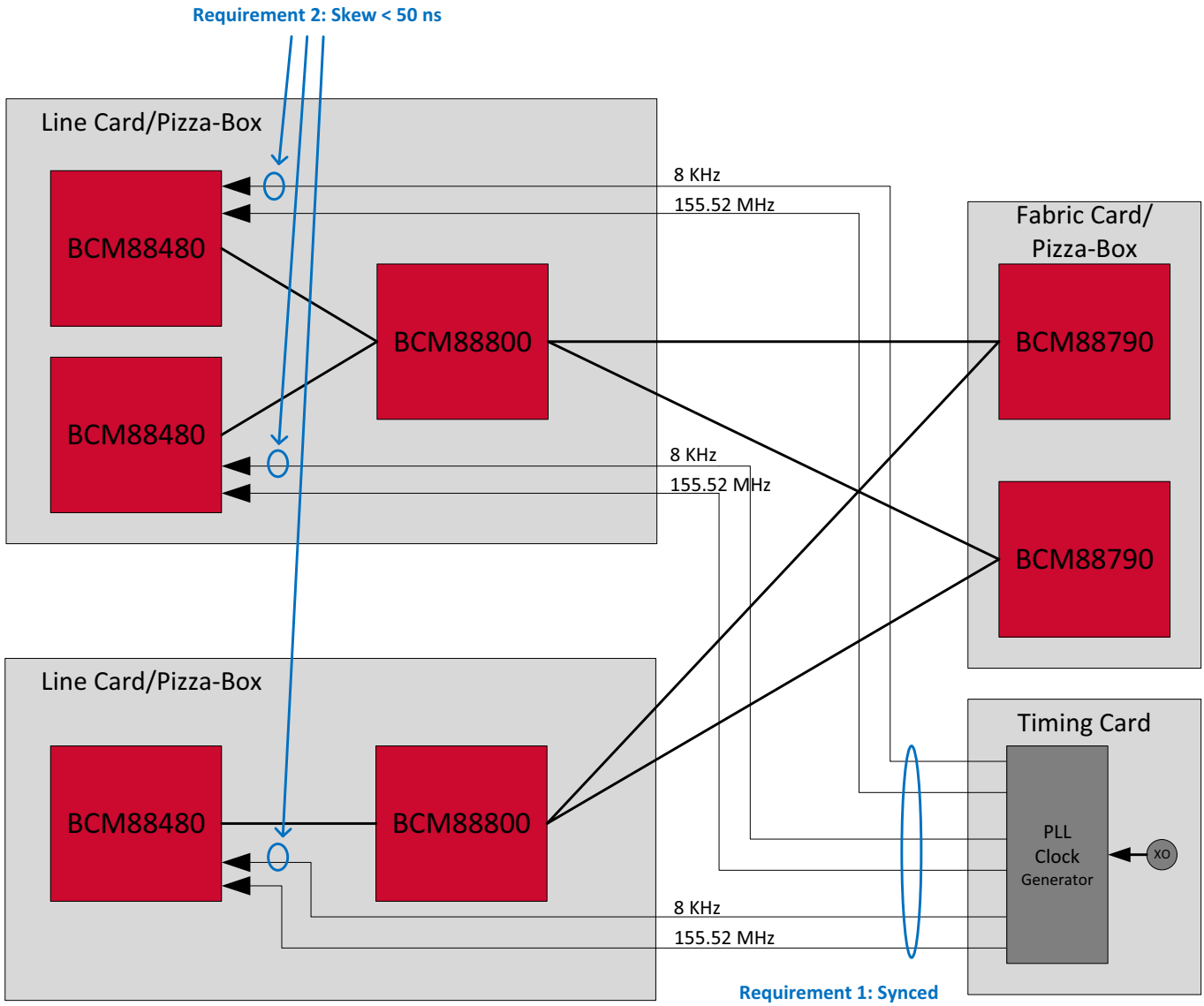
The following figures provide illustrations of the preceding requirements in two use-case examples.

Figure 49: BCM88480 FlexE Reference Clock Scheme (Pizza-Box, Single BCM88480)



NOTE: This section is not applicable for the BCM88280 or BCM88290.

Figure 50: BCM88480 FlexE Reference Clock Scheme (Chassis or Distributed Pizza-Boxes, Multiple BCM88480 Devices)



Chapter 9: Power Aspects

The power supply design must adhere to the DC specifications and the AC noise levels as defined in the device data sheet.

This section provides additional information and guidelines for power supply design for meeting the DNX16 device requirements.

The section handles supply aspects for the VDDC core supply, Blackhawk SerDes, PCIe SerDes supply, and other supply-related issues.

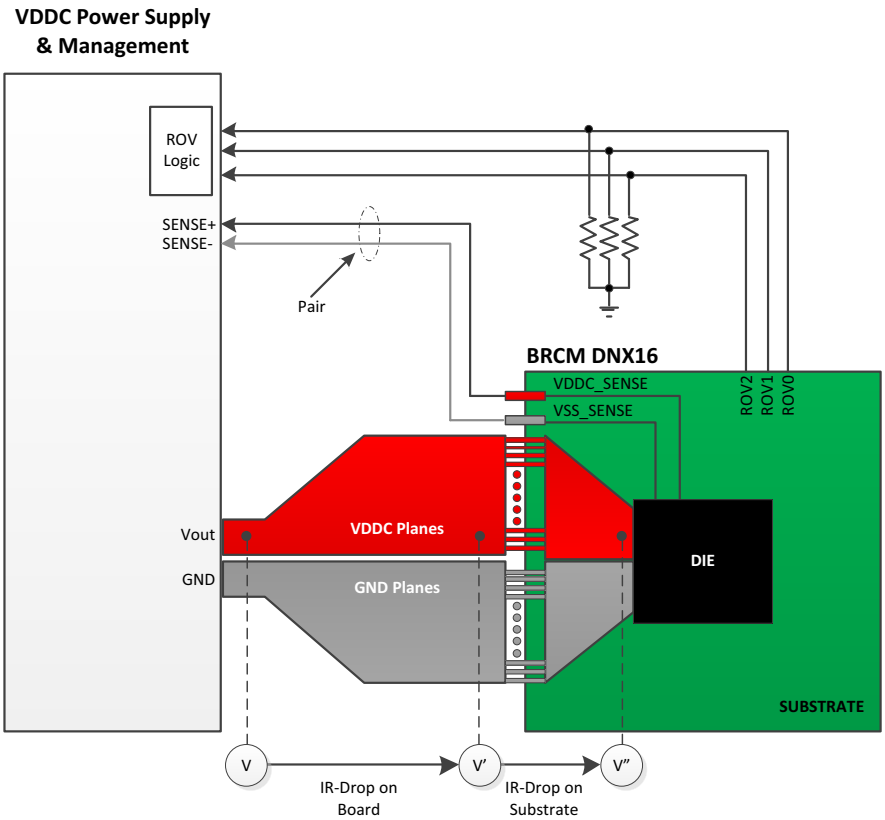
9.1 VDDC Core Supply

To meet the DNX16 device VDDC requirements, the VDDC supply design should refer to the following aspects:

- Recommended Operation Voltage (ROV)
- Power supply voltage accuracy
- IR drop and sense compensation
- Supply filtering

The following figure illustrates a conceptual VDDC power supply design scheme that handles those aspects. The following sections provide additional details.

Figure 51: VDDC Power Supply Design – Conceptual Scheme



9.1.1 VDDC Core Recommended Operation Voltage

DNX16 devices require the implementation of a programmable power supply source for the digital core supply (VDDC). The required value of VDDC is based on the ROV[2:0] pins. The DNX16 device drives the ROV pins at power-up, as described in the device data sheet (refer to the Recommended Operation Voltage and Power-up and Reset Sequence sections).

The ROV stamp is unique per each device unit. The stamp is set during device manufacturing. Different units of the same part number will have different ROV stamps. The board is required to monitor the unique ROV of each unit and set VDDC accordingly.

For each possible ROV stamp, a different VDDC is defined. For complete information, refer to the device data sheet. Setting VDDC to a value that is different from the value required by the ROV may cause the device to fail normal operation or exceed power limits.

NOTE: When multiple devices are on same board, each device requires a different VDDC supply.

NOTE: VDDC cannot be shared with any other rail with similar voltage requirements.

During device power-up and before the ROV stamp is derived by the device, an external pull-down resistor (see Figure 51) will ensure ROV = 000. When the ROV logic samples ROV = 000, the power supply should set the voltage to a nominal value, as defined in the device data sheet.

When the ROV stamp is driven by the device, the power supply should set the voltage to the value as defined in the device data sheet. (In this condition, pull-down resistors are no longer relevant.)

NOTE: The voltage transition from a nominal ROV voltage to the required ROV voltage should be smooth and should not exceed device limits.

The ROV stamp can also be read from the device registers. In the Bond Status register (ECI_BOND_STATUS), the AvsStatus field holds the ROV stamp. AvsStatus[26:24] = ROV_[2:0].

9.1.2 VDDC Core Power Supply Voltage Accuracy

Take special care when designing the VDDC supply. The device has strict voltage accuracy and voltage ripple requirements, as described in the device data sheet.

Good power-design practice is to design the power supply for better accuracy than the device requirements to have sufficient margins and to ensure the device requirements are met under all conditions.

9.1.3 VDDC Core IR Drop and Sense Compensation

The required value of VDDC (based on the ROV stamp) should be the voltage level as measured on the device die. Because the digital core of the DNX16 devices draws a large amount of current from the power supply, IR drops on the PCB and on the device package (substrate) are unavoidable. This condition, along with the tight supply margins, calls for careful attention to minimize IR drops. The effort to minimize IR drops starts at the device level and continues through the PCB design all the way to the regulator itself.

To compensate for the IR drop, DNX16 devices provide the sense pins VDDC_SENSE and VSS_SENSE. Use these pins to close the core VDDC regulator loop and enable the correct voltage at the die of the device, as shown in Figure 51.

Use both VDDC_SENSE and VSS_SENSE pair pins in power-supply implementation. Route VDDC_SENSE and VSS_SENSE pins as differential pairs based on power-supply requirements.

NOTE: For correct device operation, measure VDDC between VDDC_SENSE and VSS_SENSE.

NOTE: The power supply should implement protection against over voltage in the event of an incorrect sense indication (due to an assembly issue). Limit the voltage and do not exceed the device requirements. One common approach is to have a 50Ω pull-up resistor connected between VDDC_SENSE and VDDC.

Even with the sense implemented, take measures to control the IR drop over the board. To minimize the IR drop from the regulator output to the device pins, observe the following PCB design requirements:

- Use several layers to carry the relevant power plane/polygon in parallel.
- Use thick copper on the relevant layers used for power planes.
  For example, use 2-oz copper instead of 0.5-oz copper.
- Avoid tight via clusters that tear up the plane with their anti-pads.
- Pay close attention to how the regulator output pins interconnect with the planes.
  For example, monitor the number of stitching vias used on the board.

NOTE:

- Perform power-integrity simulations to verify that the design is appropriate. Use power integrity tools, such as Cadence Sigrity PowerDC or ANSYS SIwave-DC/PI, to analyze the IR drop as well as to detect potential hot spots.Because the regulator's sense is used to close the regulation loop at the die, *do not* place inductive filtering of any sort on this supply.
- Consider the IR drop when calculating the maximum power that is required from the power supplies (the drop can accumulate to dozens of Watts).

9.1.4 Measure and Validate VDDC

Use the following guidelines to measure and validate VDDC:

- Perform the measurement with a differential probe on the sense balls and validate the measured values by comparing them to the VDDC DC accuracy requirement specified in the device data sheet.
- Due to high current and a large IR drop, measure the DC delta on several VDDC balls and compare each delta with the simulation results. The maximum delta should be less than the DC accuracy range.
- Measure AC on the VDDC sense balls and on the other VDDC balls.

9.1.5 VDDC Supply Filtering and Decoupling

The board is required to provide a PDN target impedance as specified in the device data sheet in the Power Supply Filtering section. Note that each device may have its own requirements. For example, the PDN target impedance for the BCM88690 is different than the PDN target impedance of the BCM88790.

NOTE: When using the SVK as an example for the VDDC design, perform a power design simulation to make sure the PDN target impedance requirement in the data sheet is met. Using the same amount of capacitors and capacitance values as in the SVK may not always guarantee that the requirements are met.

NOTE: The BCM88690 VDDC rail also includes FAB_VDDC and U_VDDC. The board should connect VDDC, FAB_VDDC, and U_VDDC to the same rail.

9.2 Blackhawk SerDes Supply

All SerDes supplies are sensitive to supply noise and require careful design to prevent noise generated elsewhere on the board from getting to the device. A good SerDes supply design also prevents noise generated on one SerDes core from getting into the other SerDes core. The data sheet specifies the magnitude of tolerable supply noise per each rail.

To improve power integrity, some of the power planes are shared inside the package. Pins with the same name must be connected to the same power filtering segment.

9.2.1 Blackhawk SerDes Supply – PVDD0P8

The PVDD0P8 pins are the PLL supplies for the octets. The rails are sensitive and should be treated accordingly.

The design guidelines regarding these pins are as follows:

- Power-supply:

  Supply the PVDD0P8 from a dedicated power-supply source that is different from the source that supplies the TVDD0P8 and RVDD0P8, or from any other 0.8V rail on the board. The reason for using a different source is to reduce the noise coupling between the supplies and to allow more flexibility in adjusting the voltage rails due to IR drop (the IR drop on the TVDD0P8 and RVDD0P8 is much higher than on PVDD0P8).
- Segmentation:

  As a good practice, create multiple segments (for example, two to four), each with its own filter.

  Creating multiple segments reduces the current in the ferrite and eases the routing.

  It is possible to separate each Blackhawk octet into a different segment (for example, the approach used on the BCM56980 SVK). However, this approach has not been validated on any DNX16 device SVK.
- Filter per segment:

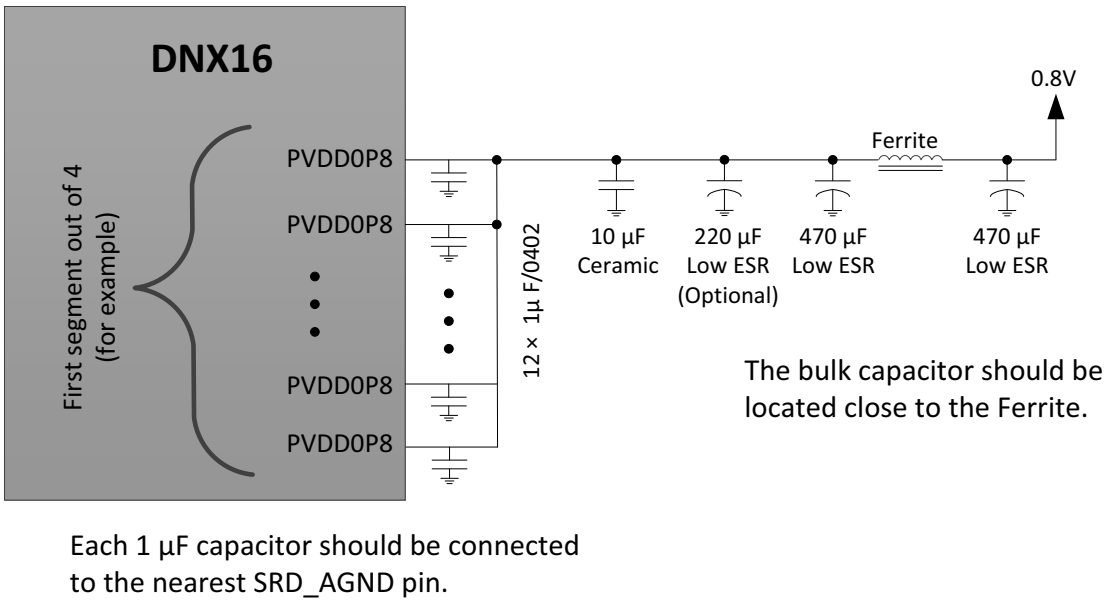
  For the recommended filter, see the following figure.

  Place one capacitor (1 µF, 0402) near each PVDD0P8 pin of the device.

  The capacitor placement is very important, and extra care should be taken in its placement. Connect each capacitor to the nearest SRD_AGND pin.
- Layout layers:

  On the layout, to keep noise out of the PVDD0P8 planes, the adjacent layers above and below the PVDD0P8 planes should be grounded on the relevant PVDD0P8 areas.

Figure 52: Blackhawk SerDes PVDD0P8 Supply Filtering



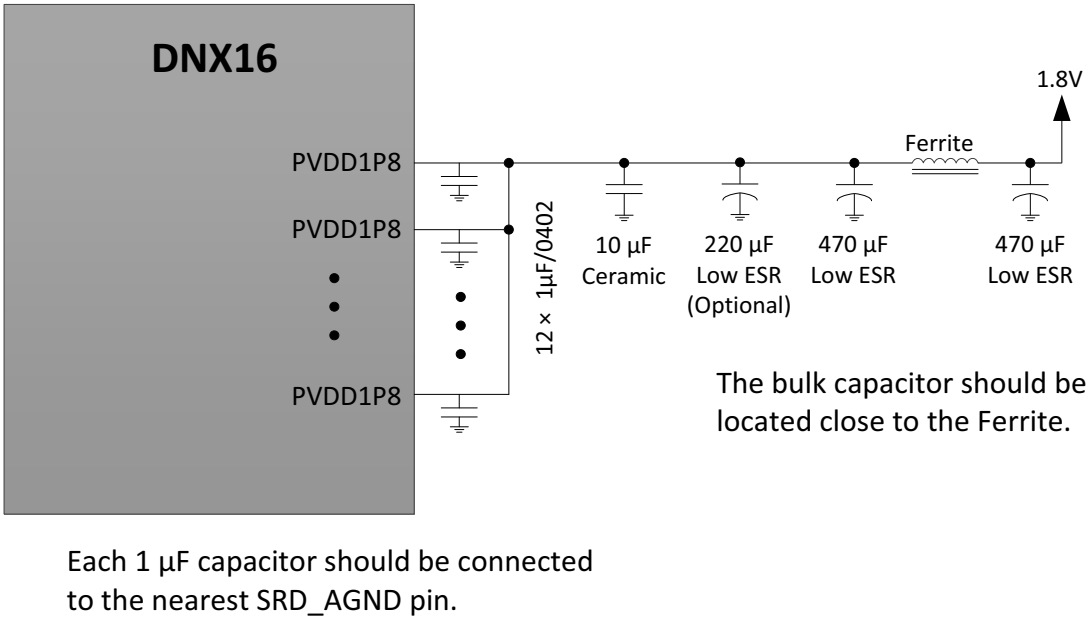
9.2.2 Blackhawk SerDes Supply – PVDD1P8

The PVDD1P8 pins are the PLL supplies for the octets. The rails are sensitive and should be treated accordingly.

The design recommendations regarding these pins are as follows:

- Segmentation:
  All SerDes PVDD1P8 pins of the device can share one segment.
  Separating the pins into a small number of segments is allowed if it eases the routing.
- Filter per segment:
  For the recommended filter, see the following figure.
  Place one capacitor (1 µF, 0402) near each PVDD1P8 pin of the device.
  The capacitor placement is very important, and extra care should be taken in its placement. Connect each capacitor to the nearest SRD_AGND pin.
- Layout layers:
  On the layout, to keep noise out of the PVDD1P8 planes, the adjacent layers above and below the PVDD1P8 planes should be grounded on the relevant PVDD1P8 areas.

Figure 53: Blackhawk SerDes PVDD1P8 Supply Filtering



9.2.3 Blackhawk SerDes Supply – TVDD0P8 and RVDD0P8

The TVDD0P8 and RVDD0P8 pins are the TX and RX supplies for the octets.

The design guidelines regarding these pins are as follows:

- Approach 1

  This approach creates large layers connected directly to the power-supply outputs without ferrite beads between the power-supply outputs and the DNX16 pins (similar to the VDDC design concept). This approach was adopted on all the latest DNX16 SVK platforms (BCM88790, BCM88690, and BCM88800).

  Pending the specific design, this approach can have a the following variations:
  - One rail for the entire device TVDD0P8 and RVDD0P8 pins.
  - One rail for the entire device TVDD0P8, and a second rail for tor entire device RVDD0P8 pins.
  - One rail for half of the device TVDD0P8 and RVDD0P8 pins, and a second rail for the other half.
- Approach 2

  This approach separates the TVDD0P8 and RVDD0P8 pins into many segments (for example, more than 10 segments), with each segment having its own ferrite bead. This is a more traditional approach, and has been the recommended approach in previous DNX HWDG documents.

  The following guidelines apply to this approach:
  - TVDD0P8 of the same octet must share the same power segment.
  - RVDD0P8 of the same octet must share the same power segment.
  - TVDD0P8 and RVDD0P8 of the same octet do not need to share the same power segment.
  - Due to the large current, choose the selected ferrite bead so that it has a very-low DC resistance (DCR).
- Filter per segment

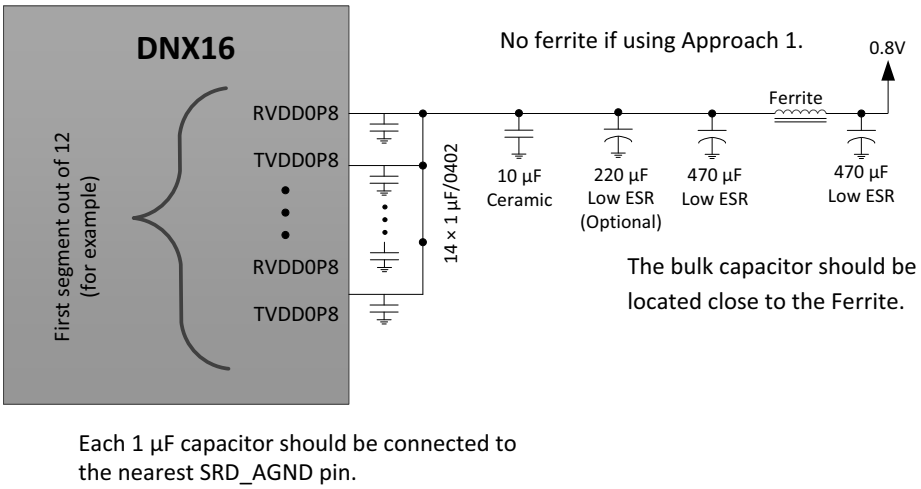
  For the recommended filter, see the following figure. (When using approach 1, there is no ferrite bead.)

  Place one capacitor (1 µF, 0402) near each TVDD0P8 and RVDD0P8 pin of the device. The capacitor placement is very important, and extra care should be taken in its placement. Connect each capacitor to the nearest SRD_AGND pin.
- Layout layers

  Required: TVDD0P8 and RVDD0P8 planes must not be adjacent to the core power supply (VDDC) to avoid supply-noise coupling from high-power VDDC planes to the SerDes planes.

  Recommended: Ground the adjacent layers above and below the TVDD0P8 and RVDD0P8 planes on the relevant areas.

Figure 54: SerDes TVDD0P8 and RVDD0P8 Supply Filtering



9.2.4 Blackhawk SerDes Supply – TVDD1P2

TVDD1P2 pins are the TVDD 1.2V supply for the octets.

The design guidelines regarding these pins are as follows:

- Segmentation

  All the SerDes TVDD1P2 pins of the device can share one segment.

  Separating the pins into a small number of segments is allowed if it eases the routing.
- Filter per segment

  For the recommended filter, see the following figure.

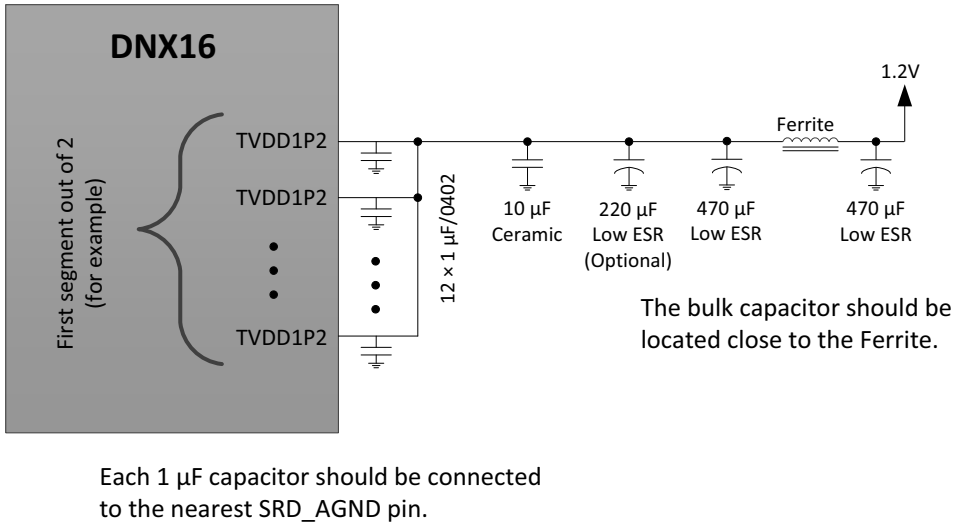
  Place one capacitor (1 µF, 0402) near each TVDD1P2 pin of the device.

  The capacitor placement is very important, and extra care should be taken in its placement. Connect each capacitor to the nearest SRD_AGND pin.
- Layout layers

  Required: The TVDD1P2 plane must not be adjacent to the core power supply (VDDC) to avoid supply-noise coupling from high-power VDDC planes to the SerDes planes.

  Recommended: Ground the adjacent layers above and below the TVDD1P2 planes on the relevant areas.

Figure 55: SerDes TVDD1P2 Filtering



9.2.5 SRD_AGND

Connect each capacitor of PVDD0P8, PVDD1P8, TVDD0P8, RVDD0P8, and TVDD1P2 to the nearest SRD_AGND pin.

Connect the SRD_AGND pins on the board with the VSS pins.

9.3 Falcon16 SerDes Supply

The information about SerDes supply for Falcon16 is the same as it is for Blackhawk. See Section 9.2, Blackhawk SerDes Supply, and the information applies to all rails: PVDD0P8, PVDD1P8, TVDD0P8, RVDD0P8, and TVDD1P2.

NOTE: When creating large layers connected directly to the power-supply outputs without ferrite beads along the way (as described in approach 1 in Section 9.2.3, Blackhawk SerDes Supply – TVDD0P8 and RVDD0P8), sharing the Blackhawk rails with the Falcon16 rails is allowed.

9.4 PCIe SerDes Supply

As mentioned in Section 9.2, Blackhawk SerDes Supply, all SerDes supplies are sensitive to supply noise and require careful filtering to prevent noise generated elsewhere on the board from getting to the device. The SerDes PLL supplies are the most sensitive supplies. The data sheet specifies the magnitude of tolerable supply noise per each rail.

The PCIe SerDes have two power rails: PCIE_PVDD0P8 and PCIE_RTVDD0P8.

For PCIE_PVDD0P8 guidelines, follow the information in Section 9.2.1, Blackhawk SerDes Supply – PVDD0P8.

For PCIE_RTVDD0P8 guidelines, follow the information in Section 9.2.3, Blackhawk SerDes Supply – TVDD0P8 and RVDD0P8.

NOTE: When creating large layers connected directly to the power-supply outputs without ferrite beads along the way (as described in approach 1 in Section 9.2.3, Blackhawk SerDes Supply – TVDD0P8 and RVDD0P8), sharing the Blackhawk rails with the PCIe rails is allowed.

9.5 HBM Power Supply

9.5.1 BCM88690 HBM Power Supply

All HBM-related supplies are sensitive to supply noise and require careful filtering. The HBM PHY PLL supplies are the most sensitive supplies. The data sheet specifies the magnitude of tolerable supply noise per each rail.

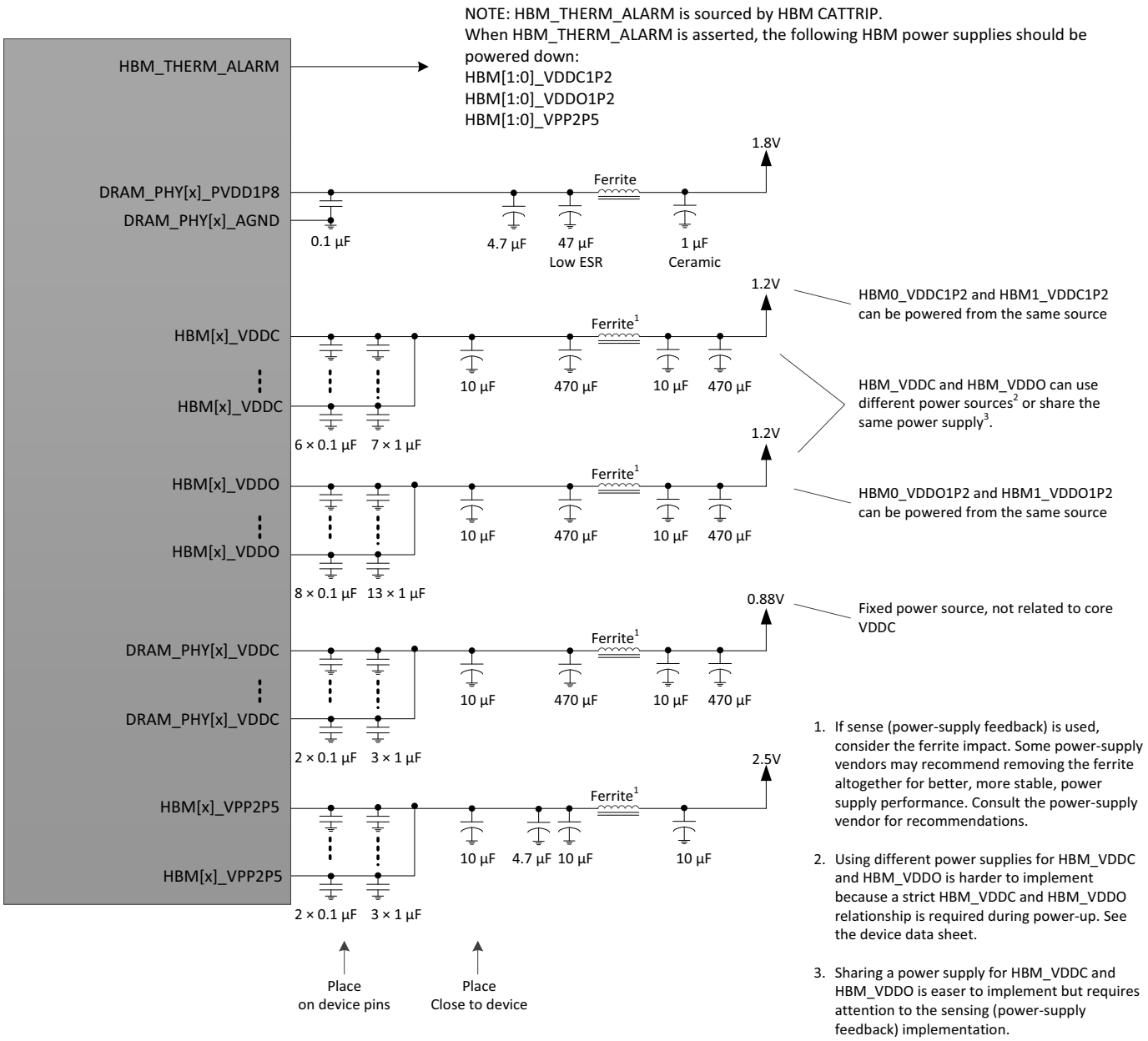
Capacitor placement rules:

- On the board, place decoupling capacitors under the device as close as possible to power vias, and spread the capacitors symmetrically for HBM0 and HBM1. Use a decoupling capacitor package of 0402 or less. See Table 30 and Figure 56.
- For each rail (DRAM_PHY* and HBM*), place a low-frequency capacitor (4.7 µF to 10 µF) as close as possible to the device. See Figure 56.
- Place DRAM_PHY0_PVDD1P8 capacitors between DRAM_PHY0_PVDD1P8 and the closest DRAM_PHY0_AGND.
- Place DRAM_PHY1_PVDD1P8 capacitors between DRAM_PHY1_PVDD1P8 and the closest DRAM_PHY1_AGND.

Table 30: HBM Power Supply Decoupling Capacitors

Power Rails		No. of Pins	0.1 µF	1 µF	Notes
HBM PHY	DRAM_PHY[1:0]_PVDD1P8	2 × 1	2 × 1	—	PHY PLL supply.
	DRAM_PHY[1:0]_VDDC	2 × 5	2 × 2	2 × 3	—
HBM	HBM[1:0]_VDDC	2 × 13	2 × 6	2 × 7	—
	HBM[1:0]_VPP2P5	2 × 5	2 × 2	2 × 3	—
	HBM[1:0]_VDDO	2 × 21	2 × 8	2 × 13	HBM and BCM88690 HBM I/Os.

Figure 56: HBM Power Supply Filtering



Route the HBM power plane as follows:

- Keep solid GND shapes above and below the HBM-related power plane.
- Make sure HBM power planes are not adjacent to the core power supply (VDDC) to avoid supply noise coupling from high-power VDDC planes to HBM planes.
- Provide adjacent ground vias for all the power vias (that is, add a GND via near each power via).

9.5.2 BCM88800 HBM Power Supply

The information in Section 9.5.1, BCM88690 HBM Power Supply, also applies to the BCM88800 device *except* for the following:

- The BCM88800 includes only one HBM die, while the BCM88690 includes two HBM dies.
- The BCM88800 has HBM_VSS pins. Connect the capacitors of the HBM_VDDC, HBM_VDDO, and HBM_VPP2P5 filters to the HBM_VSS.

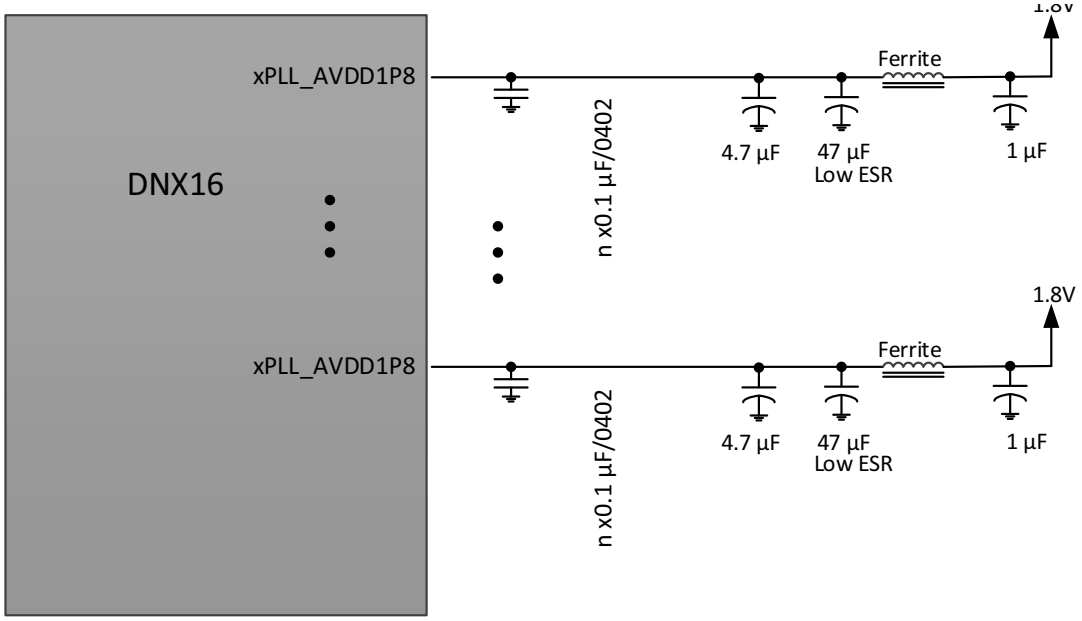
9.6 PLL AVDD1P8 Supply

The DNX16 devices have many PLLs, and each requires an analog AVDD1P8 supply of 1.8V. Separate these planes from any other 1.8V supplies.

Separate and filter each xPLL_AVDD1P8 supply. Each filter should have a ferrite bead and bulk capacitors, and place a 0.1-µF capacitor near each xPLLx_AVDD1P8 pin.

When an AGND pin is present (for example, on the BCM88690, C_PLL_AGND and U_PLL_AGND), be sure to connect the capacitor to this AGND pin.

Figure 57: PLL 1.8V Supply



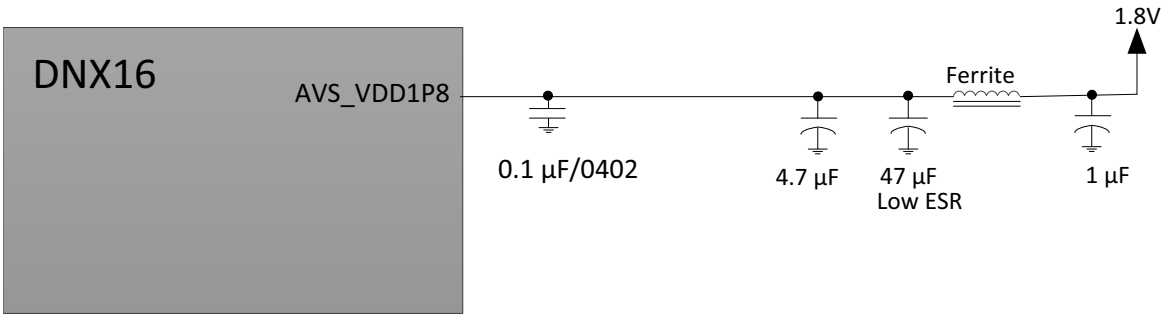
9.7 AVS_VDD1P8 Supply

Separate the AVS_VDD1P8 supply from any other 1.8V supplies.

The filter should have a ferrite bead and bulk capacitors, and place a 0.1-µF capacitor near the AVS_VDD1P8 pin.

NOTE: This pin is available only on the BCM88790.

Figure 58: AVS_VDD1P8 Supply



9.8 Supply Pull-Down Load Resistors

It is recommended to add a 50Ω pull-down resistor to each of the DNX16 power rails. The resistors help to discharge the supply rails on power-down.

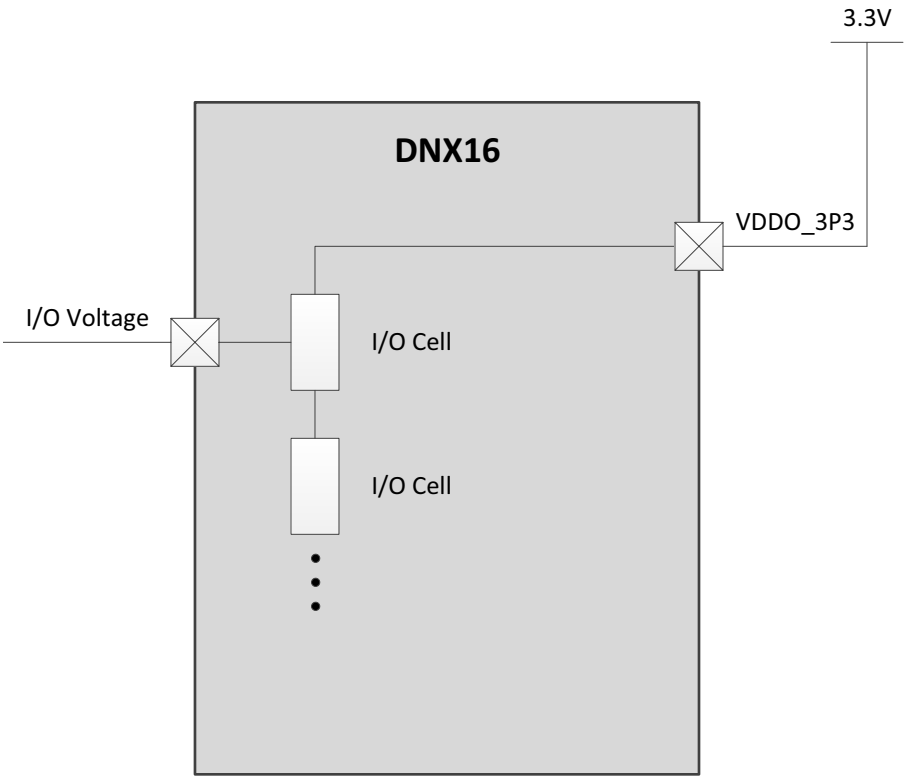
9.9 I/O Fail-Safe Considerations

NOTE: This section is applicable to the BCM88790 and BCM88690 devices. It does not apply to the BCM88800 or BCM88480 devices.

DNX16 devices have some I/O pins that are part of the VDDO_3P3 rail domain but are fail-safe only to 1.8V (and are not fail-safe to 3.3V). These pins are listed in each device data sheet in the Fail-Safe Considerations section.

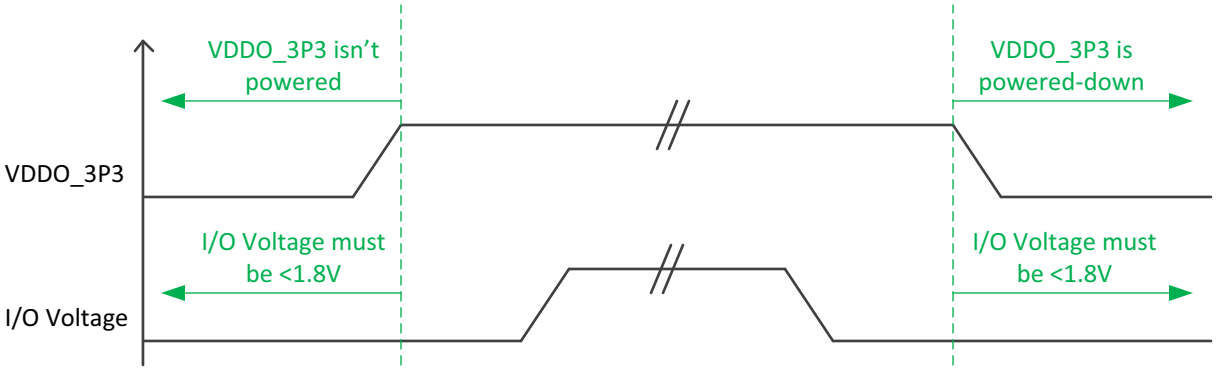
The following figure illustrates the structure of these I/O pins.

Figure 59: Supply Voltage and I/O Pad Voltage



The following figure shows the required power-up sequence for such I/O.

Figure 60: Required Power-up Sequence



The I/O voltage can be sourced by a peer device, by an external pull-up resistor, or by an internal pull-up resistor on the peer device.

Common approaches for addressing this power-up sequence requirement include:

- If an external pull-up resistor is present, connect it to VDDO_3P3.
- Power the peer device from VDDO_3P3 or from a voltage rail that is powered-up after VDDO_3P3 (and is powered-down before VDDO_3P3 is powered-down).

An alternate approach is to place a buffer (or FET) between the DNX16 and the peer device. The buffer (or FET) should be powered by VDDO_3P3.

9.10 GDDR6-Related Power Supply

9.10.1 BCM88480 GDDR6-Related Power Supply

Place bulk capacitors as close as possible to the BCM88480 device and the GDDR6 devices. Place the BCM88480 device and GDDR6 devices as close as possible to each other. Place decoupling capacitors as close as possible to the device pins.

The BCM88480 SVK supports a socket for the BCM88480 device. If a socket is not required, the GDDR6 devices can be placed closer to the BCM88480.

The following two tables provide information about the power supplies for the BCM88480 DRAM interface and GDDR6 device. Refer to the BCM88480 data sheet.

Table 31: BCM88480 Power Supplies

BCM88480 Supply	Number of Pins	Nominal Voltage	Comments
DDR_VDDO	93	1.3V	DDR I/O supply
DDR_VDDC	12	0.88V	DDR PHY core supply
DDR0_AVDD1P8	1	1.8V	DDR0 PLL filtering, has DDR0_AVSS (GND)
DDR1_AVDD1P8	1	1.8V	DDR1 PLL filtering, has DDR1_AVSS (GND)
DDR0_V_ANALOG	2	1.8V	DDR0 analog power
DDR1_V_ANALOG	2	1.8V	DDR1 analog power

Table 32: GDDR6 Power Supplies

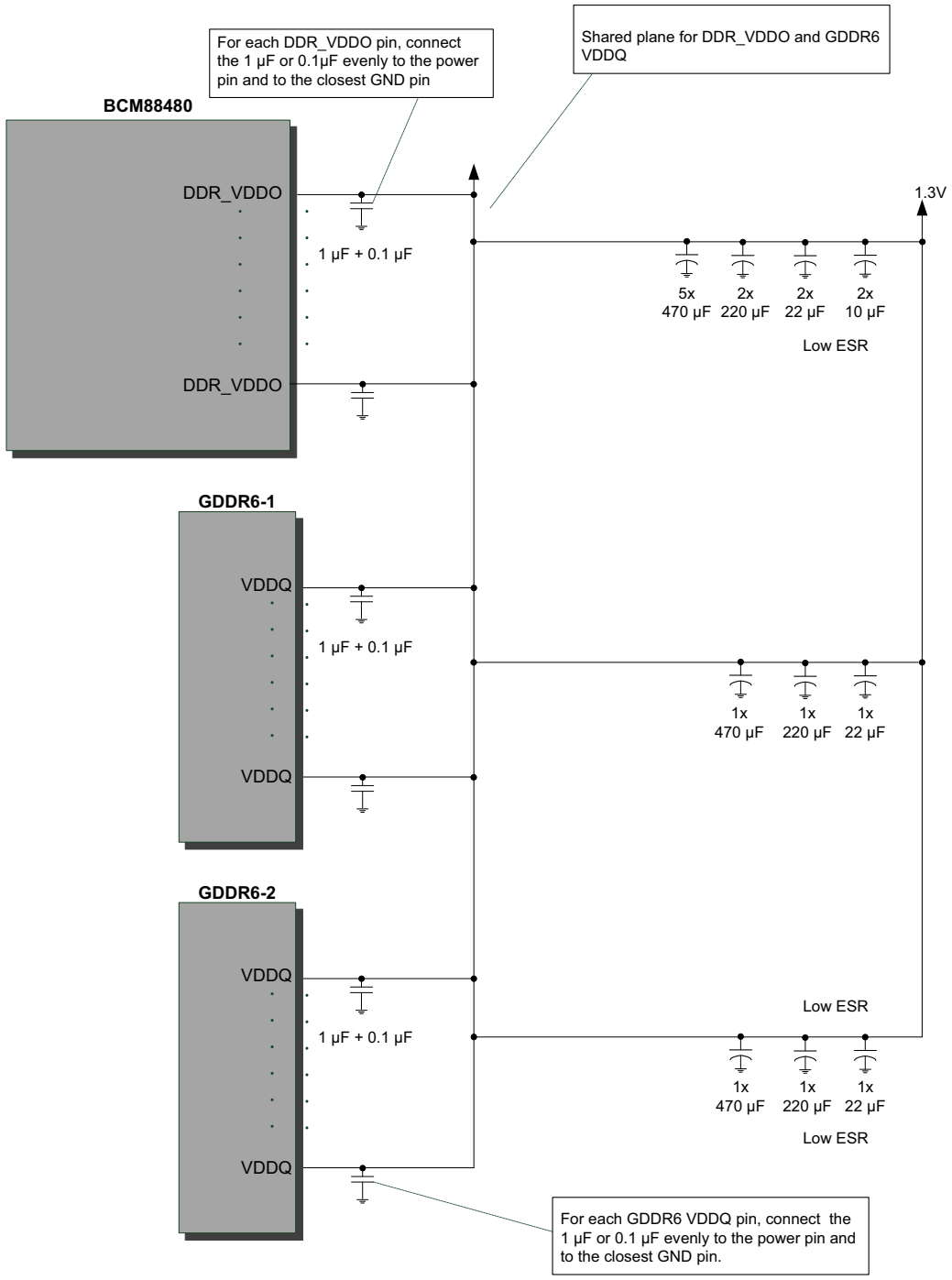
GDDR6 Supply	Number of Pins	Nominal Voltage	Comments
VDDQ	28	1.3V	I/O supply
VDD	12	1.3V	—
VPP	4	1.8V	—

9.10.1.1 BCM88480 DDR_VDDO and GDDR6 VDDQ

The BCM88480 device DDR I/Os require a supply of 1.3V connected to the DDR_VDDO pins and GDDR6 VDDQ pins. The supply plan is shared between the BCM88480 GDDR6 interface I/Os and the GDDR6 devices.

NOTE: For GDDR6 VDDQ decoupling and filtering information, refer to the memory vendor's guidelines.

Figure 61: DDR_VDDO and VDDQ Power Filter

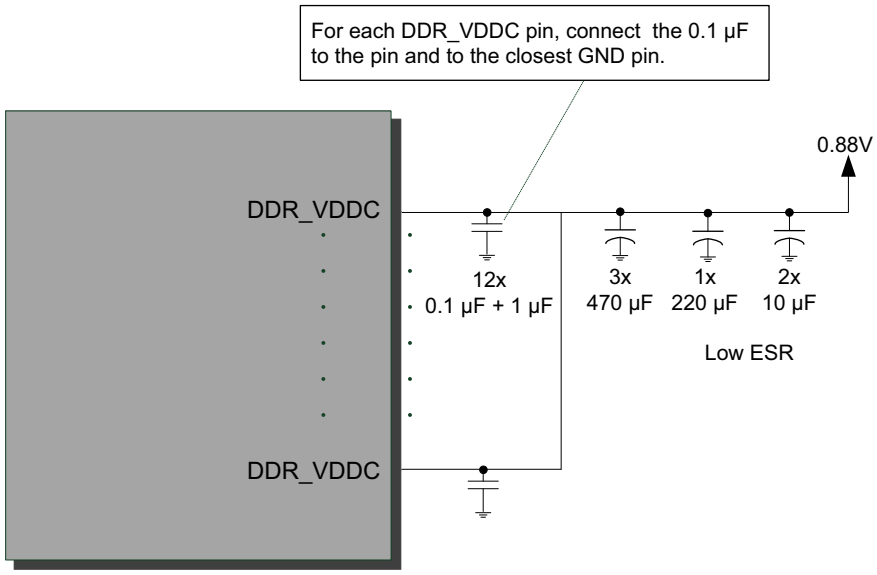


9.10.1.2 BCM88480 DDR_VDDC

The BCM88480 device DDR core logic requires a supply of 0.88V connected to the DDR_VDDC pins.

DDR_VDDC is not related to the device VDDC, and DDR_VDDC should not be sourced from the device VDDC.

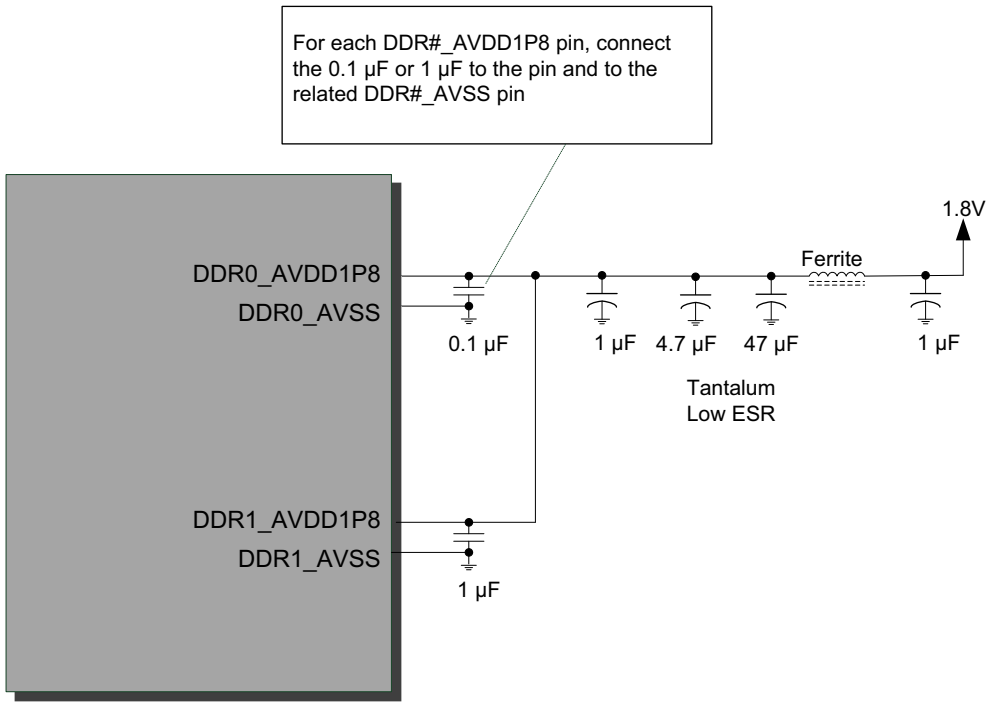
Figure 62: DDR_VDDC Power Filter



9.10.1.3 BCM88480 DDR#_AVDD1P8

The BCM88480 device DDR PLLs require an analog AVDD1P8 supply of 1.8V. These pins can share 1.8V planes. Filter the plane to achieve a power supply with very low noise. As shown in the following figure, a 0.1-µF capacitor must be placed between each _AVDD1P8 pin and the related _AVSS pin. For example, a capacitor must be placed between DDR0_AVDD1P8 and DDR0_AVSS.

Figure 63: DDR#_AVDD1P8 Power Filter



The BCM88480 SVK uses one filter for both DDR#_AVDD1P8.

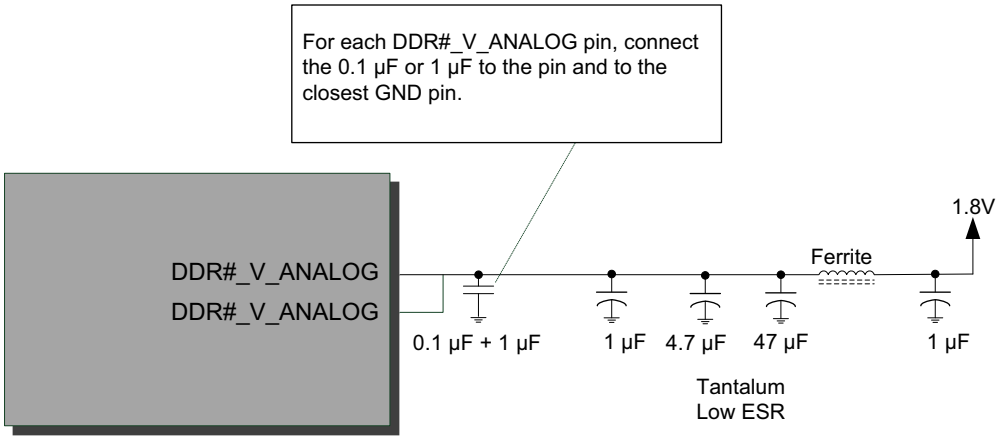
9.10.1.4 BCM88480 DDR#_V_ANALOG

The BCM88480 device has a per-interface analog supply, DDR#_V_ANALOG, that requires a 1.8V supply. These planes should be separated from other 1.8V planes to achieve very low noise supply.

As the following figure shows, a 0.1-µF capacitor must be placed between the _V_ANALOG pin and the related GND pin.

NOTE: DDR0_V_ANALOG and DDR1_V_ANALOG can be filtered together.

Figure 64: DDR#_V_ANALOG Power Filter



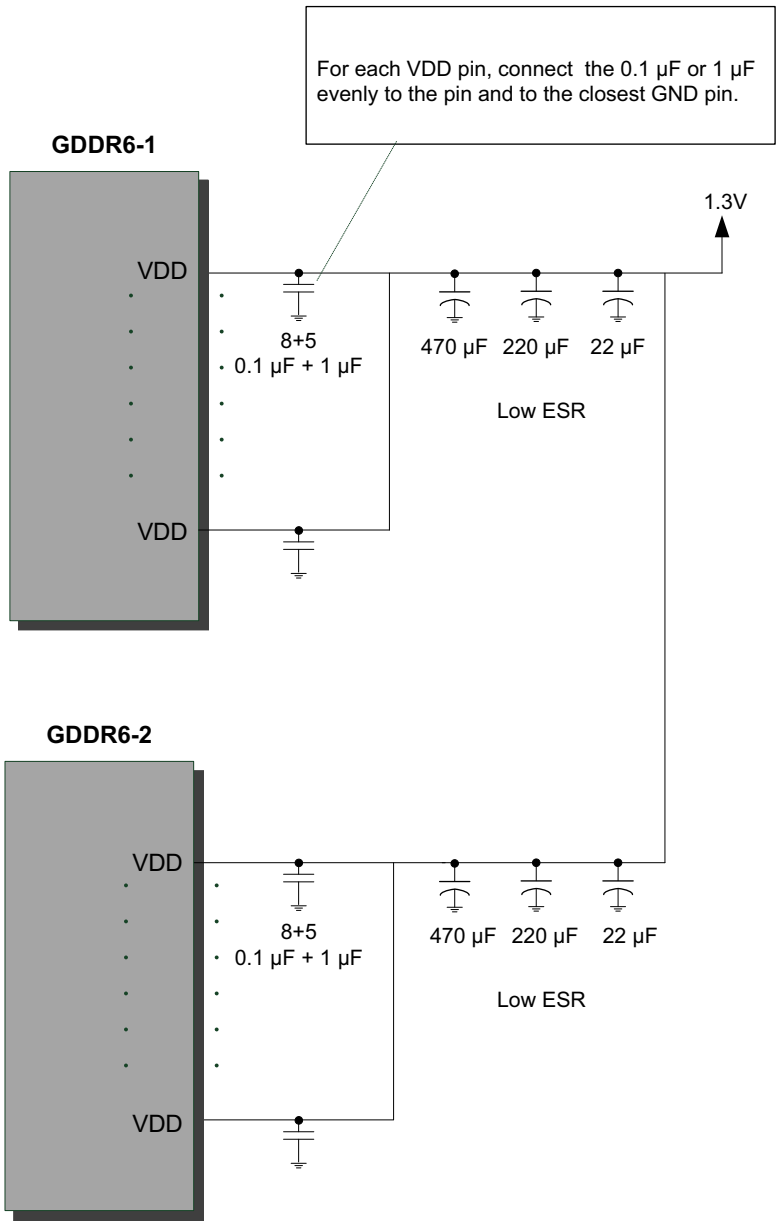
9.10.1.5 GDDR6 VDD

NOTE: For GDDR6 VDD decoupling and filtering information, refer to the memory vendor's guidelines. Some vendors provide PCB PDN requirements rather than capacitor recommendations. The following section is for reference only.

The BCM88480 SVK connects the GDDR6 device VDD and VDDQ. Refer to the memory vendor's guidelines to determine whether to follow the SVK example or to separate the rails.

The BCM88480 SVK connects the VDD of both GDDR6 devices so they are fed from the same supply and have the same power plane.

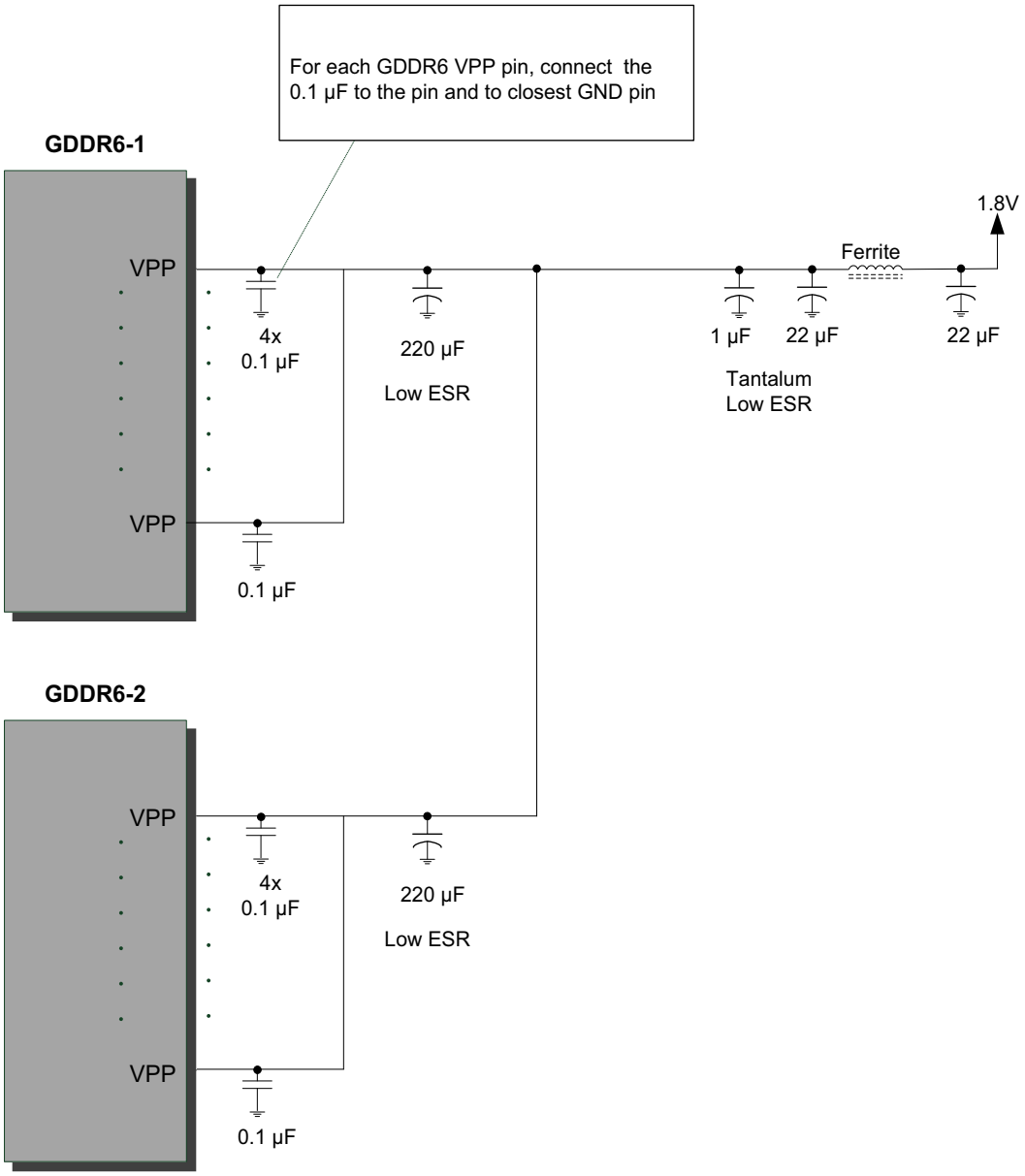
Figure 65: VDD Power Filter for GDDR6 Devices



9.10.1.6 GDDR6 VPP

For GDDR6 VPP decoupling and filtering, refer to the memory vendor's guidelines.

Figure 66: VPP Power Filter for GDDR6 Devices



9.10.2 BCM88290 GDDR6-Related Power Supply

The BCM88290 GDDR6 interface and power design is similar to the BCM88480 design. For the BCM88290, use the information in Section 9.10.1, BCM88480 GDDR6-Related Power Supply, but consider the following differences between the BCM88290 and BCM88480 devices:

- The BCM88290 uses only one GDDR6 device (DDR1), while the BCM88480 uses two GDDR6 devices.
- The number of power pin supplies in the BCM88290 is reduced by about half when compared with the BCM88480.

The following table shows the BCM88290 power supply information compared with the BCM88480 power supply information in Table 31, BCM88480 Power Supplies.

Table 33: BCM88480 Power Supplies vs. BCM88290 Power Supplies

BCM88480				**BCM88290**	
BCM88480 Supply	**Number of Pins**	**Nominal Voltage**	**Comments**	**Number of Pins**	**Comments**
DDR_VDDO	93	1.3V	DDR I/O supply	55	—
DDR_VDDC	12	0.88V	DDR PHY core supply	6	—
DDR0_AVDD1P8	1	1.8V	DDR0 PLL analog power	—	—
DDR1_AVDD1P8	1	1.8V	DDR1 PLL analog power	1	DDR_AVDD1P8
DDR0_V_ANALOG	2	1.8V	DDR0 analog power	—	—
DDR1_V_ANALOG	2	1.8V	DDR1 analog power	2	DDR_V_ANALOG

Chapter 10: Miscellaneous Interfaces

10.1 RESCAL

The DNX16 devices include RESCAL blocks that are used for the SerDes (Blackhawk and PCIe) calibrations.

Each RESCAL pin requires a 4.53-kΩ resistor with ≤1% accuracy connected to SRD_AGND. Analog circuits connected to the RESCAL pins are sensitive to trace-routing parasitics. Use 0402 or 0201 resistors, and place them in the via array on the bottom side of the device. Adjacent to each RESCAL pin is a corresponding SRD_AGND pin making for easy placement of each resistor.

10.2 LCPLL Calibration Resistors

The LCPLL requires an external calibration resistor. The calibration resistance is 5-kΩ or 4.99-kΩ ±1%. Connect the resistor between the RES_P and RES_N pins.

Place these resistors under the BGA and use very short traces with minimal resistance to the device pins.

10.3 DISCONNECT

The DISCONNECT pin is an input to the DNX16 FE/FAP devices.

When asserted high, the FE/FAP device stops sending reachability message cells to the FAP/FE devices, which is useful for a hardware implementation of an FE/FAP graceful shutdown when a card is unplugged or powered-down while traffic is running.

For applications with fabric, the DISCONNECT should be driven by a circuit that detects card removal (for example, by using a short pin) or power-down. For normal operation, set the DISCONNECT to low.

Implement a debounce circuit as long as it does not filter or delay the removal event.

For applications without fabric, set the DISCONNECT to low. Examples for these types of applications include systems with BCM88690 devices connected in mesh and BCM88690 stand-alone systems.

10.4 INT_N

The interrupt output, INT_N, is an active low, pseudo open-drain.

Pseudo open-drain means that the device has an internal weak pull-up, and a large external pull-up is required on-board.

From the user's perspective, it does not make a difference that the pin is a *pseudo* open-drain and not a *true* open-drain. Use a 4.7-kΩ pull-up resistor on-board.

10.5 I²C

10.5.1 Using I²C (Software Aspects)

The information in this section is from the `pcie_gen3_fw_install.txt` file in the `DNX_PCIe_gen3_Firmware_02_05_02` firmware package and is included in this application note for reference only.

Using the I²C has the following prerequisites:

- Board level: The CPU (on which the SDK is running) should be an I²C initiator of an I²C bus to which the DNX device is connected.
- SDK level: The SDK should be built with CPU I²C support, meaning the FEATURE_LIST in the `make.local` file should contain CPU_I2C.
  - The user should know the I²C bus number (because Linux numbers the I²C buses).
  - The user should know the DNX device I²C number in the bus (0x44 by default, set by the PUC pins).

The following command is an example of configuring pciephy firmware handling to use I²C with a specific bus number (0 in this example) and specific device number (0x44 in this example):

```
BCM.0> pciephy fw access i2c 0 0x44
```

After issuing this command, test whether I²C device access works by using the `pciephy fw version` command or write the firmware directly to the flash memory by using the `pciephy fw load <firmware file>` command.

If the BDE you are using with the SDK supports I²C access to the device internally, it is not necessary to specify the I²C bus and device numbers. In this case, you can use the `pciephy fw i2c_bde` command instead of the `pciephy fw access i2c <bus id> <device id>` command.

If the SDK does not recognize any switch device using PCIe, the `pciephy` command will not work. In this case, a dummy device must be added using the following SOC properties, where the PCIe device and revision IDs may be replaced with the IDs of the relevant device:

```
extra_unit_min=0
extra_unit_max=0
extra_unit.0=1
pci_override_dev.0=0x8690
pci_override_rev.0=1
```

10.5.2 I²C NACK Response

The information in this section is applicable when using the DNX16 I²C interface.

The I²C responder should generate an ACK after every incoming byte (command, address, or data) on the data line. It should generate a NACK only if it is busy and unable to handle incoming data.

A DNX16 device responds with a NACK after the fourth data byte in a 32-bit write transaction even though the device successfully performs the write to the described address location. This NACK causes software drivers to assume the write operation failed.

Depending on the Linux I²C driver, the Linux kernel, and the syslog configuration, a syslog error might be reported for I²C write operations to the device.

The NACK responses might also affect the speed of writing a PCIe firmware version when using I²C. Whether the speed is impacted depends on the Linux I²C driver.

As a workaround for these issues, configure the software driver to ignore the NACK responses (and consider them as ACKs).

Chapter 11: Hardware Connectivity for IEEE 1588 and Time-of-Day User Applications

This section addresses the hardware connectivity of the DNX device for the following use cases:

- A network node that implements IEEE 1588.
- A network node that uses time-of-day (ToD) for user applications, such as OAM-DM and telemetry applications.

Topics that are not related to hardware connectivity are beyond the scope of this document.

NOTE: The IEEE 1588 solution is a licensed application (BroadPTP™ and BroadSync® or OneSync™). Contact your Broadcom representative for additional details.

Table 34: Broadcom IEEE 1588 and ToD User Application Options for DNX-Based Platforms

Opt.	1588 Mode	1588 Engine	Platform	Timing FW Types
1	Ordinary/boundary clock	External (non-Broadcom)	Any[a]	BroadSync
2	Ordinary/boundary clock	DNX BroadPTP	Pizza-box with a single DNX device	BroadPTP
			Pizza-box with two DNX devices	BroadPTP + BroadSync
3	Transparent clock	None	Any[a]	BroadSync
3a	Transparent clock	None	Pizza-box with a single DNX device[b]	None
4	ToD for user applications (no IEEE 1588)	None	Any[a]	BroadSync or BroadPTP (for syncing ToD)

a. *Any* includes the following platforms:
   – Pizza-box with a single DNX device or multiple DNX devices
   – Chassis with multiple DNX devices

b. This option applies when all timestamping occurs internally on the DNX device and not on any external PHYs, and no user applications require ToD.

11.1 Ordinary/Boundary Clock – External IEEE 1588 Engine

This section provides information about Option 1 from Table 34.

Opt.	1588 Mode	1588 Engine	Platform	Timing FW Types
1	Ordinary/boundary clock	External (non-Broadcom)	Any[a]	BroadSync

a. *Any* includes the following platforms:
– Pizza-box with a single DNX device or multiple DNX devices
– Chassis with multiple DNX devices

In this mode, the network node includes an external (non-Broadcom) IEEE 1588 engine that synchronizes all of the DNX devices using the BroadSync[0] interface.

For detailed information about BroadSync operation, refer to the *Broadcom BroadSync Solution: Description, Performance Test Results, and Implementation* application note (1588-AN2xx).

The same connectivity is used regardless of whether a single DNX device or multiple DNX devices are present on the node. Examples of different platforms and DNX device configurations that use the same connectivity include the following:

- A pizza-box with a single DNX device
- A pizza-box with two or more DNX devices connected over their fabric interfaces (usually referred to as a back-to-back or mesh DNX connectivity)
- A chassis with a fabric backplane

The timestamping can be performed by the DNX devices or by external PHYs (if the PHYs are present on the board and are synced to the IEEE 1588 engine). The external IEEE 1588 engine is responsible for syncing the PHYs (directly to the external PHY and not by using the DNX device).

Interoperability of the DNX device and the external PHY is guaranteed only if the external PHY is a Broadcom PHY that supports IEEE 1588.

PTP packets should reach the IEEE 1588 engine through the DNX NIF port connected to the CPU. Another option, which is more complicated, is to use the PCIe interface.

NOTE:

- As a general guideline, support DNX device synchronization over the BroadSync interface for any device in a system that might support IEEE 1588, OAM-DM (ITU-T Y.1731), or telemetry applications.
- Sync all DNX devices in the node by using the BroadSync interface, including the DNX devices that do not perform the timestamping. Using the BroadSync interface to sync all DNX devices supports flows between all DNX devices in same node, including cases of traffic flows between a DNX device with an external PHY and a DNX device without an external PHY (in both directions). This behavior is a basic design requirement.
- Sync all DNX devices in the node by using the BroadSync interface even if the timestamping is performed only by external PHYs. This guidance is provided because future line cards in this system might not include external PHYs on all the DNX ports, and the DNX device will be required to perform the timestamping. As noted previously, in such a node, all the DNX devices must be synced to the system clock.

The following list highlights some of the connectivity guidelines for this option:

- OCXO is recommended but not required.
- TS[0]_SYNC: 4 kHz, synced in frequency and phase to the system IEEE 1588 clock.
- TS[0]_BIT_CLK: Synced to TS_SYNC[0] and must be an integer multiple of it.
- TS[0]_TIME_VAL: The required message rate is 4K messages per second.

Figure 67, Figure 68, and Figure 69 demonstrate the board connectivity described in this section. As the figures show, from the point-of-view of the DNX devices, the connectivity is identical (BroadSync timeReceiver).

Figure 67: Pizza-Box with a Single DNX Device

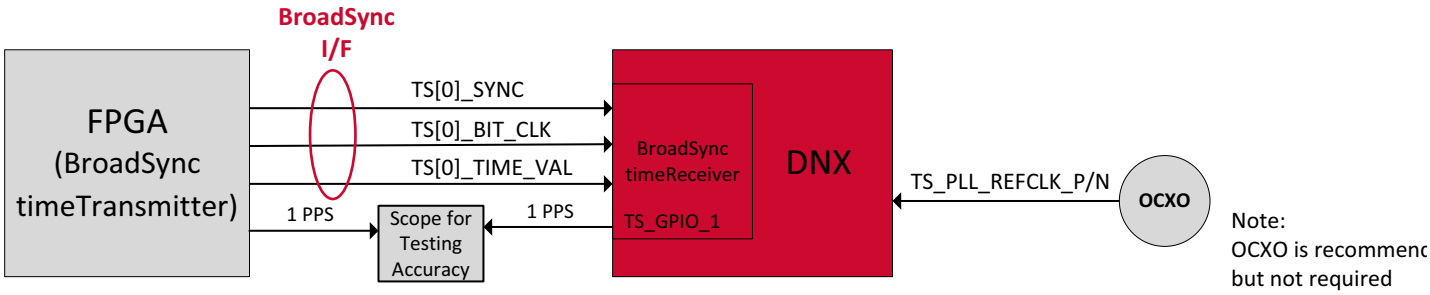


Figure 68: Pizza-Box with Two DNX Devices (Back-to-Back)

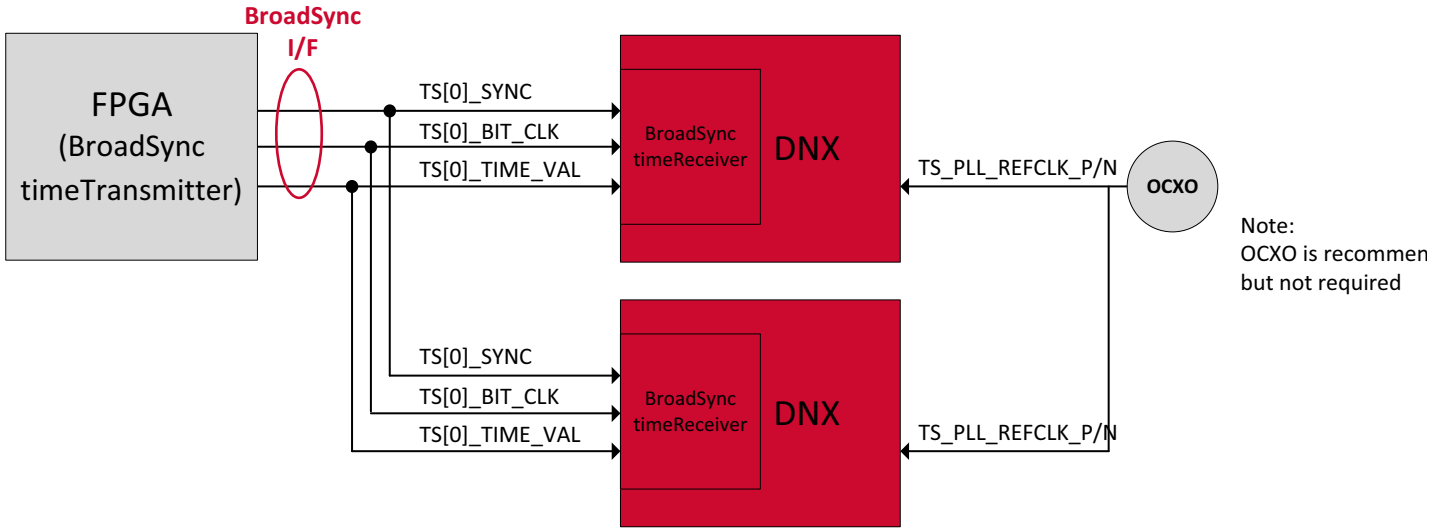
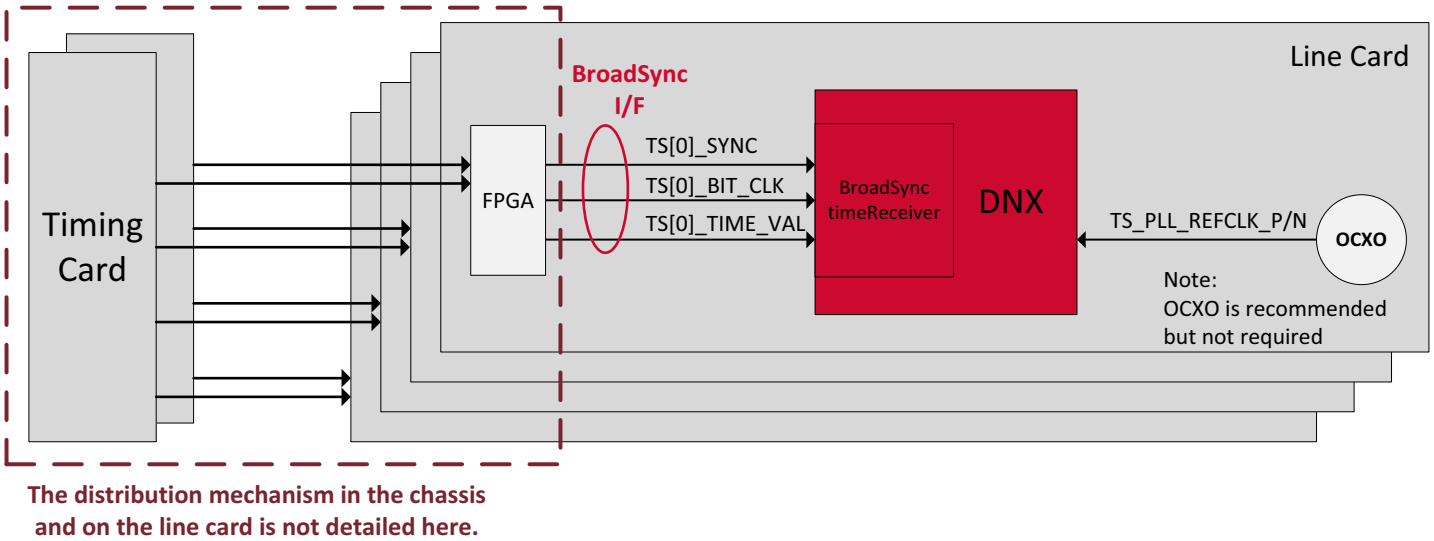


Figure 69: Chassis, Including Fabric FEs or Mesh



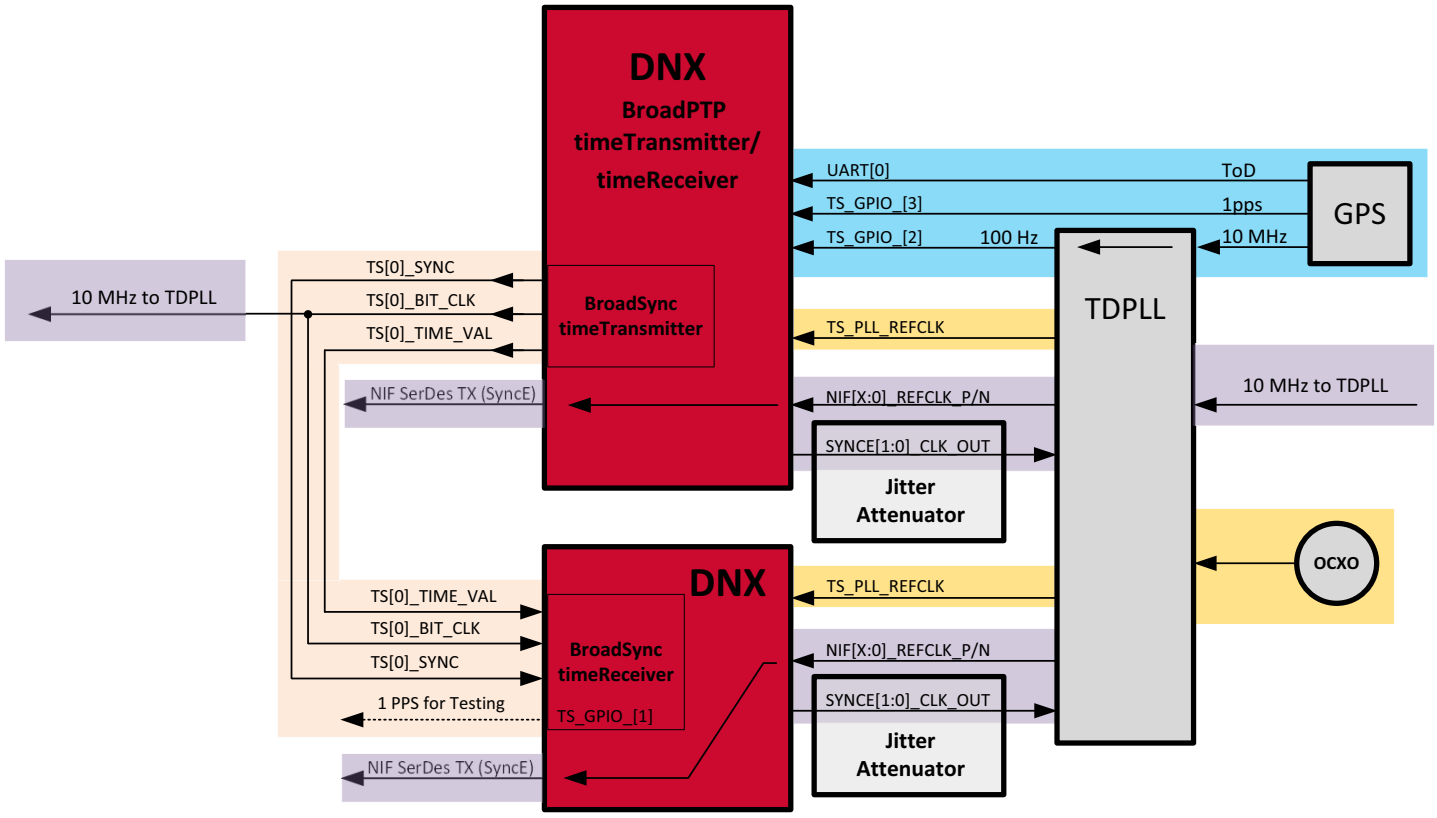
11.2 Ordinary/Boundary Clock – DNX BroadPTP (Pizza-Box)

This section provides information about Option 2 from Table 34.

Opt.	1588 Mode	1588 Engine	Platform	Timing FW Types
2	Ordinary/boundary clock	DNX BroadPTP	Pizza-box with a single DNX device	BroadPTP
			Pizza-box with two DNX devices	BroadPTP + BroadSync

The following figure is a superset of all the connectivity options required for the different use-cases. A detailed discussion of the different use-cases is beyond the scope of this document.

Figure 70: BroadPTP on a Pizza-Box with a Single- or Dual-DNX Device



The following list highlights some of the connectivity guidelines for this option:

- TDPLL must be present on the pizza-box for the proper operation of BroadPTP. The TDPLL drives the DNX NIF[X:0]_REFCLK_P/N and the TS_PLL_REFCLK. In addition, the DNX recovered clocks (SYNCE[1:0]_CLK_OUT) should be connected to the TDPLL.
- In the relevant use cases, the GPS loads the IEEE 1588 timing to the DNX device as follows:
  - ToD value – Loaded using the UART[0] interface.
  - 1 PPS signal – Connected to TS_GPIO_[3].
  - 10 MHz – Connected to the TDPLL that divides the rate to 100 Hz and is connected to TS_GPIO_[2].
- For a pizza-box with two DNX devices (connected back-to-back), BroadSync can be used for syncing between the two devices. The device running BroadPTP also uses the BroadSync timeTransmitter and syncs the second device, which uses the BroadSync timeReceiver. Both devices should be connected to the same OCXO and TDPLL.

  The number of clients supported by this configuration is the same as the number supported by the configuration of a single DNX device.

  This scheme (of the first DNX device syncing the second DNX device using BroadSync) is supported by the design but has not been tested in the lab.

  This scheme is valid only for a pizza-box platform with two devices connected back-to-back on the same PCB. This scheme is *not* valid in the following cases:
  - This scheme is *not* valid if the two devices are not on the same PCB.
  - This scheme is *not* valid for syncing two separate pizza-boxes or for a disaggregated chassis.
  - This scheme is *not* valid for syncing devices over different line cards within a chassis.

11.3 Transparent Clock

This section provides information about Option 3 and Option 3a from Table 34.

Opt.	1588 Mode	1588 Engine	Platform	Timing FW Types
3	Transparent clock	None	Any[a]	BroadSync
3a	Transparent clock	None	Pizza-box with a single DNX device[b]	None

a. *Any* includes the following platforms:
– Pizza-box with a single DNX device or multiple DNX devices
– Chassis with multiple DNX devices

b. This option applies when all timestamping occurs internally on the DNX device and not on any external PHYs, and no user applications require ToD.

In this mode, the network-node is operating in transparent-clock mode, meaning no IEEE 1588 software is present – neither externally to the DNX device nor internally on it.

The IEEE 1588 timestamping is performed by the DNX device, by external PHYs, or by both (on different ports):

- Single DNX device time-stamping only – BroadSync is not required.
- DNX devices time-stamping only – An external element (such as an FPGA) is required to provide BroadSync to all the DNX devices.
- A mix of timestamping provided by Broadcom external PHYs and DNX devices – An external element (such as an FPGA) is required to provide BroadSync to all the DNX devices and, in addition, to sync the Broadcom external PHYs (directly to the external PHY and not by using the DNX device).
- BRCM external PHYs time-stamping only – Refer to the external PHYs requirements. BroadSync to the DNX is not required.

When BroadSync is required, see Section 11.1, Ordinary/Boundary Clock – External IEEE 1588 Engine.

Option 3a is a pizza-box with a single DNX device. All timestamping is performed internally on this DNX device and not on any external Broadcom PHY, and no user applications require ToD. In this case, using BroadSync is not necessary, and a simple XO is sufficient for the transparent-clock implementation (an OCXO is not necessary).

11.4 ToD for User Applications

This section provides information about Option 4 from Table 34.

Opt.	1588 Mode	1588 Engine	Platform	Timing FW Types
4	ToD for user applications (no IEEE 1588)	None	Any[a]	BroadSync or BroadPTP (for syncing ToD)

a. *Any* includes the following platforms:
– Pizza-box with a single DNX device or multiple DNX devices
– Chassis with multiple DNX devices

Some user applications (that are not related to IEEE 1588) require using ToD. Examples include OAM-DM and telemetry applications. If ToD is required by an application, it is provided to the DNX devices using one of the following features:

- BroadSync. For information, see Section 11.1, Ordinary/Boundary Clock – External IEEE 1588 Engine.
- BroadPTP. For information, see Section 11.2, Ordinary/Boundary Clock – DNX BroadPTP (Pizza-Box).

Chapter 12: Thermal Aspects

Cooling high-power devices can be challenging because it often requires electrical, thermal, and mechanical integration in the application system design. For more information about thermal design aspects for the DNX16 devices, refer to the *Thermal Considerations for High-Power Switching Devices* application note (StrataDNX-StrataXGS-AN1xx) listed in Related Documents.

The application note describes various aspects of thermal design and testing to avoid and address thermal-related issues that may occur with application systems, such as the following:

- Temperature reference points
- Design considerations
- Temperature cycles, accelerated testing, and warpage

In addition to the application note, *Thermal Design Checklist for High-Power Switching Devices* (StrataDNX-StrataXGS-Thermal_Design_Checklist_v1.0) is available on the ESP to help avoid or resolve high-power thermal challenges.

12.1 Heat Sink Considerations

12.1.1 Heat Sink Selection

The DNX16 device must be used with a heat sink. The end-use thermal environment combined with the operating mode of the device dictates the required thermal characteristics of the heat sink (such as size, thermal resistance, and so on).

12.1.2 Heat Sink Attachment

The heat sink used with the DNX16 device is required to be mechanically mounted and held in place by a clamp or fixture to the printed circuit board.

The contact between the heat sink and the package is maintained by using an externally applied mechanical force.

The exact configuration of the mounting scheme and clamping mechanism is at the user's discretion.

NOTE:

- An adhesive-based or taped-based attachment scheme is not allowed.
- The heat sink must not be clamped to the package (the package/heat sink combination must not be freestanding).

Use a non-adhesive thermal interface material, such as phase change film or thermal grease, between the heat sink and the package top to maintain a low thermal resistance path between the package and the heat sink. This material is known as *TIM2*.

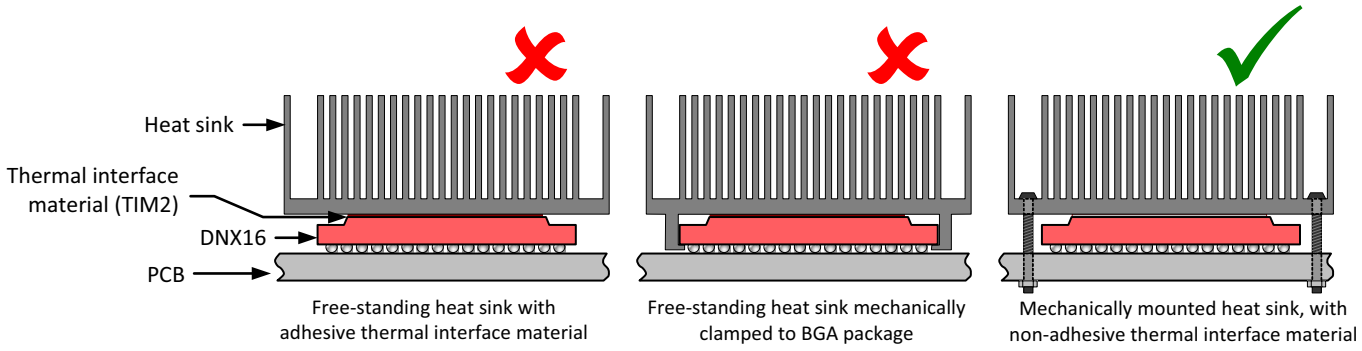
Customers are responsible for the system thermal design, including TIM2 decisions and should perform tests. Decisions should be made based on the test results.

The following TIM2 information is provided as a reference and can be used as a starting point for designs:

- BCM88790:
  - Fujipoly GR80A (0.5-mm thick) or Dexerials EX10000F7
  - 27 mm × 27 mm, centered within the package (top and bottom + left and right)
- BCM88690:
  - Fujipoly XR-Um or XR-m (0.3-mm thick)
  - 36 mm × 36 mm, centered within the package (top and bottom + left and right)
- BCM88800:
  - Fujipoly PG80A (0.5-mm thick) or XR-m (0.3-mm thick)
  - 34 mm × 34 mm, centered within the package (top and bottom + left and right)

NOTE: The heat sink design for DNX16 devices does not require a pedestal. A flat-bottom heat sink should be sufficient.

Figure 71: Heat Sink Attachment Considerations



The applied operating force between the heat sink and the package must be sufficient to meet the thermal interface material manufacturer's recommendations, but it must not exceed the maximum long-term compressive load on the package.

During the heat sink mounting process, the force between the heat sink and the package must not exceed the maximum short-term compressive load. (This amount of force is allowed for a maximum of 60 seconds at room temperature.)

The following table provides information about the long-term and short-term compressive load.

Table 35: Recommended Maximum on Package Compressive Load

	Long-Term Compressive Load		Short-Term Compressive Load (< 60 Seconds, Room Temperature)	
Device Part Number	**Lbf**	**Kgf**	**Lbf**	**Kgf**
BCM88790	65	29.5	120	54.4
BCM88690	111.2	50.4	444.7	201.7
BCM88800	71.8	32.6	287.4	130.4
BCM88480	36.2	16.4	100	45.4
BCM88290	28.4	12.9	113.6	51.5

NOTE: For applications requiring a compression force greater than the maximum values defined in Table 35, customers should perform system-level compression force reliability tests.

For system-level compression force reliability test guidelines, refer to the *Broadcom BGA Product System Load Force Qualification Test Recommendation* document (BRCM-Load-Force-Qual-Requirement_9-27-2018) on the Engineering Support Portal (ESP).

Additional items that must be addressed when choosing a heat sink are as follows:

- The applied operating force must be evenly distributed across the package top.
- Tooling holes and/or clamp locations must be selected to minimize PCB warpage.
- The clamping mechanism must not clamp to package underside.
- The clamping structure must withstand the user's mechanical testing requirements, such as shock and vibration.

12.2 Thermal Monitoring

12.2.1 Continuous-Operation Thermal Monitoring

For continuous-operation thermal monitoring of the die junction temperature, use the PVTMONs. Take the die junction as the maximum value of all PVTMONs, while considering the PVTMON accuracy and the delta between the PVTMON and the die hot-spot.

NOTE:

- Because each PVTMON is at a different location on the die, there is no expectation that all PVTMONs will provide the same or similar readings.
- Take note of the delta between the PVTMON and the die hot-spot. When this information is not available on the device data sheet, use a 5°C margin as a safeguard.
- Using the thermal diodes for continuous thermal monitoring is not necessary. Doing so may result in reading a value that is higher than the actual temperature of the die. Note that the thermal diodes are not available on other Broadcom switch product lines and are no longer available on DNX07 and later DNX generations.

12.2.2 Power-Down or Pre-SDK Temperature Reading

The thermal diodes are useful for temperature reading when the PVTMONs are not accessible, such as in the following cases:

- The device is powered-down. For example, if the power-down is due to an over-temperature event, it might be necessary to verify the temperature before powering-up again.
- The SDK is not loaded. This might occur during the power-up sequence of an industrial device, below 0°C. See Section 12.4, Industrial-Grade Devices.

NOTE: Use the diode temperature reading for reference only. (The read temperature accuracy is affected by the diode accuracy, the board layout, the board signals and power noises, the external reader accuracy, and so on.)

12.3 BCM88690 and BCM88800 HBM Thermal Design Considerations

The BCM88690 and BCM88800 are multi-die devices. The BCM88690 package contains the main ASIC die from Broadcom and *two* HBM dies from the memory vendor. The BCM88800 package contains the main ASIC die from Broadcom and *one* HBM die from the memory vendor.

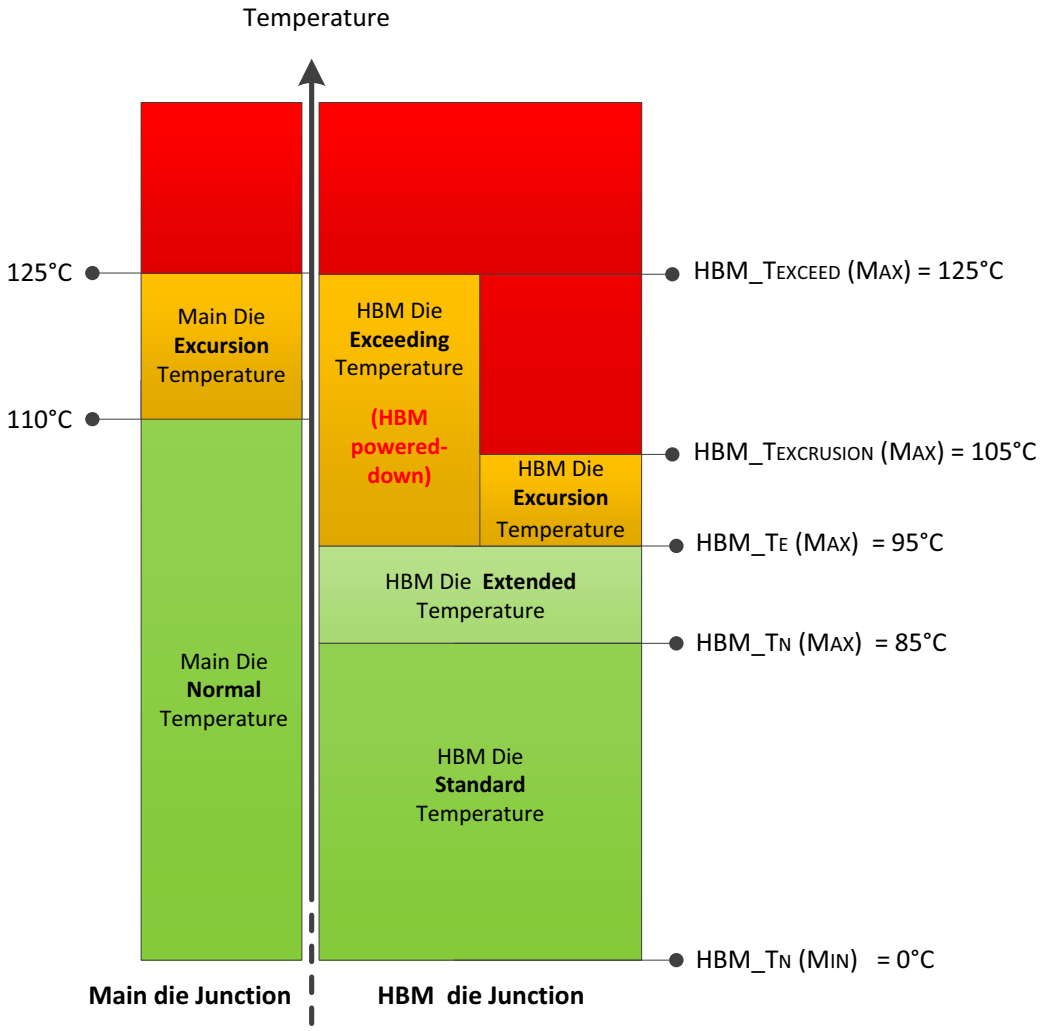
This section addresses the thermal challenges introduced by the multi-die device.

NOTE: The HBM operation point is set to a specific frequency. Changing or reducing the HBM frequency is not supported.

12.3.1 Temperature Operating Regions

The following figure provides information about the different temperature operating regions for the BCM88690/BCM88800 main die and HBM dies.

Figure 72: Temperature Operating Regions



NOTE: For more details on HBM_T_N, HBM_T_E and HBM_T_{EXCEED}, refer to the BCM88690 data sheet (88690-DS1xx) or the BCM88800 data sheet (88800-DS1xx).

Table 36: HBM Die and Main Die Temperature Regions

Region	Description
HBM standard temperature region	HBM standard operating conditions.
HBM extended temperature region	This region requires more refresh cycles. The additional cycles are handled by the BCM88690/BCM88800 without the need for user SW intervention. Expect a reduction in performance by a few percentages in extended mode.
HBM die exceeding temperature region	This region requires the HBM supplies to be *powered-down*. The amount of time the HBM operates in this region must be *less than 96 hours per year*. For detailed operating conditions, refer to the device data sheet.
HBM die excursion temperature region	The HBM is still operational in this region. The amount of time the HBM operates in this region must be less than 360 hours per year. (For additional details about the requirements, refer to the device data sheet).
Main die normal temperature region	Normal operating condition of the main die.
Main die excursion temperature region	An excursion due to thermal failure.

For details about the temperature ranges and requirements, refer to the device data sheet and the thermal application note (StrataDNX-StrataXGS-AN1xx). See Related Documents.

12.3.2 Thermal Simulations

When performing the system thermal simulations, make sure that all the BCM88690/BCM88800 thermal requirements regarding junction temperature are satisfied for both the main die and the HBM die.

As detailed in the BCM88690/BCM88800 data sheet, the following model is available:

Block thermal model

12.3.3 Thermal Supervision

While the device is operational, monitor the junction temperatures of the main die and the HBM dies constantly to prevent the HBM dies from working in non-operational conditions. Any exception can cause permanent damage to the device, reduce device reliability, and degrade device performance. Refer to the device data sheet for information about operating condition requirements.

If the HBM temperature monitor is enabled by the SOC property, the SDK enables and disables traffic to the HBM in response to temperature and shuts down the HBM power supplies if the HBM temperature limits are exceeded. For details, see Section 12.3.3.1, HBM Temperature Monitor Enabled.

If the HBM temperature monitor is disabled by the SOC property, customers should implement their own thermal management mechanism that determines when to enable/disable traffic to HBM memory or when to power up/down the HBM power supplies. Predefined API functions can be used to facilitate customer-based thermal monitoring implementations.

An SOC property controls the supported number of HBM dies, which is either two HBM dies or none. When two HBM dies are enabled, turn both HBM power supplies on or off as described in the device data sheet. To power-down memory, turn off the HBM power supply. Memory power-down mode states are not supported.

For details about the SOC properties and APIs available for temperature monitoring, refer to the DNX16 *Hardware HBM Configuration and Debugging Guide* (DNX16-AN2xx).

12.3.3.1 HBM Temperature Monitor Enabled

This section describes the user-related and driver-related steps involved in monitoring and controlling the HBM temperature when HBM temperature monitoring is enabled. If temperature monitoring is disabled, customers are responsible for implementing their own mechanism.

12.3.3.1.1 HBM Temperature Monitoring

While monitoring the HBM temperature, the *driver* handles the following events:

- If the temperature rises from the standard temperature range to the extended temperature range (see Figure 72), the HBM die updates the HBM controller with a higher refresh rate requirement. The controller automatically applies the required refresh rate with no software or user intervention and with minimal impact on HBM read/write throughput.
- If the temperature rises above the high threshold, the driver disables the flow of new traffic through HBM memory. Data already in the memory starts to drain until the memory is empty. This sequence takes a few seconds.
- If the temperature drops below the low threshold, the driver enables the flow of traffic through HBM memory.
- If the temperature rises above the HBM high die temperature (the predefined temperature point in the HBM extended region), the driver initiates the shutdown process for the HBM power supplies.

  To enable the HBM power supply shutdown, the user should implement a callback function that shuts down the HBM power supplies, per platform.

12.3.3.1.2 HBM Power-Up after Shutdown

If the HBM power supplies are shut down by the driver due to high temperature, the *user* should do the following:

- Do not violate the HBM_T_E value (even though the HBM is powered-down).
- Estimate the HBM temperature before restoring power to it. If the temperature is within the standard range or extended range, continue to the next step.
- Restore power to the HBM power supplies.
- Call the API to perform an HBM partial initialization. This step reenables the HBM operation.

After the HBM operation resumes, continue monitoring the HBM temperature.

12.3.3.2 Temperature Thresholds

The device supports user-defined, per-product thresholds for thermal avoidance to ensure proper temperature conditions exist for both the main die and the HBM dies. The thresholds are active when temperature monitoring is enabled, and the thresholds are triggered in high-temperature conditions.

When the system temperature rises above the high threshold (set by using a SOC property), it is assumed that the HBM die will soon be outside of working conditions. When the temperature is above the high threshold, new traffic does not flow through the HBM memory and flows through the on-chip buffer (OCB) only. If the temperature continues to rise and reaches the max HBM temperature (set by using a SOC property), the HBM power is shut down. If data is in the HBM memory when the temperature reaches the high threshold, it should be drained from the HBM memory before the temperature reaches the max HBM temperature. Draining the memory can take few seconds, so the high threshold should be set at a level that allows HBM memory to drain before the temperature reaches the maximum HBM temperature.

If the system starts to cool before reaching the maximum HBM temperature and drops below the low threshold, HBM is enabled, and new traffic is allowed to flow through HBM memory. The low threshold should be programmed a few degrees lower than the high thresholds to prevent oscillations.

For information about the SOC properties that define the thresholds and the maximum HBM temperature, refer to the *DNX16 Hardware HBM Configuration and Debugging Guide* (DNX16-AN2xx).

12.3.4 CATTRIP

The HBM defines the CATTRIP temperature point (HBM_$T_{CATTEMP}$). Avoid the CATTRIP temperature point in all cases. The HBM CATTRIP indicates that the HBM die temperature has reached a level at which catastrophic damage may occur unless power is reduced. If CATTRIP is reached, HBM_THERM_ALARM is asserted. In this condition, power down the HBM power until the temperature is reduced to operational conditions. When the temperature is low enough, HBM power-up can be applied.

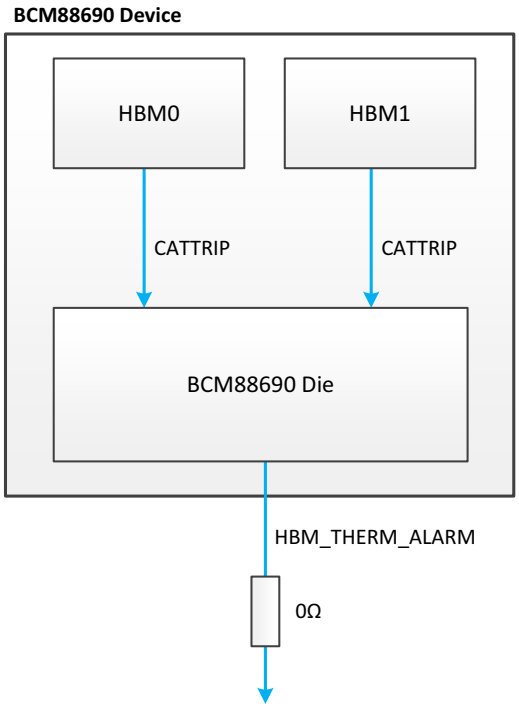
NOTE: For HBM_$T_{CATTEMP}$ details, refer to the BCM88690 data sheet (88690-DS1xx) or BCM88800 data sheet (88800-DS1xx).

The board should support HBM power-up and HBM power-down during BCM88690/BCM88800 operation; otherwise, full BCM88690/BCM88800 device power-down is required.

During normal operation, if the HBM temperature passes the high temperature threshold, the temperature monitor stops data flow to the HBM. If the temperature continues to increase and crosses the maximum HBM temperature threshold, the temperature monitor powers down the HBM (see Section 12.3.3.2, Temperature Thresholds). The high and maximum temperature thresholds are both lower than the CATRRIP temperature point, which is why it is assumed that CATTRIP is not asserted during normal operation. Any CATTRIP assertion during normal operation is considered abnormal and unexpected. This behavior requires the system to immediately reduce the temperature (for example, a full device power down) and might require a full device reset.

As the following figure shows, if CATTRIP is asserted, an interrupt is triggered, and the HBM_THERM_ALARM pin is asserted. It is possible to mitigate this condition by responding to a software interrupt, but keep latency limitations in mind. Mitigation is also possible via board support by using the ball indication HBM_THERM_ALARM to power down the HBM power supplies. Connect HBM_THERM_ALARM using a 0Ω resistor to the power supply control logic.

Figure 73: BCM88690 CATTRIP and HBM_THERM_ALARM



NOTE: The BCM88800 includes only HBM0 (and not HBM1).

12.4 Industrial-Grade Devices

Some DNX16 devices are defined as *industrial* grade (for information, refer to the data sheet for the relevant device). The following sections provide guidelines on how to approach the data sheets requirements and provides a recommended power-up sequence when powering up an industrial-grade device below the 0°C junction temperature.

12.4.1 BCM88800 Industrial-Grade Information

The BCM88800 has an industrial-grade part number. The BCM88800 data sheet provides all the relevant requirements for operating with $T_{Ambient}$ as low as –40°C.

The main die (that is, the entire device except the HBM) can be powered up and taken out of reset at T_J = –40°C. However, as specified in the data sheet, the operating temperature of the SerDes cores (PCIe, Blackhawk, and Falcon16) is higher than –40°C.

The following table provides the recommend power-up sequence to meet the device requirements when using the BCM88800 under conditions where T_A < 0°C.

Table 37: Recommended Power-up Sequence for BCM88800 Industrial-Grade (when T_A < 0°C)

Step No.	Topic	Details
1	Power-up at –40°C	Power-up the device while keeping the HBM unpowered, and take out of reset. This step can take place when $T_A \geq$ –40°C, and similarly, when $T_J \geq$ –40°C. The device power consumption after taken out of reset and before SDK loading is ~ 40W. The device power consumption can be increased to ~55W using a SDK function that loads a special image through the I^2C interface and turns on some blocks for heating. For details on the usage, refer to the *Traffic Manager Programming Guide* (88690-PG211 or later revisions, Chapter 26: Industrial-Grade (ITemp)). See also Section 10.5, I^2C.
2	Reaching PCIe operating temperature $T_J \geq$ PCIE_T_N min. (–20°C)	Wait for the device $T_J \geq$ –10°C (see the notes for this step) and reinitiate the PCIe block operation by performing a full device reset cycle (SYS_RST_N and PCIE_RST_N) to make sure the PCIe block configuration was not set while the device temperature was below –20°C. After the reset cycle, load the SDK with the `ext_ram_freeze_boot` SOC property set to prevent the HBM init from being performed during the `bcm_init`. (It will be invoked later by calling the `bcm_switch_dram_init()` API.) The device power consumption after the driver loading is ~ 50W to 60W. If required, you can operate some internal blocks to increase the power consumption to achieve the required temperatures of the next steps (for example: snake or service activation test [SAT]). **NOTE:** The following notes apply to this step: ■ Device T_J can be read using the thermal diode and the die temperature in the thermal simulations. ■ 10°C are taken as a margin for differences between the thermal indications and the actual temperature of the PCIe block.

Table 37: Recommended Power-up Sequence for BCM88800 Industrial-Grade (when T_A < 0°C) (Continued)

Step No.	Topic	Details
3	Reaching Blackhawk and Falcon16 operating temperature $T_J \geq$ BLACKHAWK_T_N min. (–10°C) $T_J \geq$ FALCON16_T_N min. (–10°C)	Wait for the device $T_J \geq$ 0°C (see the notes for this step) and re-initiate the Blackhawk and Falcon16 ports (disable and enable ports and re-initiate link-training) to make sure the SerDes block configuration was not set below –10°C. **NOTE:** The following notes apply to this step: ■ Device T_J can be read using the min. {PVTMONs} and the die temperature in the thermal simulations. ■ 10°C are taken as a margin for differences between the thermal indications and the actual temperature of the SerDes cores.
4	Reaching HBM power-up temperature $T_{J_HBM} \geq$ HBM_T_N min. (0°C)	Wait for the HBM $T_J \geq$ 0°C (see the notes for this step) before allowing the board to power-up the HBM. After the HBM rails are powered, call the `bcm_switch_dram_init()` API to invoke the HBM init. **NOTE:** The following notes apply to this step: ■ The thermal simulation model provides the HBM T_J separately from the device T_J. ■ The HBM temperature monitors are not operational before the HBM is powered-up, so for determining the HBM temperature, read the device junction temperature (PVTMONs) and make the proper calculation for the HBM junction temperature.

Make sure the following signals and interfaces are available for read and control by the Host CPU that manages the BCM88800 power-up:

- Thermal diode (THERM_DIODE_N, THERM_DIODE_P, THERM_DIODE_VDD3P3)
- I2C (I2C_SCL, I2C_SDA)
- PUC_26 (PCIE_QSPI_ENABLE)

NOTE: An I-temp system can perform this power-up sequence for all T_A temperatures, which means the same sequence is still valid when powering up at T_A > 0°C. (In other words, for simplicity, all cases can use the same procedure.)

12.4.2 BCM88790 Industrial-Grade Information

The industrial-grade part in the BCM88790 series is not available yet. However, when the part is available, the recommended power-up sequence is identical to the steps described in Table 37 with the following changes:

- Ignore all HBM information because the BCM88790 does not include an HBM.
- Ignore all Falcon 16 information because the BCM88790 does not include any Falcon16 SerDes.
- The device power consumption after the driver loading is ~17W to 25W instead of 50W to 60W.
- Use PUC_07 (PCIE_QSPI_DISABLE) instead of PUC_26 (PCIE_QSPI_ENABLE).
- The SDK function that loads a special image through the I^2C interface and turns on some blocks for heating is not currently available for the BCM88790 device.

12.4.3 BCM88480/BCM88280/BCM88290 Industrial-Grade Information

The recommended power-up sequence for the BCM88480/BCM88280/BCM88290 device is identical to the steps described in Table 37 with the following changes:

- Ignore all HBM information because the BCM88480/BCM88280/BCM88290 does not include an HBM.
- BCM88480/BCM88280/BCM88290 device power consumption after being taken out of reset and before SDK loading is ~15W.

Chapter 13: PCB Land Pattern and Reflow Process Recommendations

For information about the printed circuit board (PCB) land pattern design and reflow process recommendations, refer to the documents listed in the following table. These documents are available on the Engineering Support Portal (ESP).

Device	Document
BCM88790	Refer to the *BCM88790 SMT Recommendations* (BCM88790-SMT-Guide_9-07-2017)
BCM88690	Refer to the *BCM88690 SMT Recommendations* (BCM88690_SMT-Guide-r0_4-05-2018)
BCM88800	Refer to the *BCM88800 SMT Recommendations* (BCM88800_SMT-Guide-r0_1-08-2020)
BCM88480	For reflow process recommendations, refer to *Pkg Reflow Process Guidelines for Surface Mount Assemblies* (PACKAGING-AN105-R).
BCM88290	For reflow process recommendations, refer to *Pkg Reflow Process Guidelines for Surface Mount Assemblies* (PACKAGING-AN105-R).
	For PCB pad design recommendations, use the following guidelines: ■ Use an NSMD pad instead of an SMD pad. ■ Use the following guidelines for the NSMD pad rules: – Pad diameter: 0.406 mm (16 mils). – Solder mask opening: 0.50 mm (20 mils). – Pad-to-via dog-bone trace width: < 0.271 mm (< 10.7mils). – Via-in-pad: The via capture pad diameter must not exceed 0.406 mm (16 mils). – Via capping: The via must be capped by the BGA pad (VIPPO). – Via plug: Use a via plug to avoid hole opening on the BGA pad. ■ No special recommendations exist for stencil designs. Customers can use the same stencil design rules they applied to other BGA devices with a size of less than 55 mm.

NOTE: The SVK is available before the SMT guidelines are finalized. Differences may exist between the SVK SMT and the latest SMT guidelines. Always follow the SMT guidelines and not the SVK.

Appendix A: Known Issues for SVK Platforms

A.1 BCM88790 SVK

The SVK is the validation platform for the device. During the device bring-up and validation, the hardware recommendations might be modified. As a result, contradictions exist between the recommendations in this document and the SVK design.

Table 38 describes the known contradictions between the BCM88790 SVK platform (BCM988790R_SVK_PB_design_example_03) and this version of the *Hardware Design Guidelines*.

When using the SVK as a design example, be aware of the items and recommendations in the following table.

Table 38: BCM88790 SVK Platform versus Hardware Design Guide Recommendations

No.	Subject	Issue	Recommendation
1	VDDC capacitors	The SVK includes a total of 18 × 470-µF + 4 × 1500-µF capacitors. Installing this many capacitors is unnecessary as long as the design meets the PDN requirements as detailed in the data sheet.	Perform a PDN simulation and install capacitors as required to meet the data sheet requirements.
6	TVDD1P2 power supply	The SVK uses an 18A power supply for design simplicity (the same module across many rails). There is no need for such a high-power module.	Refer to the data sheet for the expected power consumption.

A.2 BCM88690 SVK

The SVK is the validation platform for the device. During the device bring-up and validation, the hardware recommendations might be modified. As a result, contradictions exist between the recommendations in this document and the SVK design.

The following table describes the known contradictions between the BCM88690 SVK platform (BCM988690R_SVK_PB_design_example_02) and this version of the *Hardware Design Guidelines*.

When using the SVK as a design example, be aware of the items and recommendations in the following table.

Table 39: BCM88690 SVK Platform versus Hardware Design Guide Recommendations

No.	Subject	Issue	Recommendation
1	PVDD0P8 separation from T/RVDD0P8	The SVK shares the PVDD0P8 supply with T/RVDD0P8, while the recommendation is to have separate supplies.	Use separate supplies.
3	Via-in-pad structure	The via-in-pad structure violates the SMT recommendation because the via-in-pad design requires the via barrel (wall) metal to be fully covered by ball pad metal. For information about the SMT recommendations, refer to the BCM88690 document listed in Table 46.	Follow the SMT recommendation and validate and optimize the via structure and BGA break-out in 3D SI simulations.
4	VDDC PDN	The SVK PDN has not been simulated to meet the data sheet requirements.	Perform a PDN simulation and install capacitors as required to meet the data sheet requirements.
5	HBM power/GND vias	The SVK layout does not include a GND via near each HBM power via.	Follow the *Hardware Design Guidelines* recommendation.

A.3 BCM88480 SVK

The SVK is the validation platform for the device. During the device bring-up and validation, the hardware recommendations might be modified. As a result, contradictions exist between the recommendations in this document and the SVK design.

The following table describes the known contradictions between the BCM88480 SVK platform (BCM988480R_SVK_PB_design_example_02) and this version of the *Hardware Design Guidelines*.

When using the SVK as a design example, be aware of the items and recommendations in the following table.

Table 40: BCM88480 SVK Platform versus Hardware Design Guide Recommendations

No.	Subject	Issue	Recommendation
1	VDDC PDN	The SVK PDN has not been simulated to meet the data sheet requirements.	Perform a PDN simulation and install capacitors as required to meet the data sheet requirements.
2	BCM88480 GDDR6 rails PDN	The SVK PDN has not been simulated to meet the data sheet requirements.	Perform a PDN simulation and install capacitors as required to meet the data sheet requirements.
3	GDDR6 length	The following issues exist: ■ Due to the socket requirement and the placement of the GDDR6 devices, the CK length is 4.2 inches, but it should be less than 4 inches. ■ The address/control group skew is 425 mils instead of 250 mils.	Follow the HWDG length requirements. On customer boards, a socket for the BCM88480 is not required, and the GDDR6 devices can be placed closer to the BCM88480.
4	GDDR6 CA10 connectivity	CA10 is connected to a test point and not to the GDDR6 devices. As result, the SVK does not support 24 Gb and 32 Gb GDDR6 devices.	On customer boards, if large GDDR6 devices are expected, connect CA10 between the BCM88480 and the GDDR6 devices. Otherwise, keep CA10 as terminated. For more information, see Section 7.4.1.2, CA[10].
5	SMT recommendation	The SVK is available before the SMT guidelines are finalized. Differences may exist between the SVK SMT and the latest SMT guidelines.	Follow the information in Section 13, PCB Land Pattern and Reflow Process Recommendations.
6	GDDR6 trace clearance	GDDR6 RESET_N clearance rules were not applied.	Follow GDDR6 clearance rules as required in Section 7.6.9, Trace Clearances.
7	PCB routing guidelines	The SVK did not back drill some low-layer address/control signals.	Back drill all DDR signals.
8	GDDR6 VREFC setting	The SVK defines VREFC as 0Ω to GND. Section 7.5.3, GDDR6 Device VREFC Pin, defines VREFC as 240Ω to GND by default.	Use 240Ω to GND unless the memory vendor does not allow this value and requests another value for the internal VREFC setting.

A.4 BCM88290 SVK

The SVK is the validation platform for the device. During the device bring-up and validation, the hardware recommendations might be modified. As a result, contradictions exist between the recommendations in this document and the SVK design.

The following table describes the known contradictions between the BCM88290 SVK platform (see Related Documents) and this version of the *Hardware Design Guidelines*.

When using the SVK as a design example, be aware of the items and recommendations in the following table.

Table 41: BCM88290 SVK Platform versus Hardware Design Guide Recommendations

No.	Subject	Issue	Recommendation
1	VDDC PDN	The SVK PDN has not been simulated to meet the data sheet requirements	Perform a PDN simulation and install capacitors as required to meet the data sheet requirements.
2	BCM88290 GDDR6 rails PDN	The SVK PDN has not been simulated to meet the data sheet requirements.	Perform a PDN simulation and install capacitors as required to meet the data sheet requirements.
3	GDDR6 CA10 connectivity	CA10 is connected to a test point and not to the GDDR6 devices. As result, the SVK does not support 24 Gb and 32-Gb GDDR6 devices.	On customer boards, if large GDDR6 devices are expected, connect CA10 between the BCM88480 and the GDDR6 devices. Otherwise, keep CA10 as terminated. For more information, see Section 7.4.1.2, CA[10].
4	PCB routing guidelines	The SVK did not back drill some low-layer address/control signals.	Back drill all DDR signals.
5	GDDR6 VREFC setting	The SVK defines VREFC as 0Ω to GND. Section 7.5.3, GDDR6 Device VREFC Pin, defines VREFC as 240Ω to GND by default.	Use 240Ω to GND unless the memory vendor does not allow this value and requests another value for the internal VREFC setting.

Appendix B: DNX Hardware Reviews

The following table describes the DNX hardware reviews that should be maintained for each DNX16 device design.

All reviews are initiated only after the project is officially approved.

Table 42: DNX16 Hardware Reviews

Design Phase	Review Name	Review Details
Board Schematics	Schematics Review	Purpose: To review the electrical schemes (DNX device section only) and ensure they meet Broadcom device requirements. Customer Role: Complete the scheme's review checklist file (available on the ESP), upload the completed file to the support site for review, and specify any particular areas of concern. **NOTE:** Customers assume the responsibility of thoroughly reviewing their design. Broadcom performs only a sample review of the schemes, validates that the checklist concept is understood by the customer, and answers any doubt or concern raised by the customer. The review and support is limited to the DNX device section only.
	SerDes Allocation Review	Purpose: To verify the SerDes allocation meets the constraints of the specific part number. Customer Role: Complete the SerDes allocation spreadsheet (available on the ESP). **NOTE:** The review covers the features of the DNX device but not the physical connectivity in the system. The connectivity in the system should be handled by the customer during the schematics review and following layout, if any lane swap was done.
	IEEE 1588 and BroadSync Review	Purpose: To validate the IEEE 1588 and BroadSync design. Customer Role: Provide application requirements, system block diagram, and detailed IEEE 1588 and BroadSync connectivity information. Upload this information to the support site for review and specify any particular areas of concern.
	System Thermal Design and Simulation Reviews	Purpose: Review the customer system thermal design and thermal simulation results. Usually performed for devices that are higher than 200W. Customer Role: Provide all information required by the `StrataDNX-StrataXGS-Thermal_Design_Checklist` file. Upload this information to the support site for review and specify any particular areas of concern. **NOTE:** This process is available for special cases only. Contact your Broadcom support team.

Table 42: DNX16 Hardware Reviews (Continued)

Design Phase	Review Name	Review Details
PCB Layout	Layout Review	Purpose: To validate the layout review process that was made by the customer and address key areas, if needed. Customer Role: Complete their internal layout review process, verify all data sheet and *Hardware Design Guidelines* requirements are met, and verify all required simulations (SerDes SI, DRAM SI, and PI) were performed. **NOTE:** The Broadcom support teams do not perform an actual layout review. The layout review is performed by the customer. For any concerns or questions regarding the layout, the customer is expected to open a discussion with Broadcom.
	Signal-Integrity and Power-Integrity Simulation Reviews	Purpose: To validate the simulation processes the customer made for SerDes SI (COM and IBIS-AMI), DRAM SI, and power integrity. Customer Role: Provide all simulation results, including SerDes COM simulation results and IBIS-AMI simulation results, upload any relevant information to the support site for review, and specify any particular areas of concern.
	BCM88480 DRAM Connectivity Logic Check	Purpose: To validate the connectivity for DRAM interfaces. Customer Role: Provide the board net-list in `*.tel` format (as a minimum, provide the GDDR6 section) and upload the information to the support site for review.
	BCM88480 DRAM Routing Length Check	Purpose: To validate the routing length of the DRAM interfaces. Customer Role: Provide the routing lengths in the *DNX16 BCM88480 GDDR6 Length* Excel spreadsheet provided by Broadcom and available on the ESP and upload the completed spreadsheet to the support site for review.

Related Documents

The references in this section may be used with this document.

NOTE: Broadcom provides customer access to technical documentation and software through its Engineering Support Portal (ESP) at esp.broadcom.com.

For document numbers and file names, replace the “xx” or “X.X” with the largest number available in the repository to ensure that you have the most current version of the document.

Table 43: Device Data Sheets

Document Name	Number
BCM88790 data sheet: *Self-Routing Switching Element with 50-Gb/s SerDes*	88790-DS1xx
BCM88690 data sheet: *9.6-Tb/s Integrated Packet Processor, Traffic Manager, and Fabric Interface Single-Chip Device*	88690-DS1xx
BCM88800 data sheet: *4.8-Tb/s Integrated Packet Processor and Traffic Manager Single-Chip Device*	88800-DS1xx
BCM88480 data sheet: *800-Gb/s Integrated Packet Processor and Traffic Manager Single-Chip Device*	88480-DS1xx
BCM88280 data sheet: *360-Gb/s Integrated Packet Processor and Traffic Manager Single-Chip Device*	88280-DS1xx
BCM88290 data sheet: *100-Gb/s Integrated Packet Processor and Traffic Manager Single-Chip Device*	88290-DS1xx

Table 44: Errata

Document Name	Number
BCM88790 Device Errata	88790-ER1xx
BCM88690 Device Errata	88690-ER1xx
BCM88800 Device Errata	88800-ER1xx
BCM88480 Device Errata	88480-ER1xx
BCM88280 Device Errata	88280-ER1xx
BCM88290 Device Errata	88290-ER1XX

Table 45: Pin Lists

Document Name	File Name
BCM88790 PinList and Package Info	BCM88790_PinList_and Package_info_RevX.X.zip
BCM88690 Final PinList	BCM88690_PinList_RevX.X.zip
BCM88800 PinList	BCM88800_PinList_RevX.X.zip
BCM88480 PinList	BCM88480_PinList_vX.X.zip
BCM88290 Pin List	BCM88290_PinList_vX.X.zip

Table 46: Application Notes

Document (or Item) Name	Number or File Name
SerDes Configuration and Debugging Guide for StrataDNX™ 16-nm and 7-nm Devices	DBG16S-AN1xx
StrataDNX GDDR6 Interface Configuration and Debugging Guide	StrataDNX-GDDR6-AN1xx
StrataDNX HBM Configuration and Debugging Guide	DNX-HBM-AN1xx
Thermal Considerations for High-Power Switching Devices	StrataDNX-StrataXGS-AN1xx
Thermal Design Checklist for High-Power Switching Devices	StrataDNX-StrataXGS-Thermal_Design_Checklist_vX.X
Thermal Design Checklist	Customer-System-Device_Thermal-Design-Checklist vX.xlsx
Broadcom BGA Product System Load Force Qualification Test Recommendation	BRCM-Load-Force-Qual-Requirement_9-27-2018
BCM88790 SMT Recommendations	BCM88790-SMT-Guide_9-07-2017
BCM88690 SMT Recommendations	BCM88690_SMT-Guide-r0_4-05-2018
BCM88800 SMT Recommendations	BCM88800_SMT-Guide-r0_1-08-2020
Pkg Reflow Process Guidelines for Surface Mount Assemblies[a]	PACKAGING-AN1xx
BCM88790 Shadow Moire Data	BCM88790_Shadow_Moire_data_vX
BCM88690 Shadow Moire Data	BCM88690_Shadow_Moire_data_vX
BCM88800 Shadow Moire Data	BCM88800_Shadow_Moire_data_vX

a. Applicable for the BCM88480, BCM88280, and BCM88290 only.

Table 47: GDDR6 Length Check

Document Name	File Name
DNX16 Reference Length for GDDR6	DNX16 GDDR6 Length Excel vX.X.zip

Table 48: SerDes Allocation Checkers

Document Name	File Name
BCM88690 SerDes Allocation Checker vX	BCM88690_SerDes_Allocation_Checker_x.zip
BCM88800 SerDes Allocation Description Rev xx	BCM88800_Serdes_Allocation_xx.zip
BCM88280, BCM88480 SerDes Allocation Description Rev xx	BCM88480_Serdes_Allocation_xx.zip

Table 49: SVK Design Examples

Document Name	File Name
BCM88790 SVK Design Example (Schematics and Layout)	BCM988790R_SVK_PB_design_example_xx.zip
BCM88690 SVK Design Example (Schematics and Layout)	BCM988690R_SVK_PB_design_example_xx.zip
BCM88800 SVK Design Example (Schematics and Layout)	BCM988800R_SVK_PB_design_example_xx.zip
BCM88480 SVK Design Example (Schematics and Layout)	BCM988480R_SVK_PB_design_example_xx.zip
BCM88290 SVK Design Example (Schematics and Layout)	BCM988290R_SVK_PB_design_example_xx.zip

Table 50: Schematics Review Checklists

Document Name	File Name
BCM8879X Schematics Review Checklist	BCM8879x_sch_review_checklist_X.X.zip
BCM8869X Schematics Review Checklist	BCM8869x_sch_review_checklist_X.X.zip
BCM88800 Schematics Review Checklist	BCM88800_sch_review_checklist_X.X.zip
BCM88480 Schematics Review Checklist	BCM88480_sch_review_checklist_X.X.zip
BCM88290 Schematics Review Checklist	BCM88290_sch_review_checklist_X.X.zip

Table 51: IBIS-AMI Models

Document Name	File Name
Blackhawk 16-nm PAM4 IBIS-AMI Model	BRCM_AMI_Blackhawk_16nm_PAM4_vX.X.zip
Blackhawk 16-nm NRZ IBIS-AMI Model	BRCM_AMI_Blackhawk_16nm_NRZ_vX.X.zip
Falcon 16-nm IBIS-AMI Model	BRCM_AMI_falcon16_vX.X.zip
PCIe Gen3 16-nm IBIS-AMI Model	BRCM_AMI_PCIeGen3_16nm_vX.zip

Table 52: Package Models

Document Name	File Name
BCM88790 Blackhawk Package Model – Shortest	BCM88790 Blackhawk Package Model - Shortest -vX.zip
BCM88790 Blackhawk Package Model (Part 1/2)	BCM88790 Blackhawk Package Model - Longest-vX.zip.xxx
BCM88790 Blackhawk Package Model (Part 2/2)	BCM88790 Blackhawk Package Model - Longest-v1.zip.xxx
BCM88790 PCIe Package Model	BCM88790 PCIe Package Model - vX.zip
BCM88690 Blackhawk Package Model	BCM88690 Blackhawk Package Model - vX.zip
BCM88690 PCIe Package Model	BCM88690 PCIe Package Model - vX.zip
BCM88800 PCIe Package Model	BCM88800 PCIe Package Model - vX.zip
BCM88800 Blackhawk Package Model	BCM88800 Blackhawk Package Model - vX.zip
BCM88480 Blackhawk Package Model	BCM88480 Blackhawk Package Model - vX.zip

Table 53: IBIS Models

Document Name	File Name
BCM8879X REFCLK IBIS Model	BCM8879X_refclk_X.X_ibis.zip
BCM8869X REFCLK IBIS Model	BCM8869X_refclk_X.X_ibis.zip
BCM88480 Family GDDR6 I/O IBIS Models VX.X Preliminary	BCM8848x_gddr6_ibis_XvX_preliminary.zip

Table 54: Thermal Models

Document Name	File Name
BCM88790 Block Thermal Model	BCM88790_Block_Thermal_Model_vX.zip
BCM88690 Block Thermal Mode	BCM88690_Block_Thermal_Model_vX.zip
BCM88800 Block Thermal Model	BCM88800_Block_Thermal_Model_vX.zip
BCM88480 Block Thermal Model	BCM88480_Block_Thermal_Model_vX.zip
BCM88280 Block Thermal Model	BCM88280_Block_Thermal_Model_vX.zip
BCM88290 Block Thermal Model	BCM88290_Block_Thermal_Model_vX.zip

